

Co-benefits of Substance Abuse Regulation on Temperature and Violent Crime*

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Abstract

Higher temperatures can increase substance abuse and exacerbate its physiological effects, raising the risk of violent behavior. Combining daily administrative crime records and temperatures in the United States, we use a triple difference approach to study how the temperature-crime relationship changed around the 2010 abuse-deterrent reformulation of the prescription opioid OxyContin across counties with varying pre-policy opioid exposure. We find that the reformulation moderated the temperature response of violent crimes. We then monetize the social benefits for avoided temperature-induced violent crimes and argue that standard welfare evaluations of substance abuse regulation understate its social value in a warming climate.

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JEL Classification: I18, K32, K42, Q51, Q54

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1 Introduction

Higher temperatures pose growing challenges for public health and economic well-being. In a welfare-maximizing framework, climate adaptation policies are evaluated through cost-benefit analysis that should include both direct and ancillary effects ([Hahn et al., 2025](#); [Hendren and Sprung-Keyser, 2020](#)). Within this framework, two broad classes of adaptation policies emerge ([Carleton et al., 2024](#)): interventions explicitly designed to reduce climate damages (e.g., cooling centers, seawalls), and policies with other primary objectives but that can nevertheless generate adaptation benefits (e.g., healthcare facilities, cash transfers, social safety nets).

A large body of evidence documents that higher temperatures increase aggression and violent behavior (e.g., [Anderson, 1987, 2001](#); [Kenrick and MacFarlane, 1986](#); [Ranson, 2014](#)). This relationship has weakened over time ([Burke et al., 2024](#)), but the interventions that account for the decline remain largely unknown ([Carleton et al., 2024](#)). Substance use is a potential explanation. Alcohol, opioids, and other substances impair cognitive and emotional functions. Higher temperatures raise their consumption ([Cohen and Gonzalez, 2024](#)) and amplify their behavioral effects through physiological and psychological channels ([Hensel et al., 2021](#); [Jhang et al., 2025](#)), leading to more violent behavior ([Luca et al., 2015](#)). As a result, policies targeting substance abuse may generate ancillary benefits by moderating the effects of temperature on violent crime.

We examine whether public policies that regulate access to addictive substances can mitigate one of the major social consequences of rising temperatures: violent behavior and crime. We combine daily crime records from the FBI’s National Incident-Based Reporting System with daily temperatures from ERA5-Land over 1991 to 2023. Our primary analysis leverages a nation-wide intervention in 2010 that reformulated OxyContin — the most widely prescribed legal opioid at the time — to make the pills resistant to crushing or dissolving, thereby curbing non-medical misuse and its addictive risks. We combine this policy variation with daily temperature fluctuations to estimate whether and how this intervention altered the temperature-crime relationship. We corroborate our analysis leveraging variation in the staggered opening of substance abuse treatment (SAT) facilities across counties between 1998 and 2016, a second policy margin that shifts local treatment capacity and that we show moves independently of the reformulation.

We begin by documenting four motivating facts that inform our design. First, higher temperatures raise crime, and the response is concentrated in those crimes that involve contact between victim and offender, and is small and statistically imprecise for crimes that involve limited interactions between victims and offenders, such as fraud and gambling. Second, using individual survey data we document that higher temperatures are positively associated with alcohol con-

sumption and heavy drinking, with effects more pronounced in counties with higher opioid use, providing suggestive evidence of “polysubstance use”, i.e., a complementarity between the two substances (Esser et al., 2019, 2021). We also show that higher temperatures exacerbate the health consequences of opioid use, increasing opioid-related emergency department visits. Third, higher temperatures physiologically accentuate the behavioral effects of substance use, by increasing violent crimes recorded as substance-involved. Fourth, the temperature–violent crime gradient is steeper in counties where opioid prescribing was higher before the reformulation. Altogether, these facts indicate that substance use is a major channel in the temperature-crime relationship.

Building on these empirical patterns, we design a triple difference strategy to estimate the extent to which substance abuse regulation moderates the effect of temperature on crime. Our baseline specification compares crime on relatively hotter or colder days within the same jurisdiction and month-year, absorbing day-of-week, week-of-year, and day-of-year effects. Because the reformulation applied nationally in a single year, identification comes from the intensity of pre-policy exposure rather than from the timing of adoption, and we absorb nationwide changes in the temperature-crime relationship by allowing for year-specific weather effects. We also allow weather effects to vary by state, so that the pre-policy differential between counties with high and low opioid prescribing levels is measured within states. In a more restrictive specification, we allow them to vary by county, which accounts for unobserved spatial differences in the weather-crime relationship at the cost of no longer identifying the pre-policy differential. The identifying assumption is that, conditional on these spatial and year-specific weather effects, the temperature-crime relationship would have evolved in parallel across counties with different pre-policy opioid prescribing levels in the absence of the reformulation. Therefore, the residual concern is limited to unobserved factors that changed how daily temperatures affect crime within jurisdiction-months after 2010, differentially by pre-policy opioid prescribing, and not to differences across counties in the level of crime, or in the extent to which crime is reported. We probe this concern with several observable county characteristics, allowing the weather-crime relationship to change after the reformulation with local labor market conditions, demographic composition, and police staffing.

Our baseline findings show that a 1°C increase is associated with 0.0085 fewer cases per 100,000 people per additional prescription per capita after the 2010 opioid reformulation, which corresponds to roughly an offset of 50% of its pre-policy value and around 0.2% of the pre-policy violent crime mean. In an annual event-study design, none of the pre-policy estimates is distinguishable from zero, providing support to our parallel trend assumption, and the attenuating effect of the policy persists over more than ten years through 2023. By unpacking the contribution of the policy to the decline of the crime sensitivity to temperatures over time, we quantify that

the policy accounts for approximately 78% of the decline. We document that the categories of violent crime where the policy offset is most pronounced are intimidation and simple assaults, while the response is muted for monetary crimes, such as fraud and gambling, that require a planning horizon and do not respond to acute episodes of intoxication (Goldstein, 1985).

We explore several mechanisms and heterogeneity dimensions through which the OxyContin reformulation attenuates the temperature-violent crime gradient, and examine how this policy interacts with other substance abuse regulations. First, we show that the reformulation reduces more strongly the impact of higher temperatures on substance-involved violent crimes (approximately 75% of the opioid-attributable gradient) than crimes with no recorded involvement (approximately 44%). This reduction is largest for alcohol-involved crimes, providing suggestive evidence of cross-substance interactions and consistent with polysubstance use.

Second, we do not find evidence that the effectiveness of the reformulation depends on the scope for substitution across substances. We examine two margins available in our data: proximity to the U.S.-Mexico border, which proxies access to illicit opioids (Evans et al., 2019; Sabia et al., 2024), and state legal access to medical marijuana (Evans et al., 2022). The moderating effect of the reformulation on the temperature-crime relationship is similar across these margins.

Third, we explore the generalizability of our findings to other policies that regulate substance abuse. Using variation in the staggered opening of SAT facilities, we find consistent evidence that increased local treatment capacity attenuates the temperature-crime gradient. We also show that the local temperatures, pre-intervention opioid prescriptions, and the OxyContin reformulation do not predict openings of SAT facilities, and that the two policies vary independently, supporting the interpretation of SAT expansion as a corroborating source of variation rather than a confound.

Fourth, we document heterogeneity consistent with biophysical adaptation. The OxyContin reformulation is more effective in colder and temperate counties and is weaker in hotter areas. This pattern is consistent with greater physiological tolerance to high temperatures in warmer climates, which weakens the interaction between heat and substance abuse, and thus limits the scope for substance abuse regulation to moderate temperature-driven crime in these areas.

Last, we examine the temporal dimension of the moderating effect. Leveraging intra-daily temperature variation, we find that the policy reduces the temperature sensitivity of violent crime in the evenings and at night, consistent with substance consumption peaking during leisure hours (Kuntsche and Labhart, 2012). We also find larger effects on weekends, when individuals have greater flexibility to adjust time use and substance consumption is highest (Yan and Kuo, 2019).

To conclude, we quantify the temperature-related monetary benefits of substance abuse policies through reductions in violent crimes. We estimate that, for a 1°C increase in average tem-

perature, the opioid reformulation generates annual social benefits of about \$132 million (in 2023 US\$) from avoided assaults, while including avoided rape and homicide raises the aggregate benefits to \$838 million. Because such adaptation benefits are excluded from standard evaluations of substance-abuse interventions, our results suggest that existing welfare analyses may understate the social value of these policies in a warming climate.

Our findings contribute to three literatures. First, we add to a well-established body of evidence on the effect of temperature on violent behavior, one of the largest socioeconomic costs of a warming climate (Carleton and Hsiang, 2016). Higher temperatures induce aggression and violent behavior through psychological, behavioral, and physiological channels (Baylis, 2020; Behrer and Bolotnyy, 2022; Cohen and Gonzalez, 2024; Evans et al., 2025; Fetzer, 2020; Heilmann et al., 2021; Mukherjee and Sanders, 2021; Ranson, 2014). Within this literature, Cohen and Gonzalez (2024) document that higher temperatures increase alcohol consumption and that this channel accounts for part of the temperature-crime gradient. We extend prior work and provide three novel pieces of evidence: the temperature–drinking gradient is sharper in counties more exposed to the opioid epidemic, consistent with polysubstance complementarities; higher temperatures raise the rate of substance-involved violent crime, and the temperature–crime gradient is steepest where opioid prescription was highest pre-reformulation.

Second, we contribute to the literature evaluating policies that moderate the impact of climate shocks on socioeconomic outcomes (see Carleton et al. (2024) and Kala et al. (2026) for a review, and the Adaptation Inventory (Climate Impact Lab, 2026) for an updated catalogue). Most prior work studies how income-based interventions — cash transfers, humanitarian aid, and earnings shocks — buffer climate impacts on human capital, food security, mortality, and conflict (Adhvaryu et al., 2024; Avdeenko and Frölich, 2025; Banerjee and Maharaj, 2020; Baysan et al., 2019; Christian et al., 2019; Fetzer, 2020; Garg et al., 2025, 2020; Macours et al., 2022; Premand and Stoefler, 2022; Sarmiento et al., 2024). Evidence on non-income interventions is much thinner: Cohen and Dechezleprêtre (2022) and Mullins and White (2020) show that health-care access attenuates temperature-mortality, and Colmer and Doleac (2026) show that restrictive gun laws attenuate the temperature–homicide gradient. We provide novel evidence on a distinct non-income intervention: regulation of substance abuse. Whereas restrictive gun laws act on the means through which temperature-induced violent behavior translates into homicide, our policy of interest acts on the source of impairment itself on a larger battery of violent crimes.

Third, we contribute to the literature on substance abuse regulation. Prior work establishes that the 2010 OxyContin reformulation sharply reduced abuse (Cicero and Ellis, 2015). This policy sits within a broader body of work on the social and economic consequences of the U.S. opioid

epidemic (Arteaga and Barone, 2026; Dave et al., 2025), which has also documented unintended costs and benefits: reformulation and prescription-monitoring regimes increased child maltreatment and reduced foster-care entry (Evans et al., 2022; Gihleb et al., 2022), and heroin substitution raised overdose deaths (Alpert et al., 2018; Evans et al., 2019). We document a complementary, previously unmeasured benefit: the opioid reformulation reduced the sensitivity of violent crime to temperature, a welfare-relevant externality that is absent from standard cost-benefit assessments of substance-abuse regulation. We do not argue that these policies were chosen, or should be evaluated, on climate adaptation grounds, but rather that the social value of this policy is larger than current evaluations imply in a warming climate.

2 Data and policy background

In this section, we describe the data for our empirical analysis. We retrieve comprehensive administrative data on reported crimes at the finest temporal and geographic scale available in the U.S. (Section 2.1), and we combine them with climate data (Section 2.2). We describe and provide background information on our substance abuse regulation policy of interest, the abuse-deterrent reformulation of OxyContin in 2010 (Section 2.3). We complement our analysis with supplementary data sources at the individual-, county-, and state-level to explore additional policies and mechanisms of the relationship between temperatures, substance abuse, and crime (Section 2.4). [Appendix Table A1](#) reports summary statistics for the main variables.

2.1 Crime data

We use data from the FBI’s National Incident-Based Reporting System (NIBRS) for the period between 1991 and 2023. NIBRS records all criminal incidents reported to individual law enforcement agencies (ORIs or jurisdictions) and includes detailed information on the characteristics of the victim (e.g., age, gender), the offender (e.g., gender and relationship to the victim), the crime (e.g., category of crime, substances involved), and the incident date and time.

Leveraging the reported offenses within each incident, we construct daily counts by crime category.¹ Broadly, we categorize crimes into two main categories: violent and monetary. The former category includes assault, which we split into intimidation, simple assault, and aggravated

¹Unlike the traditional Uniform Crime Reports (UCR) system, which records only the most serious offense in an incident (“hierarchy rule”), NIBRS allows agencies to report up to ten types of crime per incident. We treat each incident as one crime when constructing overall crime counts. When computing crime-specific counts, we count each reported crime separately. For example, if an incident involves assault and vandalism, we count each as one additional crime in each respective category.

assault², rape, and homicide; the latter includes burglary, fraud, gambling, larceny, motor-vehicle theft, and robbery. For each crime category, we compute the daily rate of reported incidents per 100,000 people.³

NIBRS reports whether the offender was under the influence of substances at the time of the offense. This information, however, is available only for alcohol and an unspecified drug category. To obtain substance-specific measures, we refine our classification of drug-involved crimes using detailed information on the types of drugs seized during the incident, including heroin, cocaine, marijuana, stimulants, and hallucinogens.⁴

A primary challenge with NIBRS data is that the number of reporting agencies increase over time, from 609 in 1991 to 13,125 in 2023, with agencies entering and exiting the sample throughout the period (Figure A1). This unbalanced structure has an important implication for identification. Because the composition of reporting jurisdictions changes across years, comparing crimes on a given calendar day across different years would confound true changes in criminal activity with changes in sample composition: a hot July 15th in 2005 is observed in a different set of jurisdictions than a hot July 15th in 2015. For this reason, we construct a panel at the jurisdiction-day level that is balanced within each year, and leverage within-month-year variation in daily temperatures to identify our estimates. Two features of the design limit the scope for reporting behavior to drive our estimates. The jurisdiction-month-year fixed effects absorb the reporting level of each jurisdiction in each month, so a gap in reporting affects the temperature coefficient only if it falls on days that are systematically warmer or colder than the rest of that month. The sample we use to estimate the policy effects starts in 2006, after most of the growth in the number of reporting agencies.⁵ In additional robustness checks, we exclude any agency that did not report crimes in all twelve months of a given year (Colmer and Doleac, 2026), which removes the partial years of entering and exiting agencies, and we only consider a balanced sample of agencies throughout the entire time period. Furthermore, we also confirm that our results are robust to aggregating

²According to NIBRS intimidation involves threats or fear without any physical contact or weapon; simple assault involves minor or no injury without a weapon; and aggravated assault involves a weapon or severe injury.

³We construct population at the jurisdiction-level combining two different sources: the ICPSR website at the University of Michigan, and the FBI's Crime Data Explorer (CDE). Jurisdiction-level population is not available for all the agencies. We test that our results are robust to the unrestricted sample, using the daily count of reported crimes as a dependent variable and estimating a PPML model.

⁴For example, a crime is classified as "heroin-involved" if the offender is reported as under the influence of drugs and heroin is seized during the incident. As with any administrative data, measurement error may arise from over- or under-reporting of specific substances, particularly in cases of polysubstance use.

⁵We also test for this directly. In a county-year panel for 2006–2023, we regress an indicator for whether the county has at least one agency reporting to NIBRS in that year on the interaction between pre-policy opioid exposure and the post-2010 indicator, with county and state-by-year fixed effects and standard errors clustered by county. We cannot reject that reporting status is unrelated to exposure after the reformulation (p -value = 0.883).

observations to the jurisdiction-month and county-day levels, which addresses concerns that the daily frequency of our data mechanically inflates precision.

Another common concern when using administrative crime data is that recorded offenses may reflect changes in policing or reporting rather than underlying criminal behavior, if, for example, police effectiveness, evidence collection, or reporting behavior varied with temperatures. In these cases, sample selection could be confounding the relationship between temperatures and crime. While we cannot directly rule out these mechanisms, prior work finds that they play a limited role in explaining the temperature-crime relationship (e.g., [Cohen and Gonzalez, 2024](#); [Evans et al., 2025](#); [Heilmann et al., 2021](#)). In our setting, our baseline specification accounts for short-run local unobserved confounders such as heat adaptive policing, local enforcement practices, or temporary staffing changes, by holding characteristics for each month-year in a given jurisdiction fixed. We also measure enforcement capacity directly, using annual counts of sworn officers per jurisdiction from the FBI’s Law Enforcement Officers Killed and Assaulted (LEOKA) program ([Kaplan, 2025](#)). In robustness checks, we allow the temperature-crime gradient to change after the reformulation with pre-policy officers per capita, so that jurisdictions with thinner patrol capacity are not driving the estimated moderation. We also show that our results are robust to alternative specifications that vary the spatial and temporal structure of fixed effects, thereby capturing different potential unobserved changes in evidence gathering, police effectiveness, and reporting behavior.

2.2 Weather data

We process weather data from the ERA5-Land reanalysis product ([Muñoz Sabater, 2019](#)), which provides hourly temperature and precipitation from 1950 to present at a 0.1° spatial resolution ($\approx 11\text{km}$). We combine weather data with 30 arc-seconds ($\approx 1\text{km}$) population density information ([Seirup and Yetman, 2006](#)) to compute the county-level population-weighted average daily temperature and total precipitation. In additional analysis, we also construct sub-daily temperature exposure during different times of the day (morning, afternoon, evening, and night).

2.3 2010 OxyContin reformulation

The central policy variation we leverage is the 2010 reformulation of OxyContin, which introduced abuse-deterrent properties that made the tablets substantially harder to crush, dissolve, or inject for non-medical use. The reformulation came at the end of two decades of rapid expansion in opioid prescriptions, with the number rising from 76 million in 1991 to more than 250 million by 2010 ([Volkow et al., 2014](#)), a trajectory shaped in large part by Purdue Pharma’s aggressive marketing of OxyContin — a controlled-release oxycodone formulation introduced in 1996 — to

physicians across the country (Alpert et al., 2022). Oxycodone is a semi-synthetic opioid analgesic indicated for moderate-to-severe pain and one of the most heavily diverted prescription opioids in the United States, with well-documented risks of dependence and non-medical misuse.

To address mounting concerns about misuse and diversion, Purdue developed an abuse-deterrent formulation that blocked non-oral routes of administration by making the tablets resistant to crushing and dissolution. The U.S. Food and Drug Administration approved the new formulation in April 2010, and nationwide distribution began in August 2010, when the original formulation was discontinued without advance public notice. The reformulation produced a sharp decline in OxyContin misuse (Cicero and Ellis, 2015; Sessler et al., 2014), but it also triggered substitution toward illicit opioids such as heroin and synthetic opioids, reflected in the subsequent rise in overdoses (Powell and Pacula, 2021). Our identification strategy treats the 2010 nationwide reformulation of OxyContin as an unanticipated, supply shock: although Purdue had been developing the abuse-deterrent formulation internally, FDA approval was secured only in April 2010, and the discontinuation of the original formulation in August was a unilateral commercial decision by Purdue rather than a regulatory mandate. Our pre-reformulation window (2006–2009) therefore precedes both FDA approval and any public knowledge of the switch.

We measure pre-reformulation exposure to prescription opioids at the county-level as the average number of Schedule II opioid prescriptions per capita over 2006–2009, obtained from the Centers for Disease Control and Prevention (Evans et al., 2022). The CDC data cover approximately 85% of retail pharmacy providers but exclude hospital dispensing. Figure A2 shows the spatial distribution of population-weighted average number of opioid prescriptions per capita in the pre-reformulation period. The sample average is 0.88 opioid prescriptions per capita (Table A1 reports weighted averages), and about 30% of counties recorded more than one prescription per resident. Although this measure captures a broader class of Schedule II opioids than OxyContin alone, the cumulative effect of Purdue’s marketing was to shift physician prescribing norms toward opioids generally, so that overall opioid prescribing rates in this period closely track exposure to OxyContin promotion (Alpert et al., 2022). Relative to the state-level OxyContin-specific measures used in prior work (Alpert et al., 2018), the CDC data provide substantially finer geographic variation across counties.

The geographic variation in prescription opioid use during the 2006–2009 period reflects the spatial reach of Purdue’s sales force and the regulatory environment facing individual prescribers, rather than underlying differences in local medical need, injury rates, or criminal activity (Alpert et al., 2022; Evans et al., 2019). This supply-driven origin of geographic variation in opioid exposure is important for our analysis: it implies that counties with higher pre-intervention opioid

prescription rates differ from lower-exposure counties primarily in the intensity of pharmaceutical marketing they received. Our design does not require counties to be comparable in levels since it accounts for differences in the level of crime, in its seasonality, and in reporting practices. Section 4 assesses that concern with pre-policy event-study estimates and with county characteristics that are allowed to change the temperature response after the reformulation.

As an alternative source of variation, we construct exposure measures from the Drug Enforcement Administration’s Automation of Reports and Consolidated Orders System (ARCOS), which records shipments of controlled substances from manufacturers and distributors to retail pharmacies. Using ARCOS, we compute alternative measures of opioid exposure based on the average number of Schedule II opioid pills distributed and the number of shipments per capita over 2006–2009. Relative to prescription counts, these data more directly capture the supply of opioids pushed into each county by pharmaceutical distributors, and are therefore even less likely to reflect local demand-side factors such as injury rates or pre-existing crime conditions. This makes ARCOS-based measures useful for addressing the concern that counties with more violent crime mechanically recorded higher opioid prescriptions due to injury-driven demand. Our baseline results are robust to using this alternative measure, consistent with the interpretation that the identifying variation captures pharmaceutical supply rather than local medical need or criminal activity.

2.4 Additional data

As part of our analysis, we incorporate auxiliary data sources, summarized below.

Alcohol consumption. We use data from the U.S. Behavioral Risk Factor Surveillance System (BRFSS), an annual health-related telephone survey conducted by the Centers for Disease Control and Prevention. The BRFSS is one of the world’s largest continuous health survey systems and provides state-level representative information on several health dimensions between 1991 and 2012.⁶ The survey includes information on whether respondents consumed alcohol in the past months, and the average number of drinks consumed per drinking day. It also reports respondents’ socioeconomic and demographic characteristics. We construct a repeated cross-section with over four million observations, which we match to weather data using the interview date and county of residence.

⁶We end the sample in 2012, the last year for which county of residence is available.

Substance-abuse treatment facilities. In complementary analysis, we study the effect of opening substance abuse treatment (SAT) facilities. These centers constitute the primary setting for the delivery of treatment and rehabilitation services for individuals with substance use disorders. Admissions are primarily for alcohol (22%), opioids (19%), and combined alcohol and secondary drug use (18%). The establishment of new clinics typically depends on (i) local assessments of unmet treatment needs or opportunities to expand available services and (ii) the ability to secure funding (SAMHSA, 2018). We collect annual data on the number of open substance abuse treatment facilities at the county-level from the U.S. Census Bureau’s County Business Patterns (CBP) for the 1998–2016 period.⁷ For our main analysis, we restrict the sample to counties with at least one SAT facility open at any point during the study period. We then construct the annual rate of SAT facilities per 100,000 people at the county level. Figure A3 reports the total number of SAT establishments over time, steadily increasing over time. Figure A4 shows their distribution across the U.S., weighted by population. Between 1998 and 2016, U.S. counties have on average 6.14 SAT facilities per 100,000 people, with 16% of counties, especially in the Midwest and North-East, hosting more than 10 facilities per 100,000 people.

3 Motivating evidence: temperatures, crimes, and substance use

Changes in temperature can affect violent behavior through multiple channels. A large literature documents direct physiological and psychological effects of temperatures on impulse controls, irritability, and aggression (Anderson, 1987, 2001; Kenrick and MacFarlane, 1986). Substance use interacts with these channels in two ways. First, higher temperatures can increase the likelihood of substance consumption itself, through physiological stress, mental health deterioration, or changes in time use (Cohen and Gonzalez, 2024; Mullins and White, 2019), and consumption can in turn raise the propensity for aggressive behavior (Barron et al., 2024; Luca et al., 2015). Second, conditional on consumption, higher temperatures can amplify the physiological consequences of substances, for instance through mechanisms such as vasodilatation and blood pressure increase, increasing the risk of violent behavior (Herrnstein and Wilson, 1985; Miczek et al., 2004; Sim, 2023). Although these two channels are difficult to separate empirically, both imply that policies regulating substance abuse can attenuate the sensitivity of violent crime to temperature.

This section documents four empirical patterns behind our hypothesis. First, higher tempera-

⁷We identify substance abuse treatment establishments in the CBP data using two NAICS codes: 621420 (“Outpatient mental health and substance abuse center”) and 623220 (“Residential mental health and substance abuse facilities”) (Bondurant et al., 2018; Swensen, 2015). The CBP reports the number of SAT establishments in each county, but only for counties with at least one active facility. Starting in 2017, the CBP only reports counties with three or more SAT facilities, hence we restrict our analysis to data through 2016.

tures increase violent crime, and the temperature–violent crime gradient has weakened since 2010. Second, higher temperatures are associated with greater alcohol consumption, more so in counties with higher opioid use. Third, higher temperatures increase substance-involved violent offenses. Fourth, the temperature–crime gradient is steeper where substance use is higher, whether use is measured by county exposure to prescription opioids before the 2010 reformulation or by state-level drinking prevalence. Altogether, these patterns organize the channels through which temperature, substance use, and crime are linked, and motivate our approach in Section 4.

Fact 1: The temperature–crime gradient is positive and has weakened over time. The temperature–crime relationship is well-established in the United States and elsewhere (Cohen and Gonzalez, 2024; Colmer and Doleac, 2026; Heilmann et al., 2021; Ranson, 2014). We re-estimate it both to anchor magnitudes and because our sample of more than thirty years is, to our knowledge, the longest and most geographically comprehensive daily panel on which this relationship has been estimated. Identification comes from day-to-day temperature variation within a jurisdiction and month-year, accounting for week-of-year, day-of-year, and day-of-week fixed effects (see Appendix B for additional details).⁸

Figure 1 displays the estimated coefficients, normalized by the population-weighted mean of each outcome. A 1°C increase in daily temperature raises the overall crime rate by 0.7% of its mean. To contextualize this magnitude, the standard deviation of daily average temperature in our sample is approximately 10°C, so a one-standard-deviation warmer day corresponds to about 7% more crime.⁹ The effect is concentrated in interpersonal offenses. Violent crime rises by 1% per °C, with similar responses for simple assault, aggravated assault, and homicide, and slightly smaller for intimidation and rapes. In Table B2, we show that monetary crimes rise by 0.5%, and fraud and gambling — where interactions with victims are limited or absent — respond weakly. Both the magnitudes and the asymmetry between interpersonal and economically motivated offenses are consistent with prior evidence (Cohen and Gonzalez, 2024; Colmer and Doleac, 2026; Heilmann et al., 2021; Ranson, 2014).

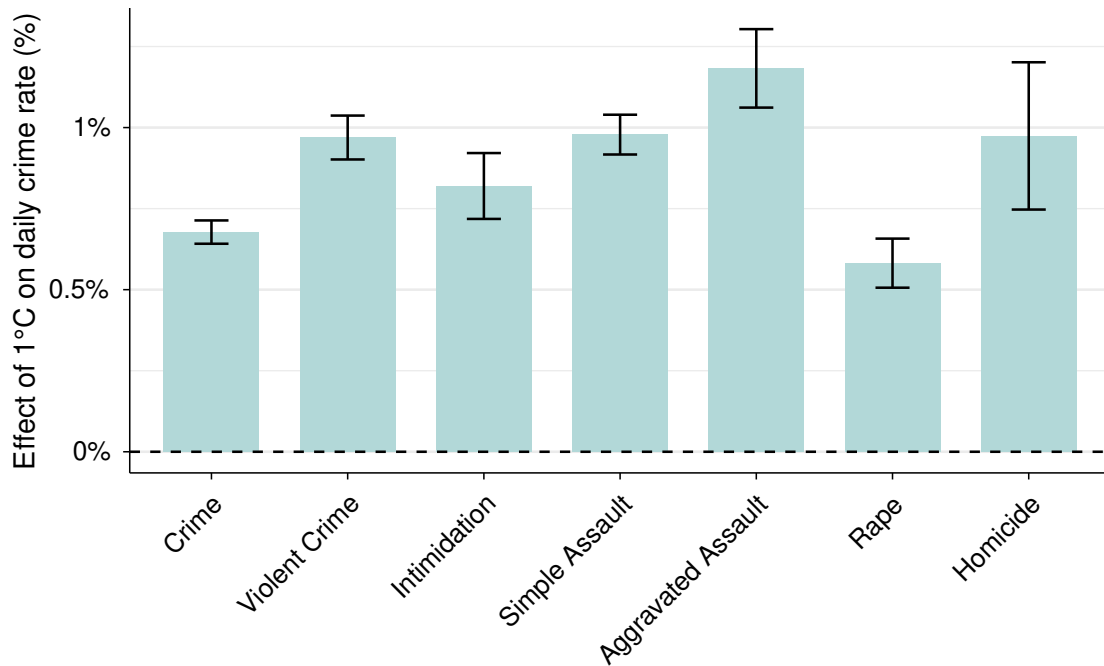
At least two mechanisms are consistent with these results. Under a physiological interpreta-

⁸In Appendix B, we test for alternative functional forms for between crime and weather and find that a linear functional form well approximates the relationship between the variables, in line with prior work (e.g., Cohen and Gonzalez, 2024; Colmer and Doleac, 2026; Heilmann et al., 2021), in particular for violent crimes, which are the main focus of this paper. We therefore adopt the linear specification throughout.

⁹We note that this coefficient captures the short-run behavioral response to day-to-day temperature fluctuations, that is, how crime on a relatively warmer day compares to crime on a cooler day in the same jurisdiction and month-year. After removing our fixed effects, the standard deviation of residualized daily temperature is 3.6°C. Our estimates do not represent the long-run response to a permanent 1°C increase in average temperatures, which may differ because of adaptation over longer horizons (Carleton et al., 2024).

tion, higher temperatures elevate stress hormones, reduce impulse control, and increase irritability, raising the probability of aggressive responses in social encounters (Anderson, 2001). Under a routine activity interpretation, higher temperatures shift time use toward outdoor and social settings, increasing the frequency of potentially conflictual interactions (Graff Zivin and Neidell, 2014). Both predict that crimes with direct confrontation respond more strongly to temperature than premeditated or opportunity-driven offenses, as we find, and both allow substance use to scale the response. What matters for our analysis is that under either interpretation, interventions that reduce substance-mediated aggression can moderate the temperature-crime gradient.

Figure 1: Effects of daily temperatures on daily crime rate by category in the U.S. through 1991 and 2023



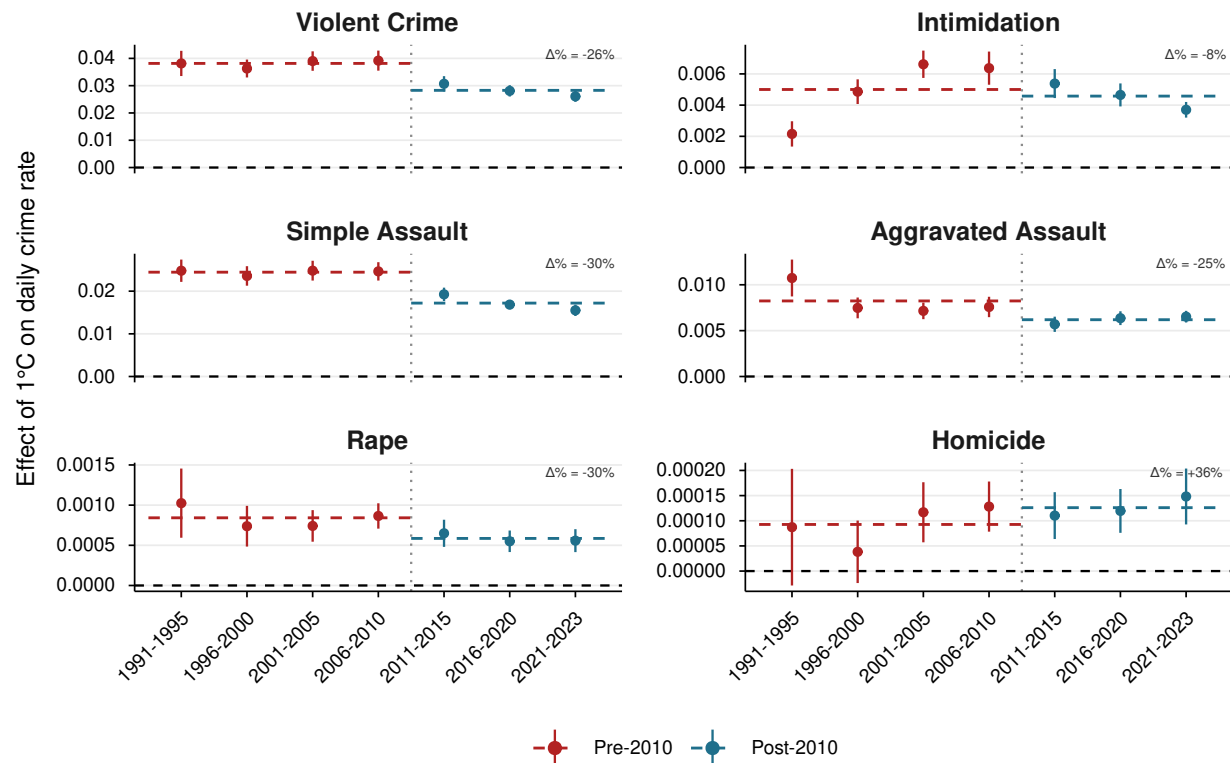
Notes: The figure plots the coefficients of temperature on crime rates (per 100,000 people) normalized by population-weighted sample mean of each outcome. The regressions control for daily total precipitation, and jurisdiction-month-year, day-of-year, week-of-year and day-of-week fixed effects. Regressions are weighted by jurisdiction population. Vertical bars show 95% confidence intervals based on standard errors clustered at the county-level. Table B1 reports the coefficients.

The estimated temperature–crime gradient is not a fixed feature of the data. Figure 2 re-estimates it separately by five-year period. For violent crime, the slope is stable across the four pre-2010 periods and then persistently lower in every period after, a decline of 26% between the pre- and post-2010 averages, with declines of similar size for simple assault, aggravated assault, and rape, while homicide, the rarest and noisiest category, is the exception.¹⁰ This evidence aligns

¹⁰We show that the decline is not an artifact of the expansion of data coverage: re-estimating on the balanced panel

with [Burke et al. \(2024\)](#), who use county-month data and estimate that the sensitivity of U.S. violent crime to temperature has fallen by about one third since the 1980s. Their approach measures the net change in sensitivity without identifying the actions that produced it. The timing of the decline in our data coincides with the substance-abuse policy we study, and in Section 4 we estimate how much of the decline that policy explains.

Figure 2: The temperature–crime gradient over time through 1991 and 2023



Notes: The figure plots the effect of a 1°C increase in daily mean temperature on daily crime rates (per 100,000 people), estimated separately by five-year period between 1991 and 2023 (the final period covers 2021–2023). Each panel refers to a different crime category: violent crime, simple assault, aggravated assault, intimidation, rape, and homicide. Dashed lines show the average of the period-specific coefficients before and after 2010, and each panel reports the percentage change between the two. The regressions control for daily total precipitation interacted with the same period indicators, and jurisdiction-by-month-by-year, day-of-year, week-of-year, and day-of-week fixed effects. Regressions are weighted by jurisdiction-location population. Vertical bars show 95% confidence intervals based on standard errors clustered at the county level.

Fact 2: Higher temperatures are associated with greater substance use and substance-related harm. Two pieces of evidence support substance use as a physiological mechanism linking temperature to violent behavior. First, using individual-level data from the Behavioral Risk Factor

of jurisdictions that report in every year yields a larger decline of 35% ([Figure B6](#)). We also show that the temperature response of monetary crimes falls as well ([Figure B5](#)).

Surveillance System (BRFSS), we document that higher temperatures raise alcohol consumption, extending [Cohen and Gonzalez \(2024\)](#). Leveraging county-level temporal variation in average temperatures over the thirty days prior to interview date, we find that a 1°C increase in daily average temperatures increases the likelihood of heavy drinking by 0.04-0.06 percentage points, corresponding to 0.6-0.8% at the mean ([Table 1](#), columns 1-3).¹¹ We also show that the effect of temperature on heavy drinking is significantly larger in high-opioid-exposure counties (column 4). In these counties, a 1°C increase in daily average temperature raises the likelihood of heavy drinking by 0.07 percentage points (0.93% at the mean), while the effect is smaller and not statistically significant in less exposed counties. This pattern is consistent with positive complementarity between alcohol and prescription-opioid consumption.¹²

Table 1: Effect of monthly temperatures on alcohol consumption using individual-level survey data

	Heavy Drinking (0/1)			
	(1)	(2)	(3)	(4)
Temperature	0.000468*** (0.0001)	0.000569*** (0.0002)	0.000425** (0.0002)	0.000336 (0.0004)
Temperature × High opioid exposure				0.000325** (0.0002)
Period	1991-2012	1991-2012	1991-2012	2006-2009
Observations	3,994,304	3,994,304	3,793,862	1,228,916
Mean Outcome	0.073	0.073	0.073	0.071
Precipitation controls	✓	✓	✓	✓
County FE	✓	✓	✓	✓
Date FE	✓	✓	✓	✓
State-Month FE		✓	✓	✓
Individual controls			✓	✓

Notes: The dependent variable is a dummy variable indicating whether the individual was a heavy drinker in the last month. Heavy drinking indicates whether in the last month an individual has consumed more than 56 drinks if male, and 28 drinks if female. Temperature is the average over the 30 days prior to the interview date. “High opioid exposure” is a dummy variable equal to one if individual lives in a county where opioid prescriptions per capita are at or above the 75th percentile. All regressions control for precipitation. “Individual controls” include education level, employment status, age, number of family member, and race from the BRFSS. Standard errors clustered at the county level in parentheses. * p<0.10, ** p<0.05, *** p<0.01.

Second, we examine the relationship between temperatures and opioid-related harm. Since

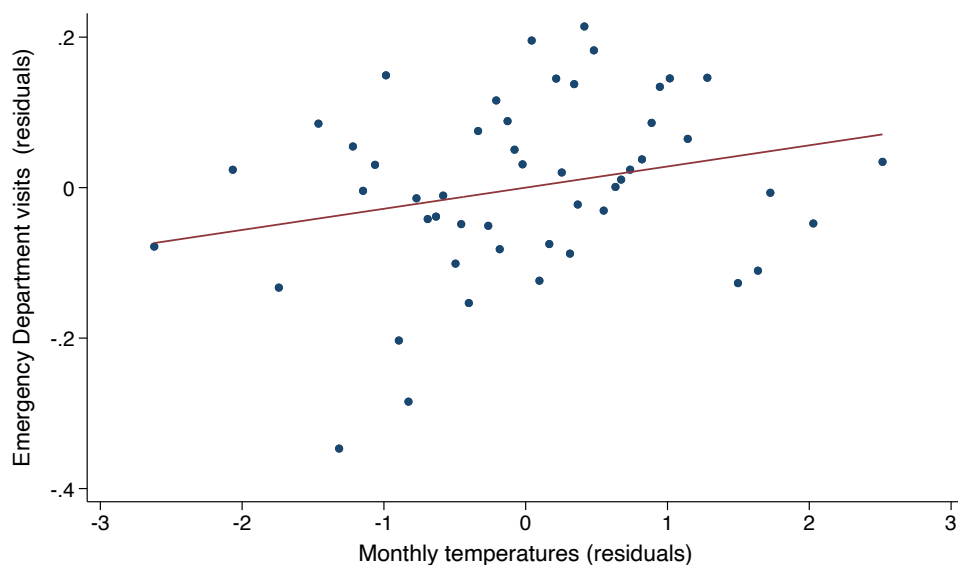
¹¹Following BRFSS “heavy drinking” is defined as more than 28 (56) drinks in the previous month for women (men). These findings are robust to alternative measures of alcohol consumption: higher temperatures increase both the probability of any alcohol use and the number of drinks in the past month ([Table C1](#)).

¹²Counties with high opioid prescription rates can differ in several dimensions we do not account for, such as occupational composition and prevalence of physical pain, that could correlate with heavy drinking, so we read this interaction as descriptive.

individual-level data on opioid use are not available, we use state-level monthly data on non-fatal opioid related emergency department visits, and show in [Figure 3](#) a positive association with temperature (p -value = 0.041), in line with prior evidence ([Chang et al., 2023](#); [Parks et al., 2023](#)).

These findings are consistent with at least three mechanisms through which temperature and substance use interact. First, under a consumption interpretation, higher temperatures may increase substance use, via impaired thermoregulation or changes in leisure times and social context. Second, under a physiological interpretation, higher temperatures may amplify the consequences of substances through vasodilatation, elevated heart rate, and changes in drug metabolism, intensifying behavioral effects ([Hensel et al., 2021](#); [Jhang et al., 2025](#)). Third, individuals who consume alcohol or opioids may select into outdoor and social settings on hot days, raising their exposure to confrontational situations. Our data do not distinguish these channels, and all three imply that policies reducing substance abuse can attenuate the temperature–crime relationship.

Figure 3: Temperature and non-fatal opioid-related emergency department visits



Notes: Figure shows binned scatterplots with 50 bins and a linear regression (coefficient = 0.0281, standard error = 0.0138, clustered at the state level) on the underlying data on the correlation net of state-month, state-year, month-year fixed effects between residualized monthly rate of emergency department visits for nonfatal opioid overdose and residualized monthly temperatures at the state level for the 2018–2021 period. Data on Nonfatal Opioid-related Overdose Emergency Department visits from [Centers for Disease Control and Prevention \(2024\)](#).

Fact 3: Temperatures increase substance-involved crime. Building on the evidence that higher temperatures increase alcohol consumption and exacerbate the consequences of opioid use, we

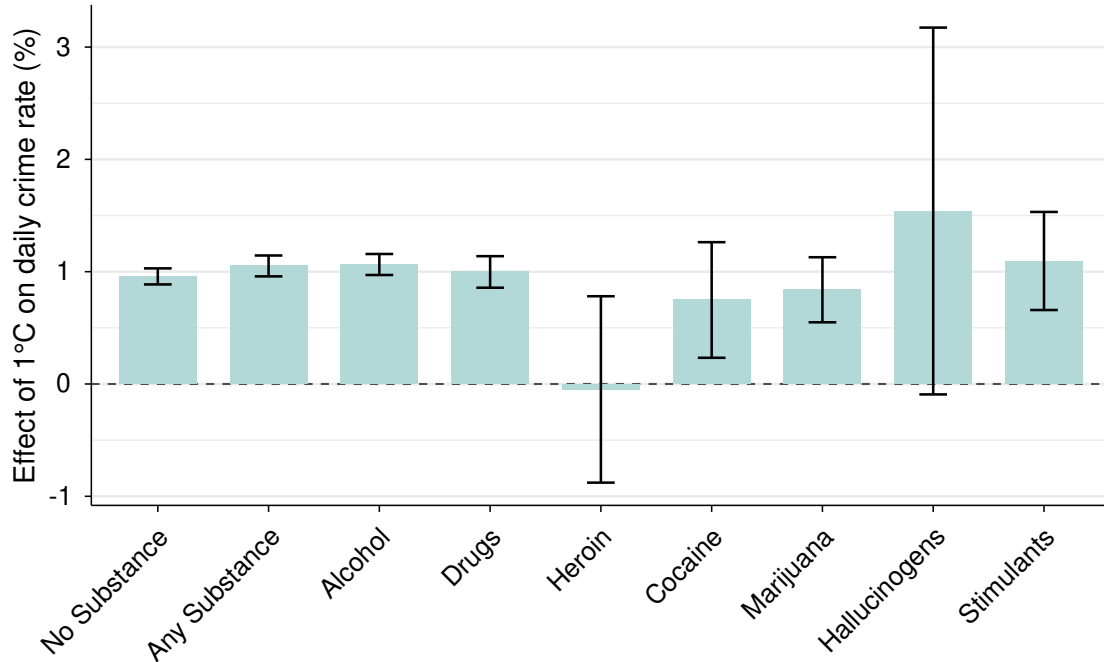
examine whether these effects translate into substance-involved violent behavior. [Figure 4](#) displays the temperature response of the violent crime rate separately by substance involvement: no substance, any substance, alcohol, drugs as a group, and specific drug types (expressed in percent of each category's population-weighted mean).

We find that a 1°C increase in daily mean temperature raises violent crimes involving any substance, alcohol-involved crimes, and drug-involved crimes by similar magnitudes (about 1.1%), in line with the aggregate gradient in [Fact 1](#).¹³ Among specific drug types, the response is positive and precisely estimated for crimes involving cocaine, marijuana, and stimulants, and positive but imprecise for hallucinogens. The exception is heroin, with the estimate essentially zero, which is the most heavily sedating substance, a pattern consistent with a channel operating through disinhibition and irritability rather than intoxication per se. We return to this finding when we examine substitution toward heroin after the opioid reformulation ([Section 5.1](#)).

Fact 4: The temperature–crime gradient is steeper where substance use is higher. Finally, we combine the evidence on substance use, temperatures, and violent behavior to ask whether the temperature–crime relationship itself varies with local substance use. [Figure 5](#) plots the marginal effect of a 1°C increase in daily mean temperature on the daily crime rate across levels opioid prescriptions per capita. We estimate these gradients on the pre-reformulation period and measure exposure as the county average over 2006–2009, so the figure describes the temperature–crime relationship before the policy we study. Because these regressions include state-specific and year-specific temperature slopes, which absorb the common effect of temperature, the plotted quantity is the additional temperature response associated with a given opioid prescription rate, relative to a county with none. At one prescription per capita, slightly above the sample average of 0.88, a 1°C increase raises the daily violent crime rate by an additional 0.38%, approximately 39% of the gradient we report in [Fact 1](#). The differential is positive in all six panels and of similar magnitude across crime categories, between 0.26 and 0.42% per prescription per capita. Estimates vary in precision, which falls with incidence: the differential is precisely estimated for violent crime and simple assault, marginal for aggravated assault and intimidation, and undetectable for homicide and rape, crimes rare enough on a daily basis that we cannot distinguish a differential effect from zero.

¹³The substance-involved estimates are only marginally larger than the corresponding estimate for crimes with no reported substance involvement (1%). We do not claim that only substance-involved crimes increase with temperature: mechanisms unrelated to substance use also raise violence on hot days, and we cannot rule them out ([Cohen and Gonzalez, 2024](#)). Our approach, however, does not require us to rule out alternative mechanisms; we ask instead whether removing access to one pharmacological input flattens the temperature–crime gradient.

Figure 4: Effect of daily temperatures on substance-involved violent crimes

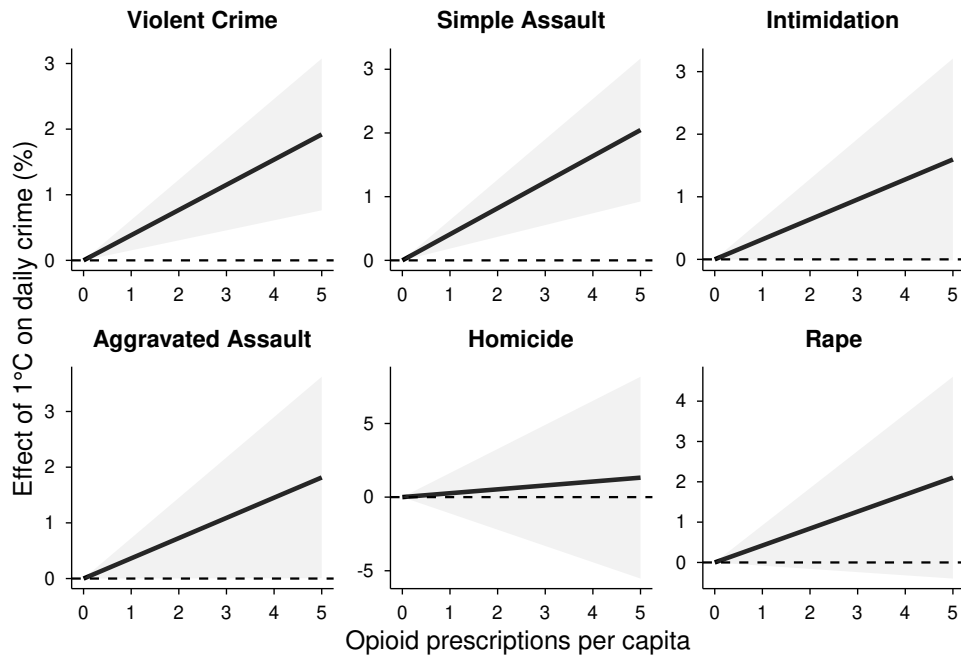


Notes: The figure plots the coefficients of temperature on crime rates per 100,000 people for different categories of crimes. The coefficients (in teal) are normalized by the weighted sample mean. The regressions control for daily total precipitation, and jurisdiction-month-year, day-of-year, week-of-year, and day-of-week fixed effects. Regressions are weighted by jurisdiction population. Confidence intervals at 95% are reported with standard errors clustered at the county-level.

We describe this pattern as motivating evidence, but purely associational. Counties with high pre-reformulation prescriptions differ from low-prescribing counties along dimensions that could independently steepen the temperature–crime gradient, including climate, air-conditioning prevalence, economic conditions, and policing capacity, and [Figure 5](#) cannot separate these from a role for opioid use itself. Our objective is to show that the interaction our empirical design exploits is present in the data and large enough to be informative.

Two interpretations are consistent with this pattern, and both motivate our design. The first is physiological: opioids reduce inhibition and impair emotional regulation, and these effects compound when temperatures independently elevate physiological stress and irritability ([Miczek et al., 2004](#); [Sim, 2023](#)), so higher-prescribing counties display steeper gradients because a larger share of their residents, potential victims as well as offenders, is in a physiological state that is more sensitive to heat-induced stress. The second operates through polysubstance use: the temperature response of heavy drinking is itself larger in high-opioid counties ([Table 1](#), column 4),

Figure 5: Differential effect of temperature on daily crime rates by county-level opioid prescriptions



Notes: The figure plots the marginal effect of a 1°C increase in daily mean temperature on the daily crime rate, expressed as a percent of the population-weighted mean of each outcome, across levels of county per capita opioid prescriptions. The sample period is 2006–2009. Each panel refers to a different crime category: violent crime, simple assault, aggravated assault, intimidation, homicide, and rape. Estimates are obtained from population-weighted regressions of daily crime rates on the interaction between daily mean temperature and county-level opioid prescriptions, controlling for precipitation interacted with opioid prescriptions, jurisdiction-by-month-by-year, day-of-year, week-of-year, and day-of-week fixed effects, and state-specific and year-specific temperature and precipitation slopes. The year-specific slopes absorb the common temperature effect, so the figure reports the differential response associated with opioid prescriptions. Shaded areas show 95% confidence intervals based on standard errors clustered at the county level.

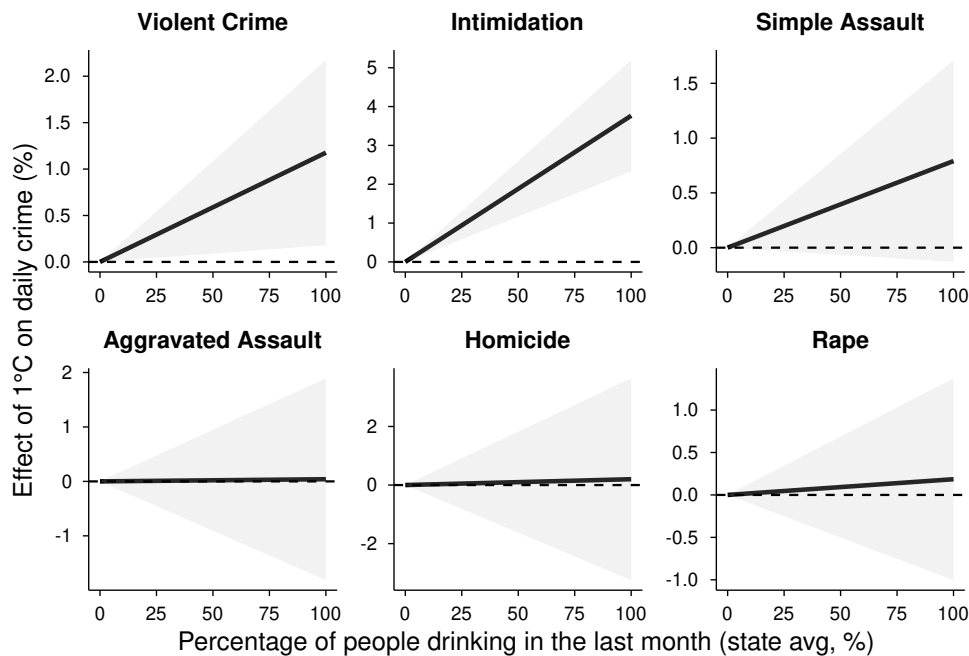
and alcohol has a more direct and better-documented link to aggression (Barron et al., 2024; Luca et al., 2015), so higher-prescribing counties may display steeper gradients partly because opioid use co-occurs with heavier drinking. While we cannot separate these two explanations, both imply that a policy reducing the abusability of prescription opioids should flatten the temperature–crime gradient.¹⁴

Consistent with the second mechanism, Figure 6 repeats the exercise for alcohol, interacting

¹⁴One specific concern about this association is reverse causality, namely that more violent crime generates more injuries, which in turn generate more opioid prescriptions. Two features of the setting make this an unlikely explanation. First, reverse causality would have to produce a steeper sensitivity of crime to daily temperature in high-prescribing counties, not just a higher level of prescribing or of crime. Second, the geography of pre-2010 opioid exposure was driven by the reach of pharmaceutical sales forces rather than by local injury or crime rates (Alpert et al., 2022; Evans et al., 2019). We also show that our baseline estimates are robust to measuring opioid exposure with DEA shipment records (Table D4), which capture the supply of opioids pushed into a county and are less likely than prescription counts to reflect demand-side factors such as injury rates or pre-existing crime levels.

daily temperature with the state-level share of adults who report drinking in the past month, measured in the BRFSS, which is representative at the state level. The differential is again positive: a ten percentage point increase in the share of drinkers is associated with a 0.12% larger temperature response of violent crime, and the estimates are most precise for violent crime and intimidation. Because drinking varies at the state level, state-specific temperature slopes cannot be included, so the comparison relies on cross-state variation; and drinking is itself an outcome of temperature, as Fact 2 documents. Both caveats reinforce our reading of these relationships as descriptive.

Figure 6: Differential effect of temperature on daily crime rates by state-level drinking prevalence



Notes: The figure plots the marginal effect of a 1°C increase in daily mean temperature on the daily crime rate, expressed as a percent of the population-weighted mean of each outcome, across state-level share of people who drank in the past month. The sample period is 1991-2012. The drinking share is constructed from the BRFSS as the sampling-weighted state-year mean of an indicator for any alcohol consumption in the past month, then averaged across years within each state. Each panel refers to a different crime category: violent crime, simple assault, aggravated assault, intimidation, homicide, and rape. Estimates are obtained from population-weighted regressions of daily crime rates on the interaction between daily mean temperature and the state-level share of people who drank in the past month, controlling for precipitation interacted with the same share, jurisdiction-by-month-by-year, day-of-year, week-of-year, and day-of-week fixed effects, and year-specific temperature and precipitation slopes. The year-specific slopes absorb the common temperature effect, so the figure reports the differential response associated with heavy drinking. Shaded areas show 95% confidence intervals based on standard errors clustered at the county level.

Collectively, these four empirical patterns point to substance use as a key mechanism linking temperatures to interpersonal crime, motivating our analysis of whether interventions that reduce

substance abuse can attenuate this relationship.

4 The role of substance-abuse regulation on temperature-crime

4.1 Empirical approach

Our baseline specification aims at quantifying the extent to which the OxyContin reformulation in 2010 reduced the impact of temperatures on crimes by attenuating the physiological consequences of substance abuse. Since this policy was mandated across the entire U.S. in 2010, we design a specification akin to a triple difference, combining temperature variation with pre-reformulation cross-sectional county-level exposure to prescription opioids before and after the policy. Our baseline specification is

$$\begin{aligned}
Y_{idmt} = & \beta_1 T_{c(i)dmt} \times \text{Opioid exposure}_{c(i)} + \beta_2 T_{c(i)dmt} \times \text{Opioid exposure}_{c(i)} \times \mathbf{1}_{t \geq 2010} + \\
& + \gamma_1 P_{c(i)dmt} \times \text{Opioid exposure}_{c(i)} + \gamma_2 P_{c(i)dmt} \times \text{Opioid exposure}_{c(i)} \times \mathbf{1}_{t \geq 2010} + \\
& + \delta_t T_{c(i)dmt} + \delta_s T_{c(i)dmt} + \theta_t P_{c(i)dmt} + \theta_s P_{c(i)dmt} + \\
& + \mu_{imt} + \phi_{dt} + \kappa_{dw} + \lambda_{wt} + \varepsilon_{idmt},
\end{aligned} \tag{1}$$

where we interact daily average temperature $T_{c(i)dmt}$ and total precipitation $P_{c(i)dmt}$ with county-level pre-2010 exposure to prescription opioids ($\text{Opioid exposure}_{c(i)}$), as measured by the population-weighted average number of Schedule II opioid prescriptions per capita, and an indicator variable which takes value of one for the post-reformulation period, starting from 2010 ($\mathbf{1}_{t \geq 2010}$).

Our coefficient of interest, β_2 , captures how the temperature-crime relationship varies between high- and low-opioid exposure counties before and after the reformulation. We include jurisdiction-month-year fixed effects, μ_{imt} , that absorb differences in crime within each jurisdiction for a given month-year (e.g., Chicago Police Department in September 2013); day-of-year fixed effects, ϕ_{dt} , that capture nationwide shocks common to all jurisdictions on a given day (e.g., September 1st); day-of-week fixed effects, κ_{dw} , that account for systematic intra-week variation in criminal activity and reporting practices (e.g., Saturday); and week-of-year fixed effects, λ_{wt} , that absorb differences common across jurisdictions in a month across weeks (e.g., fourth week of November). Since there might be residual unobserved differences in how temperature affects crime across time or space, in our baseline specification, we flexibly allow for the effect of temperature and precipitation to vary by state (e.g., temperatures in Illinois and in Massachusetts having different effects), and by year (e.g., temperatures in 2002 and 2012 having different effects).

Our specification allows us to identify the change in the temperature-crime relationship be-

tween high- and low-opioid exposure counties before and after the reformulation, leveraging within month-year variation in temperatures and crimes in a given jurisdiction, and accounting for annual and state-specific differences in the effect of weather on crimes. In a more conservative specification, we replace state-specific effects of weather with county-specific effects, effectively allowing temperature and precipitation to have different effects, e.g., in Middlesex and in Suffolk counties. We do not use this specification as baseline since it comes at the cost of being unable to identify β_1 and γ_1 , which we need to quantify the moderation role of the reformulation.

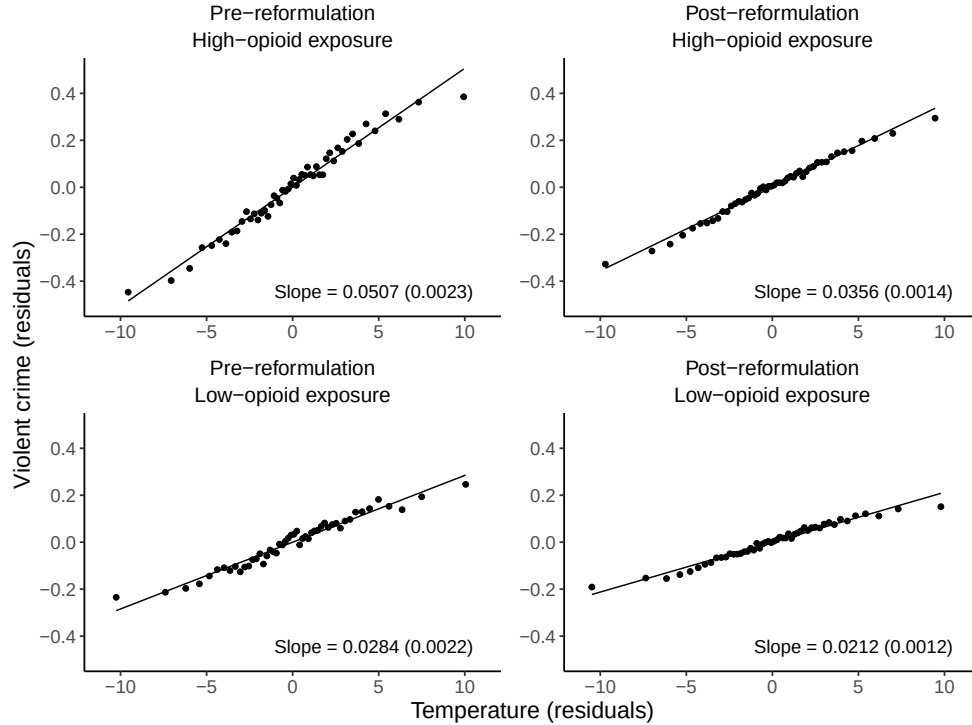
Since the jurisdiction-month-year fixed effects and the year- and state- (or county-) specific weather effects absorb any difference in the level of crime, in its seasonality, and in reporting practices, the only residual concern for the identification of β_2 is an unobserved factor that changed how temperature affects crime after 2010, differentially across counties with high and low pre-reformulation exposure to opioids. While our identifying assumption is fundamentally untestable, we provide evidence in support of its plausibility. First, we flexibly allow the effect of temperature and precipitation to shift after the reformulation with a set of pre-determined county characteristics that are plausibly correlated with exposure to prescription opioids, i.e., unemployment rate, the labor force, the rural population share, the demographic composition of the county, and the number of sworn officers per capita. Second, we rule out differential pre-policy trends between high- and low-exposure counties by estimating dynamic treatment effects in two ways. We first allow for temporally heterogeneous effects of the policy in a binned event study. We construct binary indicators that, respectively, distinguish i) pre-reformulation years (2006–2009); ii) years immediately following reformulation (2011–2013); iii) years for medium-run impacts (2014–2016); and iv) years for long-run impacts (2017–2023). We then estimate a year-by-year event study specification, where we identify differences in the temperature-crime relationship between counties with high and low pre-intervention exposure in a given year as compared to 2010.

4.2 Results

Before turning to the regression estimates, we document the variation that identifies them and that the comparison behind them is visible in the data. [Figure 7](#) plots binned scatterplots of the daily violent crime rate against daily temperature, both residualized of jurisdiction-by-month-year, day-of-year, week-of-year, day-of-week fixed effects, and county- and year-specific precipitation, separately for jurisdictions in counties above and below the population-weighted median of pre-policy opioid prescriptions (0.79 per capita), and separately before and after the 2010 reformulation. All four gradients are positive and precisely estimated, but they differ in slope. Before the reformulation, the gradient is 0.051 violent crimes per 100,000 people per °C in high-exposure

counties and 0.028 in low-exposure counties, the cross-sectional pattern of Fact 4 in levels. After the reformulation, both gradients are lower, in line with the aggregate decline in [Figure 2](#), but by different amounts: the high-exposure gradient falls to 0.036, a decline of 0.015 (30%), while the low-exposure gradient falls to 0.021, a decline of 0.007 (25%).¹⁵

Figure 7: Temperature and violent crime before and after the 2010 OxyContin reformulation, by pre-policy opioid exposure



Notes: Each panel plots a binned scatterplot of the daily violent crime rate, in crimes per 100,000 people, against daily mean temperature, for a distinct group of jurisdiction-days. Both variables are first residualized on jurisdiction $\times$ year-month, day-of-year, week-of-year, and day-of-week fixed effects, and county- and year-specific precipitation, weighting observations by jurisdiction population. Observations are then grouped into 50 equal-sized bins of the temperature residual, and each point is the population-weighted mean of the two residuals within a bin. Rows split counties at the population-weighted median of average Schedule II opioid prescriptions per capita over 2006–2009; columns split jurisdiction-days at 2010, the year of the reformulation. The line in each panel is a population-weighted linear fit, and the reported slope gives the implied change in daily violent crimes per 100,000 people per $^{\circ}\text{C}$, with standard errors clustered by county in parentheses.

[Table 2](#) reports the estimates of [Equation 1](#) for the violent crime rate and its components, allowing the temperature and precipitation effects to vary by state and by year (Panel A) or by county

¹⁵The slopes reported are descriptive, and not a substitute for [Equation 1](#). The median split coarsens the continuous exposure measure, and location- and time-specific temperature effects cannot be estimated, since they would absorb the gradients we plot, so the figure here shows the total gradient within each group.

and by year (Panel B). Before 2010, the temperature response of violent crimes is 0.017 crimes per 100,000 people per °C larger for each additional pre-policy prescription per capita (column 1). After 2010, the triple interaction offsets this differential by 0.0085 crimes per 100,000 people per °C, 50% of its pre-period value, and around 0.2% of the pre-policy violent crime mean per prescription per capita. The differential that remains after 2010 is 0.0085 and is different from zero at the 1% level. The county-specific temperature effects in Panel B absorb the pre-period differential and identify the offset only from within-county changes around the reformulation. The estimate is essentially unchanged, so the offset does not reflect cross-county differences in temperature sensitivity correlated with exposure. Across crime components, the offset is concentrated in intimidation (column 2) and simple assault (column 3), which together account for over 90% of the violent crime estimate and are both significant at the 1% level. The reformulation removes nearly all of the pre-period intimidation differential and about half of the simple assault differential. Aggravated assault displays a positive pre-period differential and a small, insignificant post-2010 change, while for homicide and rape neither coefficient is distinguishable from zero, since the rarity of these outcomes limits precision. The moderating effect of the reformulation is concentrated in high-frequency, lower-severity interpersonal crimes that involve no serious injury.

The specifications in [Table 2](#) absorb any change in the temperature response that is common to all counties and identify only the differential effect by opioid exposure. To recover the total decline of the temperature-crime gradient, we estimate a specification that does not absorb the state- and year-specific weather effects. [Table D1](#) shows that before 2010, at the population-weighted mean exposure of 0.88 prescriptions per capita, a 1°C warmer day increases the violent crime rate by 0.0405 crimes per 100,000 people, around 1% of the pre-policy mean. After 2010, this response falls by 0.0117 crimes per 100,000 people, a decline of approximately 29%, in line with the 26% decline documented in [Section 3](#). Using our estimated coefficients, we attribute roughly 78% of the decline to the triple interaction, and the remaining 22% to the common post-2010 change, which is not statistically distinguishable from zero. Under the identifying assumptions of [Equation 1](#), the reformulation therefore accounts for around three quarters of the decline in the temperature sensitivity of violent crime documented in [Section 3](#).

We also document that the reformulation offset extends beyond interpersonal violence. [Table D2](#) estimates [Equation 1](#) for monetary crimes and their components. Aggregate monetary crimes follow a similar pattern to violent crimes: a positive pre-period differential is offset after 2010 by 44% (column 1). The offset is concentrated in burglary (column 2) and larceny (column 5), where the reformulation respectively removes about 50% and 41% of the pre-period differential. Monetary crimes that require further planning such as fraud (column 3), gambling (column 4),

motor vehicle theft (column 6), and robbery (column 7) display neither a pre-period differential nor a post-2010 change. A potential explanation is the feasibility of each crime during an acute episode of intoxication, withdrawal, or craving. Opioid misuse compounds the discomfort, stress, and impulsivity induced by higher temperatures (Section 3), and it can create a recurrent need for cash or goods to finance consumption (Goldstein, 1985). Assaults, threats, break-ins, and thefts can be committed in that state, while monetary crimes such as fraud and gambling require access to financial instruments and planning horizons that an acute episode does not allow.

Table 2: Effects of opioid reformulation on temperature and crimes

	Crimes (per 100,000 people)					
		Assault				
	Violent Crime (1)	Intimidation (2)	Simple (3)	Aggravated (4)	Homicide (5)	Rape (6)
<i>Panel A</i>						
Temperature × Opioid exposure	0.0170*** (0.0040)	0.0031** (0.0013)	0.0113*** (0.0023)	0.0026** (0.0010)	0.00002 (0.0001)	0.0002 (0.0003)
Temperature × Opioid exposure × Post-2010	-0.0085*** (0.0023)	-0.0028*** (0.0008)	-0.0051*** (0.0015)	-0.0005 (0.0007)	-0.000041 (0.00007)	-0.0000616 (0.0003)
Temperature × State FE	✓	✓	✓	✓	✓	✓
Precipitation × State FE	✓	✓	✓	✓	✓	✓
$\beta_1 + \beta_2 = 0$	0.0085***	0.0003	0.0062***	0.0020***	-0.0000	0.0001
<i>Panel B</i>						
Temperature × Opioid exposure × Post-2010	-0.0082*** (0.0021)	-0.0027*** (0.0008)	-0.0048*** (0.0014)	-0.0007 (0.0007)	-0.00004 (0.00007)	0.00003 (0.0003)
Temperature × County FE	✓	✓	✓	✓	✓	✓
Precipitation × County FE	✓	✓	✓	✓	✓	✓
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓
Temperature × Year FE	✓	✓	✓	✓	✓	✓
Precipitation × Year FE	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓
Pre-policy mean outcome	3.909	0.799	2.406	0.607	0.012	0.124
Observations	38,013,909	38,013,909	38,013,909	38,013,909	38,013,909	38,013,909

Notes: The dependent variable is the number of crimes per 100,000 people. *Post-2010* is a dummy variable equal to one for the post-reformulation period, starting from 2010, and zero, otherwise. *Opioid exposure* is the county-level pre-2010 exposure to opioids. Standard errors are clustered at the county level. All regressions are weighted using jurisdiction population weights. The row $\beta_1 + \beta_2$ reports the differential in the temperature response between counties that remains after 2010, per additional pre-policy prescription per capita, obtained as the sum of the pre-2010 interaction and its post-2010 change. Stars refer to a significance test of $\beta_1 + \beta_2 = 0$, that is, of complete removal of the pre-2010 differential. The regression also includes interactions between precipitation, opioid exposure, and time indicator for post-2010 period. Significance levels are indicated as follows: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Our triple difference research design allows us to examine the dynamic effect of the policy in

moderating the temperature-crime relationship. [Figure 8](#) reports the change in the temperature response of violent crimes, in % of the pre-policy mean, for each additional prescription per capita. The left panel reports the pooled estimate for a single post-reformulation indicator covering 2010 to 2023, column 1 of Panel A of [Table 2](#). The right panel replaces that indicator with four period indicators, each estimated relative to 2010: the pre-reformulation years (2006–2009), the short run (2011–2013), the medium run (2014–2016), and the long run (2017–2023). The effect in the pre-policy time indicator is small in magnitude and not statistically significant at any conventional level, allaying concerns on differential pre-intervention trends. In the post-policy period, the short-, medium- and long-run effects are negative and statistically significant at the 95% level. The three post-reformulation estimates are not significantly different from each other, suggesting a temporally persistent effect of the policy in attenuating the effect of temperatures on violent crimes in counties with higher rates of opioid prescriptions.¹⁶

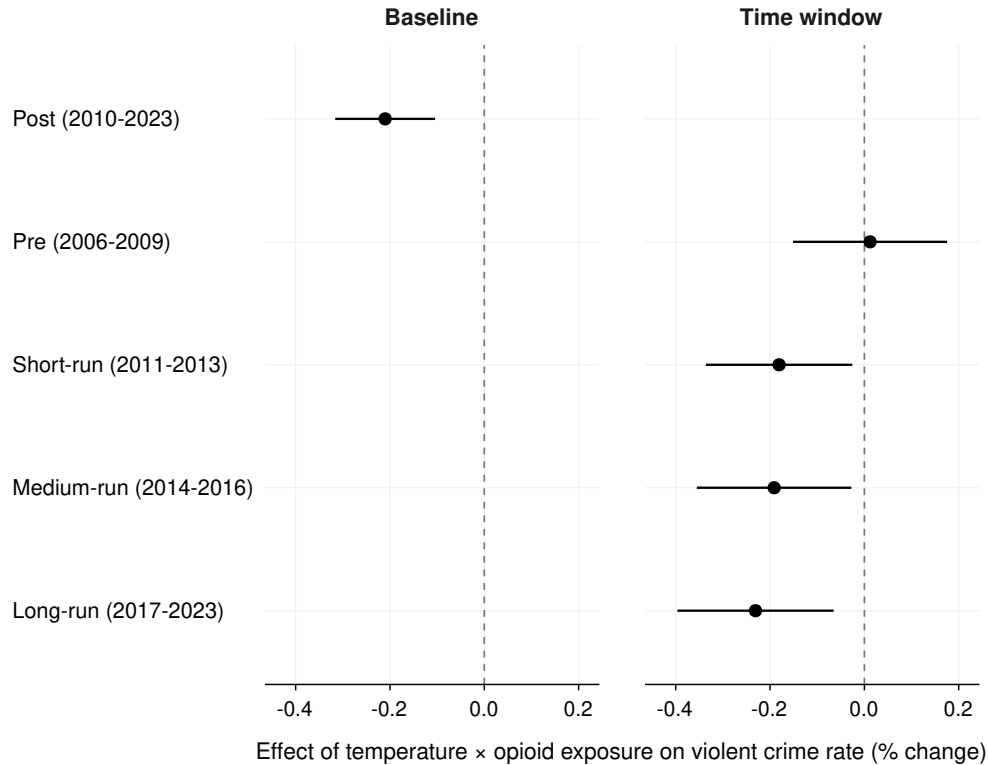
[Figure 9](#) reports the year-by-year estimates, with 2010 as the omitted reference, of the triple interaction term in an event study design for violent crimes where we account for year- and county-specific effects of temperature. None of the four pre-reformulation estimates is statistically distinguishable from zero. After the reformulation starting in 2011, for an increase in daily temperatures, the number of daily violent crimes decreases in high-exposure counties relative to low-exposure counties. The 2011 estimate indicates a statistically significant reduction of 0.0106 crimes per 100,000 people per °C, and every estimate until 2023 is negative and similar in magnitude. Given uncertainty around future climate realizations, the persistence of sizable effects over more than a decade underlines the importance of a holistic assessment of substance regulation complementing existing welfare evaluations, whose benefits must be evaluated on a long-run horizon.¹⁷

Our results are consistent with two co-existing channels through which lower opioid misuse reduces the temperature sensitivity of violent crimes, and we discuss each in turn.

¹⁶We document a similar pattern for simple assaults that show no response distinguishable from zero before 2010, and negative window estimates after 2010 of 0.23%, 0.24%, and 0.27% of its own pre-policy mean ([Figure D1](#)). Intimidation and aggravated assaults have similar but noisier responses. We also repeat the same exercise for rape and homicide cases, but the results again confirm no post-policy decrease for such violent crimes ([Figure D2](#)). We find no differential response to precipitation in any post-reformulation window ([Figure D3](#)).

¹⁷[Figure D9](#) reports the same event study for the three categories of assault. We find that the pattern for violent crimes comes mostly from simple assault. [Figure D10](#) repeats the exercise for fraud, where no estimate is distinguishable from zero as we would expect. Finally, since the reformulation took place in August 2010, the 2010 year is partially treated and [Figure D11](#) shows that when using 2009 as the reference year the estimates from 2011 onward are unchanged.

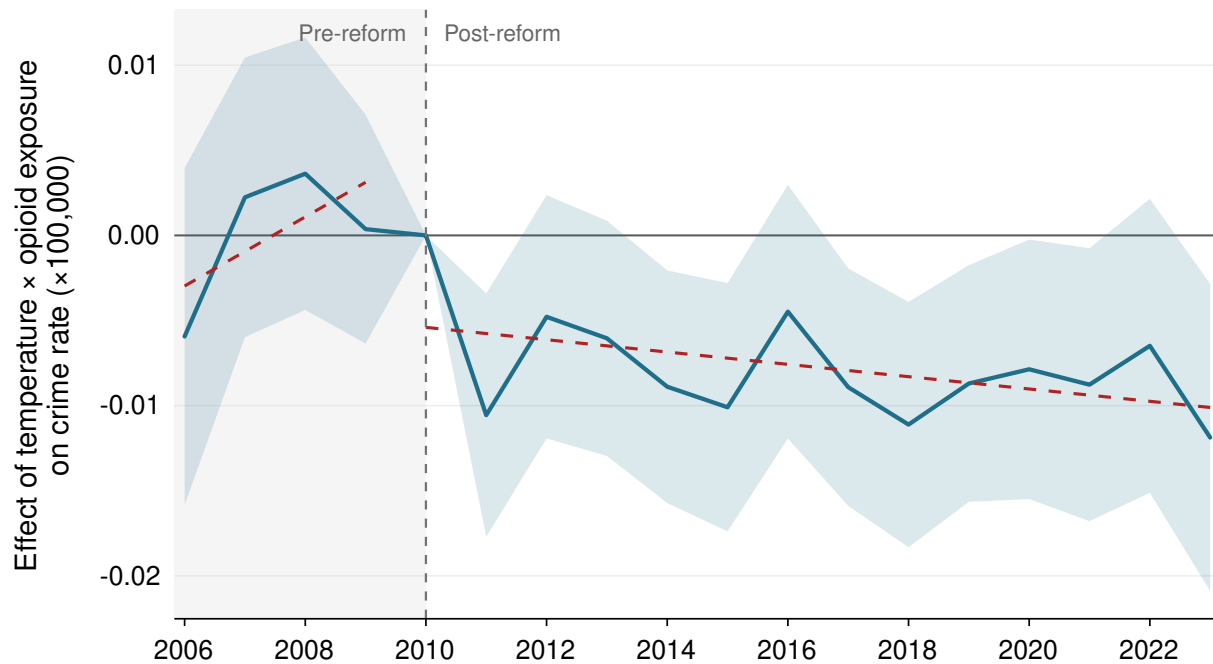
Figure 8: Dynamic effects of opioid reformulation on daily temperature-violent crimes



Notes: Each panel corresponds to a separate regression. The dependent variable is the daily crime rate per 100,000 people, normalized on the y-axis according to the average crime rate before 2010. On the left panel, the figure plots the coefficients from the triple interaction term between average daily county-level temperature, pre-intervention exposure as the population-weighted mean per capita opioid prescription, and an indicator variable that takes the value of one for the post-reformulation years, 2010-2023. On the right panel, the triple interaction is with an indicator variable (short-run) equal to one for the years immediately following reformulation, 2011 to 2013, another one (medium-run) equal to one for the years 2014 to 2016, and another indicator variable (long-run) for the years 2017-2023 on the jurisdiction-level rate of violent crimes. The regression estimates similar coefficients for precipitation and it controls for jurisdiction-month-year, day-of-year, day-of-week, and week-of-year fixed effects, and county- and year-specific temperature and precipitation effects. Regressions are weighted by the population in each jurisdiction. Bins represent the 95% confidence intervals with standard errors clustered at the county-level.

The first channel operates through the number of people in an acute state of opioid intoxication, withdrawal, or craving on a given day. Opioid intoxication and withdrawal impair cognitive function, reduce inhibition, and elevate impulsivity, so that the physiological stress induced by higher temperatures is more likely to translate into aggressive behavior (Herrnstein and Wilson, 1985; Miczek et al., 2004). By restricting access to abusable formulations of OxyContin, the reformulation reduced the share of the population in these states on any given day and dampened the behavioral response to temperatures on both margins. The channel operates through both offenders and victims: intoxication impairs situational awareness, reduces the capacity for self-protection, and increases engagement in high-risk social situations (Pallatino et al., 2021; Testa,

Figure 9: Event study of the differential effect of opioid reformulation on temperature-violent crimes



Notes: The figure plots estimates of the triple interaction between daily temperature, pre-intervention opioid exposure, and year indicators, with 2010 as the omitted reference year, in a regression where the outcome variable is the number of daily crimes (per 100,000 people) for violent crimes. The regression controls for jurisdiction-month-year, day-of-year, day-of-week, and week-of-year fixed effects, triple interaction between daily precipitation, pre-intervention opioid exposure, and year indicators, and county- and year-specific temperature and precipitation effects. Dashed red lines are unweighted linear fits through the annual estimates for 2006 to 2009 and for 2011 to 2023. The joint test of the four pre-reform coefficients (2006–2009) gives $F(4) = 1.13$ ($p = 0.340$). The regression is weighted using jurisdiction population weights. Shaded area represent the 95% confidence intervals with standard errors clustered at the county-level.

2004), and both pathways predict the sign we estimate.

The second channel operates through complementarity with other substances. Opioid misuse frequently co-occurs with alcohol consumption in patterns of polysubstance use (Compton et al., 2021; Esser et al., 2019, 2021), and alcohol has been linked to aggression more directly and robustly than opioids alone (Barron et al., 2024; Luca et al., 2015). If the two are complements — so that individuals who misuse opioids also consume more alcohol, particularly on hot days — then reducing opioid availability through the reformulation may have also indirectly reduced alcohol-mediated crimes. Section 5.1 shows that the mitigating effect of the reformulation on the temperature–crime gradient is concentrated in alcohol-involved offenses, providing suggestive evidence of cross-substance behavioral interactions.

Both channels operate through the same input, the share of the population in an acute state of misuse on a given day, and Section 5 tests their relative importance with offenses recorded as

substance-involved and with heterogeneity in access to substitutes.

Robustness. Our results are robust to a plethora of robustness tests. We exclude the Covid-19 period after 2019 (Figure D4 and Table D3, column 1); use time-invariant jurisdiction population weights (Figure D5 and Table D3, column 2); exclude agencies that do not report crimes for all the twelve months in a year in the entire sample (Figure D6 and Table D3, column 3); include only a balanced sample of agencies starting in 2006 (Figure D7 and Table D3, column 4); cluster standard errors at the state level (Figure D8 and Table D3, column 5); use alternative measures of exposure to opioids, including the number of opioid pills and the shipments per capita (Table D4, columns 1-2); aggregate the sample at the jurisdiction-month and county-day levels (Table D4, columns 3-4); drop one season at a time (Table D5); use alternative fixed effects, accounting for jurisdiction by calendar-day, jurisdiction by month-year, and date (day-month-year) fixed effects, which address unobservables at other temporal and spatial scales (Table D6); and control for county characteristics (unemployment rate, labor force, rural population share, demographic composition of the county, and the number of sworn officers per capita) that could have changed the temperature-crime relationship after the reformulation (Table D7).

5 Mechanisms and additional results

The results in Section 4 are consistent with several co-existing channels through which reduced opioid availability moderates the temperature-crime gradient, including direct reductions in opioid misuse, cross-substance complementarities and substitutability, and potential concerns on other policy confounders. In this section, we move beyond our baseline results and provide empirical tests that assess the relative importance of these channels. We examine the role of substance-involved crimes directly, study how the effectiveness of the reformulation varies with access to substitute substances, such as both legal medical marijuana and illicit opioids, study the potential confounding and interacting role of the expansion of substance abuse treatment facilities at the time, and finally document heterogeneity by climate, time of day, and day of week that speaks to the biophysical and behavioral channels through which the policy operates.

5.1 Interactions with other substances

Substance-involved crimes. A direct test of the substance abuse channel is to examine whether the reformulation attenuated the temperature-crime gradient specifically for offenses in which the offender was reported to be under the influence of substances. Because opioid misuse is often

combined with complementary substances (Compton et al., 2021), restricting access to abusable opioids could indirectly reduce behaviors sensitive to higher temperatures through complementary substance use. We re-estimate Equation 1 separately by the substance involvement recorded and report estimates in Table 3. While both crimes with no substance involved and with any substances involved (alcohol or an unspecified drug category) respond to the reformulation (respectively, columns 1 and 2), the effect is considerably larger for violent crimes with substances involved, in particular alcohol (column 3). The reformulation removes 44% of the pre-period gradient in column 1, in line with the shares removed across the crime categories in Section 4, and roughly 75% in columns 2 and 3. For other substances, the reformulation had no statistically nor economically significant effect (columns 4-6), suggesting that any potential substitution toward these substances did not strengthen their role in temperature-induced violent crimes.¹⁸

Substitution from OxyContin toward heroin followed the reformulation (Alpert et al., 2018; Evans et al., 2019), and it may have been larger among offenders, who have more severe use disorders and better access to illicit markets (Compton et al., 2021). In this case, redirecting opioid-involved crimes would leave the aggregate unchanged rather than reducing it. Column 4 provides a test of this hypothesis: if heroin substitution among offenders was behaviorally consequential, we would expect the triple interaction term for heroin-involved violent crimes to be positive and significant after the reformulation. Instead, the coefficient is economically negligible and statistically indistinguishable from zero, suggesting that even if heroin substitution occurred among offenders, it did not strengthen the temperature sensitivity of heroin-related violent crime.

Table 3: Effects of substance abuse regulation policies on temperature-violent crimes involving substances

	Violent crimes (per 100,000 people)					
	No substance (1)	Any substance (2)	Alcohol (3)	Heroin (4)	Cocaine (5)	Marijuana (6)
Temperature × Opioid exposure	0.0135*** (0.0038)	0.0036*** (0.0008)	0.0036*** (0.0007)	-0.00000112 (0.00000661)	-0.0000438 (0.0000294)	0.00000424 (0.0000463)
Temperature × Opioid exposure × Post-2010	-0.0059*** (0.0021)	-0.0026*** (0.0006)	-0.0028*** (0.0006)	0.00000967 (0.00000842)	0.0000509 (0.0000312)	0.000000261 (0.0000482)
Pre-policy mean outcome	3.4239	0.4910	0.4533	0.0002	0.0016	0.0052
Observations	38,013,909	38,013,909	38,013,909	35,749,195	35,749,195	35,749,195

Notes: Table reports estimates of our baseline specification (Equation 1) on the number of violent crimes involving substances in a jurisdiction per 100,000 people. The equation controls for jurisdiction-month-year, day-of-year, day-of-week, and week-of-year fixed effects, and state- and year-specific temperature and precipitation effects. Standard errors are clustered at the county level. All regressions are weighted using jurisdiction population weights. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

¹⁸We also note that the outcomes in columns 2-3 and 4-6 are measured differently in the crime administrative records, respectively indicating whether the offender committed the crime under the influence, and the drugs seized at the incident, but not necessarily indicating crimes committed under the influence.

Section 3 documents that the temperature response of violent crime declined after 2010, and the substance split allows us to ask where that decline is concentrated. Figure B7 plots the same split by five-year period from 1991 to 2023. Before 2010 the temperature-crime gradient is relatively constant, and the response of violent crimes with recorded substance involvement lies above the response of crimes without in each of them. The temperature response of substance-involved crimes then falls by 35% between the pre-2010 and post-2010 period averages, against 24% for crimes without. These magnitudes match the triple-difference estimates. Evaluated at the average pre-policy opioid exposure (0.88), the offsets in Table 3 imply that the reformulation lowered the temperature response of substance-involved crimes by 0.0023 crimes per 100,000 people per °C, against a decline of 0.0020 in the figure, and the response of crimes without recorded involvement by 0.0052, against a decline of 0.0079. The reformulation therefore accounts for approximately the whole decline for substance-involved crimes and for roughly 66% for crimes without.

Geographic access to illicit opioids from Mexico. Substitution toward illicit opioids required access to illicit markets, and most of the illicit opioid supply reaching the United States is imported across the southwest border. Mexican-sourced heroin was the substitute available at the time of the reformulation, and illicitly manufactured fentanyl, often pressed into counterfeit pills, became the dominant one soon after (Drug Enforcement Administration, 2020). We therefore ask whether the effectiveness of the reformulation depends on the geographic access to these markets, measured as the distance from each county to the nearest legal U.S.–Mexico border crossing.¹⁹

We re-estimate Equation 1 interacting both the temperature-opioid exposure term and its post-2010 interaction with an indicator for counties above the median distance, and interact all the time fixed effects and the state- and year-specific weather slopes with the same indicator, for all violent crimes and for both substance categories. Table 4 reports the estimates. The post-2010 differential between the two groups is small and statistically indistinguishable from zero in all three columns, and in crimes with recorded substance involvement (column 3), the estimate has a sign opposite to what substitution predicts. The moderation in the counties below the median distance is itself different from zero in columns 1 and 3, which rules out complete substitution near the border. We note that the effect of temperature on violent crimes in counties with higher prescribing opioids is stronger in counties farther away from the border, although not statistically

¹⁹We compute the distance between each county’s population centroid and U.S.–Mexico border crossings, obtained from the Bureau of Transportation Statistics (BTS, 2024). We restrict the sample to crossings accessible by pedestrians, private vehicles, or buses, and compute travel distances from the population centroid of each county to the nearest crossing. The sample median is 1,763 km. The measure is time-invariant, so the test asks whether the moderation differs by proximity to the main import corridor over the post-2010 period. It does not track the local availability of any single substance in a given year.

significant. In Section 5.3, although purely associational, we offer a potential interpretation that relates to climatic conditions and biophysical adaptation.

Table 4: Effects of substance abuse regulation policies on temperature-violent crimes by distance to the US-Mexico border

	Violent crimes (per 100,000 people)		
	Total (1)	No substance (2)	Any substance (3)
Temperature $\times$ Opioid exposure	0.0130** (0.0062)	0.0096* (0.0057)	0.0034*** (0.0011)
Temperature $\times$ Opioid exposure $\times$ Post-2010	-0.0075** (0.0032)	-0.0044 (0.0030)	-0.0031*** (0.0009)
Temperature $\times$ Opioid exposure $\times$ Far from U.S.-Mexico border	0.0092 (0.0078)	0.0087 (0.0073)	0.0005 (0.0015)
Temperature $\times$ Opioid exposure $\times$ Post-2010 $\times$ Far from U.S.-Mexico border	-0.0009 (0.0045)	-0.0020 (0.0043)	0.0011 (0.0012)
Pre-policy mean outcome	3.912	3.427	0.491
Observations	37,978,125	37,978,125	37,978,125

Notes: Table reports estimates of our baseline specification (Equation 1) on the number of violent crimes in a jurisdiction per 100,000 people, in all crimes (column 1), in crimes with no reported offender substance involvement (column 2), and in crimes with any reported offender substance involvement (column 3). “Far from U.S.-Mexico border” is an indicator for counties whose distance to the nearest U.S.-Mexico border crossing exceeds the sample median of 1,763 km. The reference group is counties below the median. Each column reports a single regression in which temperature interacted with pre-reformulation county opioid exposure is fully interacted with the post-2010 indicator and with the border-distance indicator. The equation controls for jurisdiction-month-year fixed effects, for day-of-year, day-of-week, and week-of-year fixed effects interacted with the border-distance indicator, and for state- and year-specific temperature and precipitation effects, each interacted with the border-distance indicator. Standard errors are clustered at the county level. All regressions are weighted using jurisdiction population weights. Significance levels are indicated as follows: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Legal access to medical marijuana. We examine an alternative margin through a legal therapeutic substitute. In particular, we ask whether the effectiveness of the opioid reformulation depends on legal access to medical marijuana as of 2010. We use an indicator for states with legal access to medical marijuana (from the National Conference of State Legislatures) and interact it, as above, with the temperature-opioid exposure term and with its post-2010 interaction, absorbing all fixed effects and weather slopes interacted with the indicator. Table 5 reports the estimates. Neither margin differs significantly by legal access: the interaction of the pre-period gradient with the medical marijuana indicator and the quadruple interaction with the post-2010 indicator are small in magnitude and statistically indistinguishable from zero for all violent crimes (column 1), and also when we distinguish between violent crimes with no reported offender substance involvement (column 2), and in violent crimes with any reported offender substance involvement (column 3). In column 1, the post-2010 offset is 0.21% of the pre-policy mean in states without legal access, and it remains statistically significant when evaluated within states with legal access. Legal

access to a therapeutic substitute does not restore the acute state that the reformulation removed.

Table 5: Effects of substance abuse regulation policies on temperature-violent crimes by state legal access to medical marijuana

	Violent crimes (per 100,000 people)		
	Total (1)	No substance (2)	Any substance (3)
Temperature $\times$ Opioid exposure	0.0169*** (0.0045)	0.0135*** (0.0041)	0.0034*** (0.0008)
Temperature $\times$ Opioid exposure $\times$ Post-2010	-0.0082*** (0.0025)	-0.0057** (0.0023)	-0.0025*** (0.0007)
Temperature $\times$ Opioid exposure $\times$ Medical marijuana law	0.0012 (0.0081)	0.0007 (0.0076)	0.0006 (0.0027)
Temperature $\times$ Opioid exposure $\times$ Post-2010 $\times$ Medical marijuana law	-0.0004 (0.0049)	-0.0010 (0.0045)	0.0005 (0.0024)
Pre-policy mean outcome	3.909	3.424	0.491
Observations	38,013,909	38,013,909	38,013,909

Notes: Table reports estimates of our baseline specification (Equation 1) on the number of violent crimes in a jurisdiction per 100,000 people, in all crimes (column 1), in crimes with no reported offender substance involvement (column 2), and in crimes with any reported offender substance involvement (column 3). “Medical marijuana law” is an indicator for states with legal access to medical marijuana as of 2010. The reference group is states without legal access. Each column reports a single regression in which temperature interacted with pre-reformulation county opioid exposure is fully interacted with the post-2010 indicator and with the medical marijuana indicator. The equation controls for jurisdiction-month-year fixed effects, for day-of-year, day-of-week, and week-of-year fixed effects interacted with the medical marijuana indicator, and for state- and year-specific temperature and precipitation effects, each interacted with the medical marijuana indicator. Standard errors are clustered at the county level. All regressions are weighted using jurisdiction population weights. Significance levels are indicated as follows: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Overall, our estimates rule out complete substitution along each margin. While we note that neither of our tests for substitution allows for a causal claim, we find that the moderation of the reformulation survives in counties closer to the U.S.-Mexico border and in states with legal access to marijuana. If the temperature sensitivity of violent behavior operates through the acute state produced by misused prescription opioids, neither the illicit sedating substitute nor the legal therapeutic one restored it.

5.2 Interaction with other substance-related policies

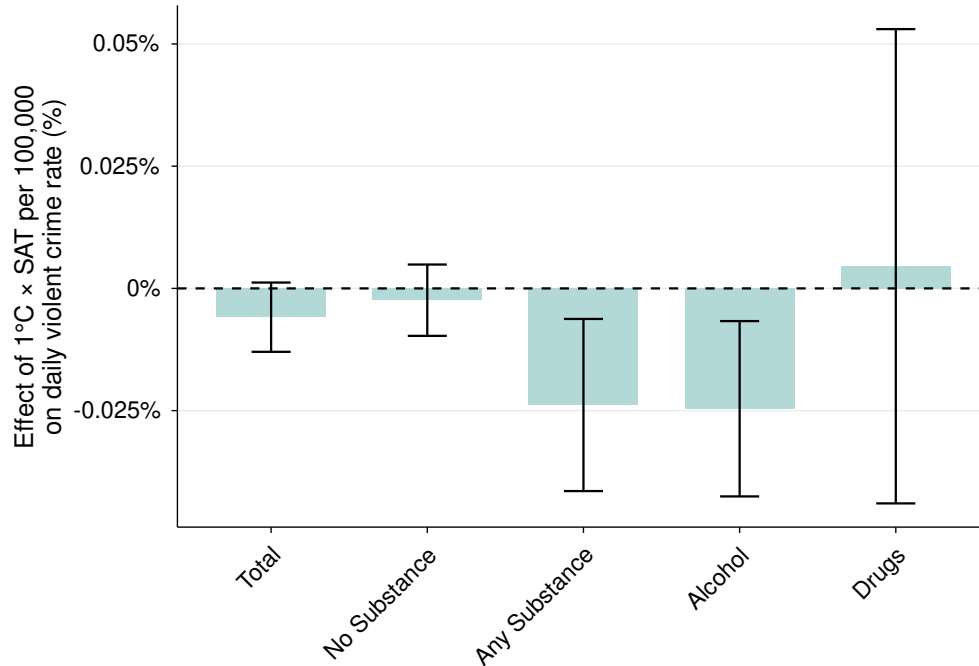
In this section, we explore the generalizability of our results to other policies that regulate substance abuse. In particular, we study the effect of an additional contemporaneous public policy aiming at governing access to addictive substances and its interaction with the opioid reformulation: the staggered expansion of substance abuse treatment (SAT) facilities at the county level. We estimate whether access to SAT facilities moderates the impact of temperature on violent crimes using a specification analogous to Equation 1 (see Appendix E for additional details). Identifica-

tion of the interaction term comes from comparing how crimes in a given jurisdiction respond to daily temperature fluctuations as county-level treatment capacity changes over time, conditional on jurisdiction-month-year and temporal factors that are common across jurisdictions by day of the week, day of the year, and week of the year.

Figure 10 presents estimates of the interaction between daily temperature and SAT availability, relative to the average crime rate in counties without any SAT establishments (tabular results in Table E1) for total violent crimes and violent crimes involving major substances. We find that opening one additional SAT establishment per 100,000 people reduces the effect of a 1°C increase in temperature on violent crime by 0.0002 cases per 100,000 people, equivalent to about 0.006% of the daily average violent crime rate in counties without SAT establishments. While the estimates on the aggregate violent crimes are imprecisely estimated, splitting violent crimes by substance involvement, we find that crimes with recorded substance involvement account for about 65% of the reduction in cases: an additional establishment reduces the temperature effect on these crimes by 0.024% of their daily average rate in counties without establishments. Considering the specific substance involved, we find that the effect is concentrated in alcohol-involved crimes (0.025% of the corresponding mean), consistent with alcohol accounting for approximately 40% of admissions in SAT facilities and with the positive association with temperature documented in Section 3. By contrast, we find no statistically or economically meaningful attenuation for crimes involving other drugs. This could be because drug-involved violent crimes are rare and they record drugs seized at the incident rather than the offender’s intoxication (Section 5.1), so the margin is measured with error.

Evaluated at the sample mean of 5.4 facilities per 100,000 people, the implied reduction in the temperature–violent crime gradient corresponds to 0.03% of the daily average crime rate in counties without any establishments, and to 0.13% of the daily average rate of substance-involved violent crimes. In Table E2, we check for the robustness of our estimates using alternative fixed effect structures. In particular, a less restrictive specification that recovers the average treatment effect of temperature (Panel B) implies that an additional SAT facility offsets approximately 1.5% of the baseline temperature effect on violent crime, or 8% at the sample mean number of facilities. Additional robustness checks are reported in Appendix E: our estimates are robust to extending the sample to counties that never host a treatment facility (Table E3); to restricting it to agencies that report crimes in all twelve months of the year (Table E4); to using a balanced panel of agencies (Table E5).

Figure 10: Effect of substance abuse treatment facilities on temperature-crime relationship



Notes: The figure plots the coefficients from the interaction term between average daily temperature and number of substance abuse treatment (SAT) facilities per 100,000 people in a regression where the outcome variable is the number of crimes per 100,000 people. The coefficients are normalized using the average number of crimes in counties that have no SAT establishments. Facility counts cover 1998–2016. The sample is restricted to the period 1999–2017 and to counties with at least one establishment opened in the same period. The regression also controls for county- and year-specific temperature and precipitation coefficients, jurisdiction-month-year, day-of-year, week-of-year, and day-of-week fixed effects. Regressions are weighted by jurisdiction-location population. Confidence intervals at 95% are reported with standard errors clustered at the county-level.

A natural question is how the opioid reformulation interacted with local treatment infrastructure expansion. The two policies may be complementary if SAT facilities help individuals manage withdrawal and transition away from opioid dependence, making supply-side restrictions more effective where treatment alternatives are accessible. Alternatively, the policies may operate through distinct channels such that their interaction is limited. To examine this, we extend our baseline specification with an indicator that distinguishes counties with at least one SAT establishment. Results are reported in [Table 6](#). For all violent crimes and for those with no substance involvement recorded, the difference in the post-2010 change is not distinguishable from zero (columns 1 and 2). For crimes with recorded substance involvement, the estimates are precisely estimated (column 3). In counties without an establishment the pre-2010 gradient is negative and moves toward zero afterwards. In counties with at least one SAT establishment, the reformulation lowers the temperature response of substance-involved violent crimes by 0.57% of the pre-policy mean per °C at one prescription per capita, removing roughly 72% of the pre-2010 gra-

dient attributable to opioid exposure in these counties. These patterns suggest complementarities between the reformulation and access to substance-abuse treatment, consistent with the idea that treatment availability enhances the effectiveness of opioid supply restrictions on the temperature-crime gradient by facilitating adjustment away from opioid dependence.

Table 6: Effects of opioid reformulation on temperature-violent crimes by SAT facility presence and substance involvement

	Violent crimes (per 100,000 people)		
	Total (1)	No substance (2)	Any substance (3)
Temperature $\times$ Opioid exposure	0.0061 (0.0078)	0.0118 (0.0079)	-0.0060** (0.0024)
Temperature $\times$ Opioid exposure $\times$ Post-2010	-0.0037 (0.0081)	-0.0081 (0.0079)	0.0044* (0.0024)
Temperature $\times$ Opioid exposure $\times$ Any SAT	0.0111 (0.0088)	0.0016 (0.0088)	0.0099*** (0.0025)
Temperature $\times$ Opioid exposure $\times$ Post-2010 $\times$ Any SAT	-0.0050 (0.0085)	0.0022 (0.0082)	-0.0072*** (0.0025)
Pre-policy mean outcome	3.909	3.424	0.491
Observations	38,013,909	38,013,909	38,013,909

Notes: The outcome variable is the number of violent crimes in a jurisdiction per 100,000 people, for all violent crimes (column 1), violent crimes with no reported offender substance involvement (column 2), and violent crimes with any reported offender substance involvement (column 3) for the period 2006–2017. “Any SAT” is an indicator equal to one for counties that host at least one substance abuse treatment establishment. The indicator is interacted with the temperature-opioid exposure term, with its post-2010 interaction, with the corresponding precipitation terms (not reported). The equation controls for jurisdiction-month-year fixed effects, for day-of-year, day-of-week, and week-of-year fixed effects interacted with the SAT presence indicator, and for state- and year-specific temperature and precipitation effects, each interacted with the SAT presence indicator. Pre-policy means are population-weighted averages over 2006–2009. All regressions are weighted using jurisdiction population weights. Standard errors clustered at the county level in parentheses. Significance levels: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Altogether, we interpret these findings as suggestive corroborating evidence for our main results rather than a primary causal claim for three main identification concerns. First, SAT placement may be endogenous to local trends in substance abuse, crime, or economic conditions. We address one component of this concern directly: facility counts and rates do not respond to local temperature differentially by opioid exposure nor after the reformulation (Table E7), so the weather realizations that identify the interaction do not themselves shift treatment capacity. Adding the leads and contemporaneous SAT facility rates to the baseline specification does not alter our estimates, suggesting that the temperature-crime relationship does not differ prior to SAT openings (Table E6).²⁰ Second, counties expanding SAT access may simultaneously adopt

²⁰We cannot estimate a traditional event study specification because in this setting treatment is a continuous facility rate that varies each year, and it is not absorbing, with the majority of the counties that open a first facility after the

other policies that moderate the temperature–crime relationship. Our jurisdiction-month-year fixed effects absorb any level shift in crimes associated with such policies, and the county- and year-specific temperature effects remove permanent differences in the gradient across counties together with common national movements in it, but none of these controls absorbs a change in the slope of crime on daily temperature specific to the counties and years in which treatment capacity expands. If a co-occurring policy flattens the temperature–crime relationship in those counties and years, our estimates would attribute that flattening to treatment capacity.²¹ Third, the stable unit treatment value assumption may fail if counties that do not expand treatment capacity respond in ways correlated with daily temperature, for instance by adjusting policing on hot days. We have no direct test of this channel, but identification from within-jurisdiction daily temperature variation limits the scope for spillovers in the level of crime to bias the interaction, and a spillover that flattened the temperature–crime relationship in counties without expansion would, if anything, attenuate the coefficient we report.

5.3 Additional results

We test for a set of alternative mechanisms that may explain the effectiveness of substance abuse policies in mitigating the temperature effects on crime.

Biophysical adaptation. Our hypothesis ties the moderating role of substance abuse regulation to a physiological channel: heat amplifies the behavioral effects of substances, and policies that reduce substance use attenuate this amplification. Two features of the underlying physiology determine how the strength of this channel varies across space. First, people in warmer areas are more physiologically and behaviorally adapted to heat (Carleton et al., 2022, 2024; Heutel et al., 2021). Second, opioids and other central-nervous-system depressants impair thermoregulation by suppressing heat dissipation and elevating the risk of hyperthermia under heat stress (Chang et al., 2023; Hensel et al., 2021; Jhang et al., 2025; Parks et al., 2023). These forces could interact so that in hotter areas, biophysical adaptation should weaken the compounding effect of higher tem-

start of our sample period returning to zero facilities at some point. The heterogeneity-robust estimators developed for staggered adoption (Callaway and Sant’Anna, 2021; Sun and Abraham, 2021) require an absorbing treatment. Estimators that allow treatment to switch off are defined for a discrete treatment and compare units whose treatment changes to units whose treatment does not (de Chaisemartin and D’Haultfœuille, 2020), which our continuous treatment does not provide.

²¹We check this directly for the state policies most often studied alongside the reformulation: prescription drug monitoring programs (Buchmueller and Carey, 2018; Grecu et al., 2019; Dave et al., 2021), pill mill laws (Rutkow et al., 2015; Mallatt, 2017) and doctor shopping laws (Meara et al., 2016; Deiana and Giua, 2021). Table D8 adds the interaction of temperature with each of these policies, and with SAT capacity, to the reformulation specification. The reformulation lowers the temperature response of violent crimes by 41% of the opioid-attributable gradient in this sample.

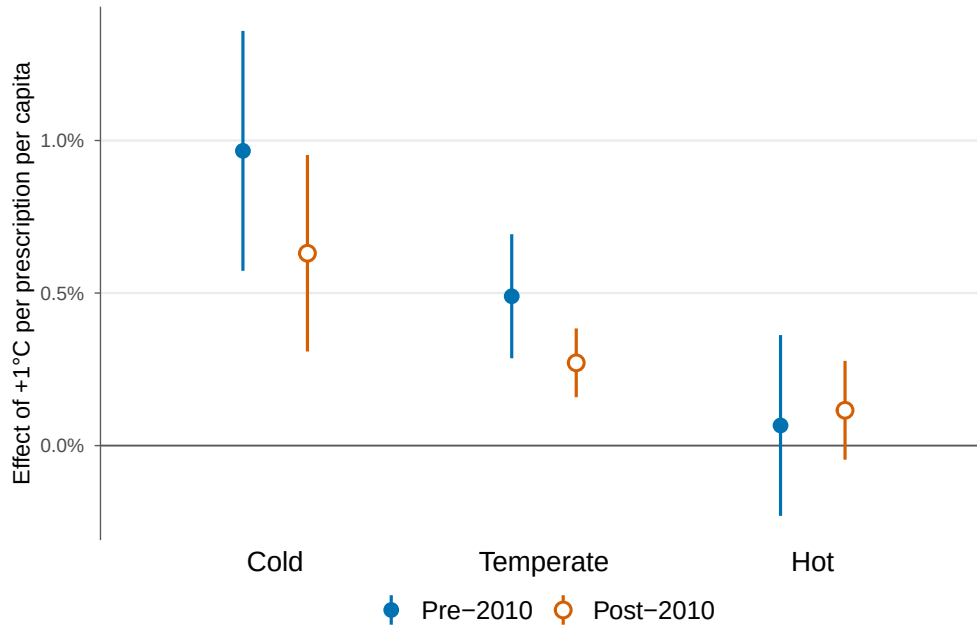
perature and opioid-induced thermoregulatory impairment, leaving a smaller opioid-attributable temperature gradient for regulation to affect.

We test this prediction by re-estimating [Equation 1](#) with both the temperature-opioid exposure term and its post-2010 interaction interacted with terciles of long-run average temperature, and with all the time fixed effects and the state- and year-specific weather slopes interacted with the same indicators. [Figure 11](#) plots the resulting opioid-attributable temperature gradient before and after the 2010 reformulation in each tercile. Before 2010, the gradient is steepest in locations least adapted to hot temperatures. A 1°C warmer day raises violent crime by approximately 0.97% of the pre-policy mean per prescription per capita in the coldest tercile, by 0.49% in the temperate tercile, and by 0.07% in the hottest tercile, where the estimate is not distinguishable from zero. We find that the post-2010 change amounts to 35% of the pre-2010 gradient in cold counties and to 45% in temperate counties. In the hottest tercile of counties, where individuals are more adapted to heat through long-run physiological and behavioral adjustments, the post-2010 change is not distinguishable from zero.

We interpret this pattern as consistent with acclimatization weakening the compounding of heat and substance impairment, but we cannot fully exclude three alternative explanations. First, there might be selective composition of opioid users in hot counties toward individuals more tolerant to heat. Second, hot counties may differ in complementary adaptation investments or substance abuse regulations that independently flatten the temperature-crime gradient. Third, there may be residual unobserved correlates of long-run climate that our tercile classification leaves unabsorbed. Each of these explanations would also predict a flatter opioid-attributable gradient in adapted areas, and a cross-sectional comparison across climates cannot separate them from biophysical adaptation.

Timing of crime. We also examine whether the moderating effect of the reformulation varies with the time of day and the day of the week ([Table D10](#)). Opioid abuse impairs sleep duration and quality ([Bertz et al., 2019](#)), and higher nighttime temperatures disrupt sleep ([Minor et al., 2022](#); [Obradovich et al., 2017](#)), which may reduce the ability to manage stress and raise the probability of aggressive responses to aversive stimuli ([Rauer and El-Sheikh, 2012](#)). Using intra-daily temperature variation and the reported time of crime occurrence, we estimate separate effects for violent crimes in the morning, afternoon, evening, and night, each with the temperature measured over the corresponding hours (columns 1 to 4). The opioid-attributable temperature gradient is present at all times of day, but the reformulation removes 63% of it in the evening and 57% at night, against

Figure 11: Effect of the opioid reformulation on the temperature-crime relationship by climate tercile



Notes: The figure plots the effect of a 1°C increase in daily mean temperature on the daily violent crime rate per unit of pre-policy opioid exposure (prescriptions per capita), before and after the 2010 OxyContin reformulation, by tercile of the county's long-run mean temperature over 1991–2023. Coefficients are expressed as a percent of each tercile's pre-policy mean violent crime rate (3.30, 3.68, and 5.61 crimes per 100,000 people in cold, temperate, and hot counties). Terciles are defined on the nationwide distribution of county mean temperature. All six estimates come from a single regression on the full sample that interacts the temperature $\times$ opioid exposure and temperature $\times$ opioid exposure $\times$ post-2010 terms of Equation 1, and the corresponding precipitation terms, with the tercile indicators. The jurisdiction-month-year, day-of-year, day-of-week, and week-of-year fixed effects and the state- and year-specific temperature and precipitation effects are all tercile-specific. Post-2010 estimates sum the pre-2010 coefficient and its post-2010 change. Table D9, column 2 reports the coefficients displayed in this figure. The regression is weighted using jurisdiction population weights. Vertical bars are 95% confidence intervals based on standard errors clustered at the county level.

39% in the afternoon and 10% in the morning, where the post-2010 change is not distinguishable from zero. These results are consistent with a mitigation of the sleep-deprivation channel through which substance abuse harms are exacerbated at higher temperatures. We also study differences across days of the week to consider changes in time use as potential mechanisms. Weekends provide greater flexibility to adjust routines and are associated with higher consumption of alcohol (Kuntsche and Labhart, 2012; Studer et al., 2014; Voas et al., 2013) and opioids (Spiller et al., 2010; Yan and Kuo, 2019). Splitting the sample by day of the week (columns 5 and 6), the opioid reformulation removes 62% of the gradient on weekends and 47% on weekdays.

Crime characteristics. We also examine whether the moderating effect of the reformulation differs across crime characteristics, using the weapon involved and the location of the incident (Ta-

ble D11). We find that the reformulation removes 43% of the pre-2010 temperature-crime gradient at one prescription per capita (column 1), while estimates are small and imprecise for violent crimes with a recorded weapon. This is in line with our findings that the reformulation moderates entirely simple assaults and intimidation, while firearm crimes are almost entirely aggravated assaults and we do not find evidence of an offset for this sub-category of violent crimes. Turning to the location of the incident, the reformulation lowers the temperature sensitivity of violent crimes at home and away from home (columns 3 and 4), and indoors and outdoors (columns 5 and 6). Crimes away from home account for 39% of the aggregate gradient and 34% of its post-2010 reduction, so the moderation we document is not limited to domestic settings. Relative to their pre-policy means, the gradient lies between 0.33% and 0.93% across the four location categories, and the reformulation removes between 44% and 49% of it in each. These shares are in line with the 50% of the opioid-attributable gradient that the reformulation removes for violent crime.

6 Monetizing social benefits of substance-abuse policies in a warming climate

Our estimates show that policies targeting substance abuse can generate ancillary social benefits by attenuating the sensitivity of violent crimes to temperatures. This mechanism is particularly salient in the context of crimes, since prior work has found little evidence of successful adaptation policies that attenuate the temperature-crime relationship (Burke et al., 2024). From a climate adaptation perspective, ignoring co-benefits can understate the net social returns of policy interventions that indirectly mitigate climate impacts (Carleton et al., 2024). A natural question is whether the implied benefits are economically meaningful and how they compare, in magnitude, to both alternative crime-reduction policies and other successful climate adaptation interventions.

We consider all categories of violent crimes in our data: intimidation, simple assault, aggravated assault, rape, and homicide, and use monetary costs associated with each crime category in \$2023 from multiple sources (Miller et al., 1996, 2021; Roman et al., 2023), with details in Appendix F. We compute the annual social costs associated with a 1°C increase in average temperatures in the U.S. under four scenarios: (i) no adaptation; (ii) accounting for heterogeneous temperature responses by long-run climate (Carleton et al., 2022); (iii) accounting for the abuse-deterrent reformulation of OxyContin; and (iv) accounting for both the OxyContin reformulation and heterogeneity by long-run climate.²² Our monetization captures the marginal social cost of

²²This exercise monetizes the violent crimes induced by a uniform 1°C increase in daily temperatures at current conditions, not the damages of permanent warming: we scale a short-run response over every day of the year, holding

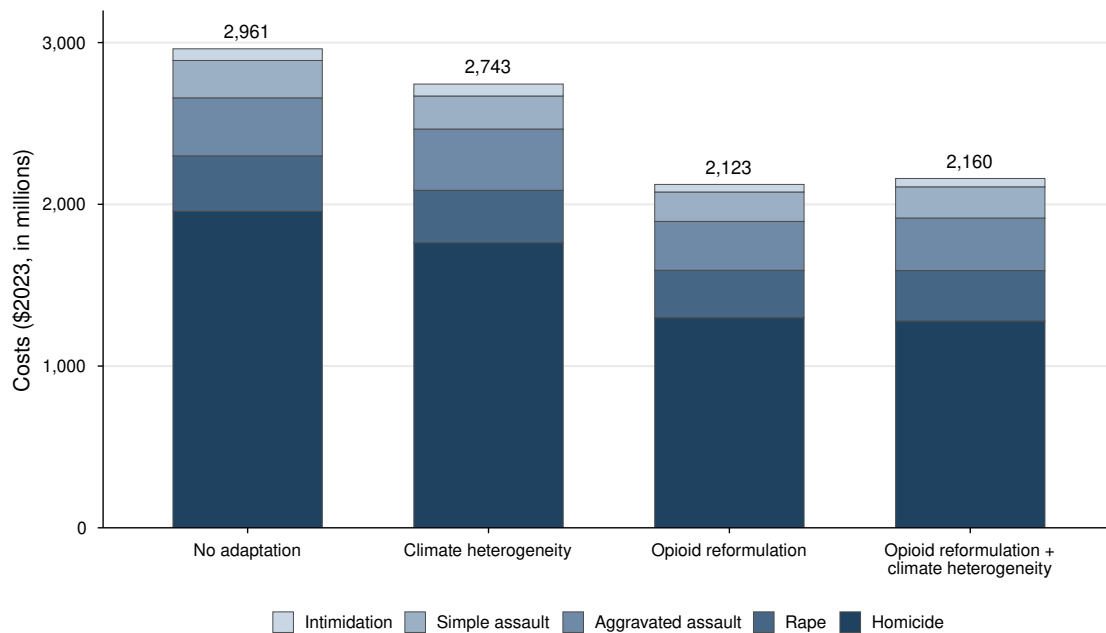
the additional violent crimes induced by a 1°C increase in temperatures, not the total social cost of crimes in the U.S., which is orders of magnitude larger. For context, applying the monetary estimate associated with each crime category to the universe of violent crimes recorded in our data in 2023 yields a total annual burden exceeding \$284 billion. Our estimates instead ask the welfare-relevant question in our context: how does the social cost of violent crime grow with temperatures, and how much can substance abuse regulation prevent?

Our monetary estimates by crime category are reported in [Figure 12](#), and the corresponding number of crimes in [Figure F1](#). The first bar switches off the moderation by the opioid reformulation: it reports the uniform temperature response had the opioid-attributable gradient remained in place. Under this scenario, a 1°C increase in temperatures is responsible for about 37,400 additional violent crimes per year, with a social cost of \$2.96 billion, of which \$1.96 billion from homicides. Allowing the temperature response to differ by long-run climate lowers this figure only marginally to \$2.74 billion. This result is in line with prior climate-crime reviews ([Burke et al., 2015, 2024](#); [Carleton et al., 2024](#)) and suggests that the physiological and psychological channels linking temperature to violent behavior remain largely unmitigated when only allowing for the effect of a temperature increase to differ by county’s sensitivity to hotter temperature. The third bar accounts for the opioid reformulation. The cost of a 1°C increase falls to \$2.12 billion, 28% lower than without accounting for the reformulation and heterogeneity in the response by climate, or about 8,600 fewer crimes per year. The benefit is precisely estimated for the three assault categories, at \$132 million per year. Adding reductions in rape and homicide, we obtain a total benefit of \$838 million across all violent crimes. The fourth bar allows for both the temperature response and the moderation by the reformulation to differ across climate terciles. The annual social cost is within 2% of the third bar, because the moderation is not distinguishable from zero in hot counties, where biophysical adaptation might be reducing the regulation effectiveness as discussed in [Section 5](#).

To examine the economic significance of these magnitudes, we benchmark the implied welfare gains against other crime-reduction and climate adaptation policies in the United States. The estimated net benefits of the policies are comparable to the creation of 1,591 new mental healthcare facilities, based on evidence that opening an additional facility reduces crime costs by approxi-

fixed the stock of adaptation, the composition of crime, and opioid exposure. Scenario (ii) and (iv) let the response differ across counties that have been historically hotter or colder, but they do not describe how a county adjusts as its own climate warms. Reading the estimates as long-run benefits, net of the benefits and the costs of adaptation, would require assuming that the differences we estimate across today’s climates carry over within a county to a warmer future.

Figure 12: Social costs of violent crimes for a 1°C increase in temperature



Notes: The figure reports the estimated annual social costs of violent crimes induced by a 1°C increase in temperature, expressed in 2023\$ (millions), based on the monetary costs for each category of crime reported in [Table F1](#). Estimates are based on the 2006–2023 sample. The “No adaptation” bar switches off the moderation by the reformulation and reports the uniform temperature response interacted with opioid exposure. The “Climate heterogeneity” bar allows the temperature-crime relationship to vary by terciles of long-run average temperature. The “Opioid reformulation” bar uses the estimates reported in [Table D1](#). The “Opioid reformulation + climate heterogeneity” bar allows for the temperature-crime relationship to vary by terciles of long-run average temperature and by opioid exposure before and after the reformulation, using estimates reported in [Table D9](#), column 1, estimated for each crime category. Bar totals sum across separate regressions; confidence intervals by crime category are reported in [Table F1](#).

mately \$527,000 per year ([Deza et al., 2022](#)). Comparing our figures to estimates of benefits from climate adaptation interventions and adaptation co-benefits from other policies, the residential air-conditioning expansion in the late 20th century generated approximately \$4.25 billion benefits annually in reduced mortality ([Barreca et al., 2016](#)), and more restrictive gun laws are estimated to yield annual social benefits of a similar order of magnitude, about \$3.38 billion per 1°C, by reducing temperature-driven homicides ([Colmer and Doleac, 2026](#)). While smaller in magnitude, our finding points to a previously neglected co-benefit of substance abuse regulation evaluated over the entire temperature distribution.

We also compare the welfare gains of the policies to fiscal costs of key U.S. federal programs addressing opioid abuse. Our annual net benefits are equivalent to 4.6 years of funding for the Comprehensive Addiction and Recovery Act (CARA), or to 53% of the SAMHSA State Opioids Response grants.²³

²³The Comprehensive Addiction and Recovery Act, signed into law in 2016, allocates about \$181 million (in \$2023)

While these calculations point to welfare gains from addressing substance abuse as a channel for mitigating the effects of rising temperatures on violent crime, they remain an incomplete welfare assessment for several reasons. First, we do not conduct a full cost-benefit analysis or compute the marginal value of public funds ([Hendren and Sprung-Keyser, 2020](#)), since it would require a comprehensive measurement of all direct and indirect policy costs. Existing evidence suggests the opioid reformulation involved more limited direct public expenditures and generated approximately \$535 million in healthcare cost savings against \$145 million of additional health-care expenditure over a five-year period ([Kumar et al., 2019](#)), or net savings of about \$78 million per year. It also produced indirect social benefits of roughly \$476 million per year in workplace cost savings from reductions in premature deaths, lost wages, absenteeism, and disability ([Kirson et al., 2014](#)). Against these it carried social costs, including substitution toward heroin use ([Evans et al., 2019](#)) and increases in child physical abuse and neglect ([Evans et al., 2022](#)), which no study monetizes, so a comprehensive welfare accounting remains infeasible. Limiting our attention to the benefits that prior work has quantified, our results add \$132 million for a 1°C increase in temperatures through reductions in temperature-induced assaults, an increase of 24%, and up to \$838 million if we also include homicides and rape. Second, our analysis is based only on the subset of jurisdictions reporting to NIBRS, which covered approximately 84% of the U.S. population in 2023. Third, administrative crime data capture only reported offenses: crimes that do not reach law enforcement would be automatically omitted from our calculation. Fourth, our monetization captures only the violent crime-related component of temperature-induced crime. We focus on these crime categories for which we find the strongest empirical evidence highlighting also the underlying physiological mechanisms, and because credible lifetime cost estimates are available for these crimes. This should be viewed as a lower bound on our implied benefits of substance abuse regulation, since our empirical results point at other categories of monetary crimes that do not require planning also responding to temperature and being similarly moderated by the policy. As such, our results demonstrate a novel and economically meaningful channel through which substance abuse regulation mitigates the social damages of temperature increases.

each year to fund programs that fight the opioid epidemic (see [here](#) for more information). SAMHSA State Opioid Response grants provide about \$1.58 billion (in \$2023) of federal funding to states, territories, and tribal entities to address the opioid and stimulant overdose crisis through prevention, treatment, and recovery support services (see [here](#)).

7 Conclusions

This paper provides evidence that public policies that reduce substance abuse generate substantial unmeasured welfare adaptation gains by attenuating the relationship between temperature and crime. Using more than thirty years of daily administrative data across the United States and a triple-difference research design, we show that the 2010 reformulation of OxyContin — an unanticipated, nationwide supply-side restriction on the most widely prescribed opioid at the time — substantially attenuated the temperature sensitivity of violent crimes.

We estimate that the opioid reformulation generates annual social benefits of about \$132 million per 1°C increase in temperature through avoided assaults, and of up to \$838 million once rape and homicide are included. We emphasize that this figure captures the marginal social cost of the additional crimes attributable to higher temperatures, not the total burden of crimes, which is orders of magnitude larger.

Our findings carry several implications for policy and for future research. For policy, they suggest that welfare evaluations of substance abuse interventions should incorporate their indirect climate adaptation value ([Hendren and Sprung-Keyser, 2020](#); [Carleton et al., 2024](#)), particularly in the context of ongoing debates about the costs and benefits of opioid supply restrictions ([Maclean et al., 2021](#)). Our results document a previously unrecognized positive externality of this policy. More broadly, our findings suggest that the social returns to substance abuse regulation are understated in settings where the physiological and behavioral consequences of substance use interact with environmental conditions, a consideration that is absent from standard cost-benefit analyses. For research, while we document that the moderating effect operates primarily through crimes that involve substances, we cannot cleanly separate the relative contributions of the offender and victim channels. Future work with individual-level data linking substance use, victimization, and temperature exposure could decompose these channels more precisely.

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Online Appendix “Co-benefits of Substance Abuse Regulation on Temperature and Violent Crime”

Filippo Pavanello, Guglielmo Zappalà

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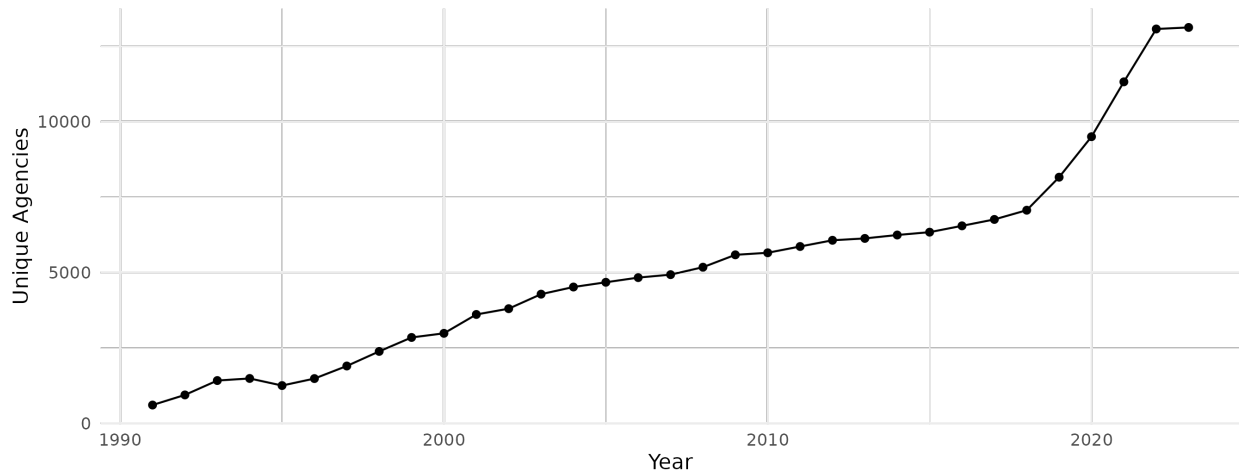
A Descriptive tables and figures

Table A1: Descriptive statistics

	1991-2023		2006-2023		2006-2009		2010-2023	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Daily crimes per 100,000 people								
Total crimes	14.1566	14.0388	13.5695	13.1235	17.5327	15.5559	12.9235	12.5650
Violent crimes	3.2737	4.8140	3.1735	4.5179	3.9086	5.3810	3.0537	4.3493
Assault	3.1500	4.7115	3.0497	4.4176	3.7753	5.2784	2.9315	4.2491
Intimidation	0.6094	1.8922	0.6035	1.8227	0.7990	2.2512	0.5716	1.7408
Simple	2.0078	3.5058	1.9335	3.2751	2.4058	3.8811	1.8565	3.1587
Aggravated	0.5685	1.6909	0.5489	1.5484	0.6070	1.7630	0.5395	1.5104
Rape	0.1137	0.7325	0.1133	0.7049	0.1236	0.7721	0.1117	0.6933
Homicide	0.0123	0.1993	0.0128	0.1903	0.0116	0.1880	0.0130	0.1907
Monetary crimes	7.6168	9.3255	7.2686	8.8146	9.3864	9.8951	6.9234	8.5763
Robbery	0.2045	0.7585	0.2001	0.7105	0.2936	0.9241	0.1849	0.6681
Motor Vehicle Theft	0.6704	1.6951	0.6514	1.5975	0.7317	1.7589	0.6383	1.5692
Burglary	1.2832	2.8486	1.1756	2.6041	1.8537	3.3976	1.0651	2.4327
Larceny	4.7574	7.0184	4.4774	6.6494	5.7067	7.0267	4.2771	6.5641
Gambling	0.0023	0.1065	0.0020	0.0907	0.0037	0.1163	0.0017	0.0857
Fraud	0.6990	2.0765	0.7620	2.0918	0.7969	2.3552	0.7563	2.0455
Weather variables								
Average daily temperature (°C)	12.8341	10.3296	13.0277	10.2931	12.0528	10.3987	13.1866	10.2671
Total daily precipitation (m)	0.0029	0.0066	0.0029	0.0067	0.0028	0.0067	0.0029	0.0067
Policy								
Opioid prescriptions (per capita)					0.8778	0.3675		
Observations	51,066,663		38,013,909		6,021,190		31,992,719	

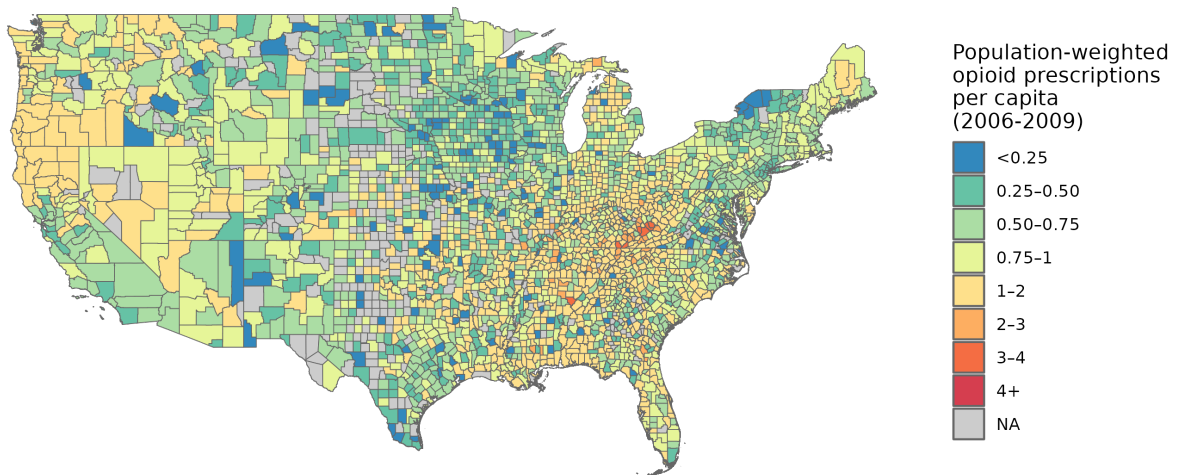
Notes: Daily crime descriptive statistics are weighted by jurisdiction-location population.

Figure A1: Number of reporting jurisdictions, 1991-2023



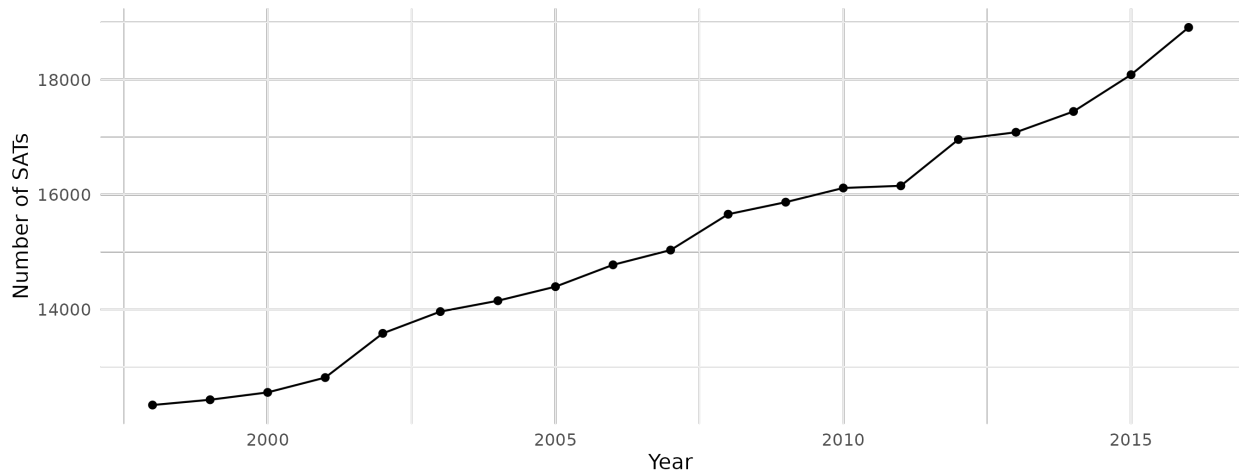
Notes: Figure shows the number of reporting agencies (ORIs) in the NIBRS data set by year.

Figure A2: Pre-reformulation opioid exposure, 2006-2009



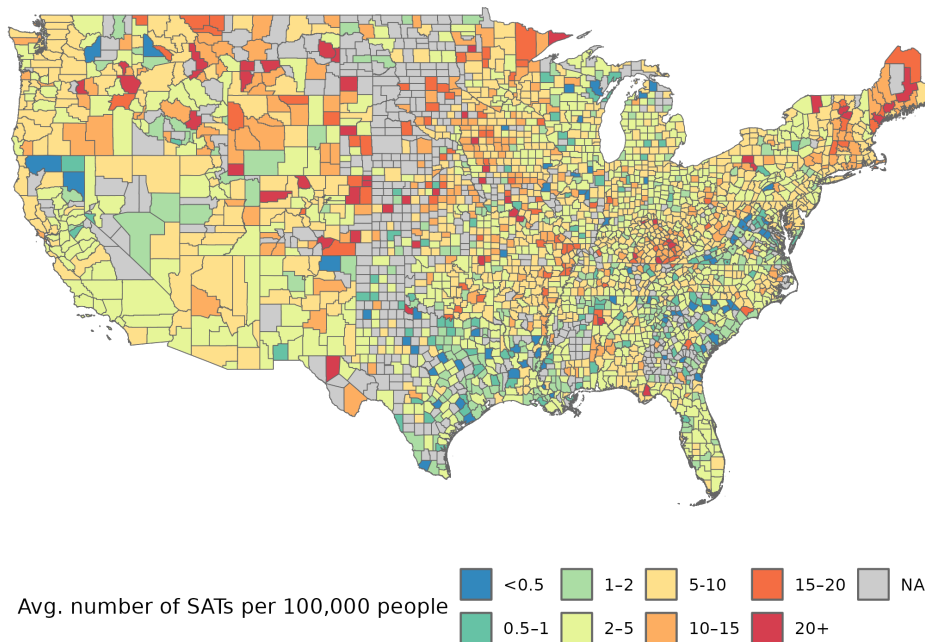
Notes: Figure shows the population-weighted average number of opioids prescriptions per capita in the pre-reformulation period from 2006 to 2009 for 2,694 counties in the CDC data set.

Figure A3: Number of substance abuse treatment facilities, 1998-2016



Notes: Figure shows the number of opened substance abuse treatment facilities in the CBS data set by year.

Figure A4: Substance abuse treatment facilities distribution, 1998-2016



Notes: Figure shows the average number of opened substance abuse treatment facilities per 100,000 people between 1998 and 2016 in the CBP data set. Sample mean is 6.14, standard deviation is 5.21.

B Temperature and crime

We model the baseline relationship between temperature and crime as follows:

$$Y_{idmt} = f(T_{c(i)dmt}, P_{c(i)dmt}) + \mu_{imt} + \lambda_{wt} + \phi_{dt} + \kappa_{dw} + \varepsilon_{idmt} \quad (\text{B.1})$$

where Y_{idmt} is the number of reported cases of crime per 100,000 people in jurisdiction i on day d of month m and year t ; μ_{imt} are jurisdiction-month-year fixed effects; λ_{wt} , ϕ_{dt} and κ_{dw} are respectively week-of-year, day-of-year (i.e., calendar day) and day-of-week fixed effects. We cluster standard errors at the county-level and use population weights at the jurisdiction level. The term $f(T_{c(i)dmt}, P_{c(i)dmt})$ is a function of average daily temperature (in °C) and daily precipitation (in m). In the baseline specification, we model the two weather variables linearly, such that

$$f(T_{c(i)dmt}, P_{c(i)dmt}) = \beta_1 T_{c(i)dmt} + \beta_2 P_{c(i)dmt}$$

The coefficients β_1 and β_2 capture the linear impact of temperature and precipitation exploiting plausibly exogenous quasi-random variation in daily weather realizations (Deschênes and Greenstone, 2007).

We conduct several robustness checks to test our relationship. This includes accounting for COVID-19 (Table B4); excluding agencies that do not report for all twelve months (Table B5); alternative fixed-effects, including leveraging interannual rather than intra-month-year variation (Table B6); clustering standard errors at the state level (Table B7); estimating without population weights (Table B8), and expressing the dependent variable as a count rather than a rate, and using PPML (Table B9).

Table B2 displays the effects of temperature and precipitation on monetary crimes. We find a positive association between temperature and property crimes like robbery, motor vehicle theft, burglary, and larceny. Notably, we do not find a significant coefficient for gambling, a crime that does not involve a physical interaction between offenders and victims.

We also explore non-linear relationship between temperature and crime using polynomials of the weather variables up to the fourth degree (Table B10 and Figure B1-B2),²⁴ temperature bins (Table B11 and Figure B3-B4), and the share of hours in a day in a temperature bin (Table B12). The estimates suggest that a linear specification is a good approximation of the temperature-crime gradient for violent crimes, but not for monetary crimes: fraud cases, for instance, increase with

²⁴We compute non-linear transformations in temperature and precipitation at the grid-cell level before weighing and averaging, in order to preserve non-linearities in the original weather data (Hsiang, 2016).

temperature only when moving from very cold to mild temperatures. Finally, we use seven days lags to control for potential displacements of crime cases (Table B13). We observe displacement effects offsetting nearly 0 to 30% of the contemporaneous effect of temperature after seven days, depending on the type of crime.

In Table B1 we also report the coefficients on precipitation, which is negatively correlated with crime rates. We estimate that a one metre increase in daily total precipitation is associated with 3.54 fewer crimes per 100,000 people. Notably, these results hold for violent crimes, especially different assault types, while precipitation is not significantly correlated with most monetary crimes (except motor vehicle thefts and burglary), suggesting that precipitation might alter time use behavior and thus reduce crimes involving social interactions.

Table B1: The relationship between temperature and violent crime (1991-2023)

	Crimes (per 100,000 people)						
	Crime (1)	Violent (2)	Assault			Rape (6)	Homicide (7)
			Intimidation (3)	Simple (4)	Aggravated (5)		
Temperature (°C)	0.0959*** (0.0026)	0.0317*** (0.0011)	0.0050*** (0.0003)	0.0196*** (0.0006)	0.0067*** (0.0004)	0.0007*** (0.000044)	0.0001*** (0.000014)
Precipitation (m)	-3.542*** (0.3219)	-2.002*** (0.1263)	-0.3478*** (0.0431)	-1.188*** (0.0914)	-0.4458*** (0.0484)	-0.0536*** (0.0181)	-0.0094* (0.0057)
Mean Outcome	14.1566	3.2737	0.6094	2.0078	0.5685	0.1137	0.0123
Observations	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily crimes per 100,000 people. Temperature is the daily average temperature measured in degrees Celsius (°C). Precipitation is the total daily precipitation measured in metres (m). Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table B2: The relationship between temperature and monetary crime

	Crimes (per 100,000 people)						
	Monetary (1)	Robbery (2)	Motor Vehicle Theft (3)	Burglary (4)	Larceny (5)	Gambling (6)	Fraud (7)
Temperature (°C)	0.0394*** (0.0011)	0.0015*** (0.0001)	0.0024*** (0.0002)	0.0096*** (0.0004)	0.0248*** (0.0007)	0.000006 (0.000006)	0.0010*** (0.0001)
Precipitation (m)	0.2611 (0.2220)	-0.0156 (0.0266)	0.0850* (0.0472)	0.4961*** (0.0769)	-0.2586 (0.1645)	-0.0019 (0.0034)	-0.0439 (0.0518)
Mean Outcome	7.6168	0.2045	0.6704	1.2832	4.7574	0.0023	0.6990
Observations	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of crimes per 100,000 people. Temperature is the daily average temperature measured in degrees Celsius (°C). Precipitation is the total daily precipitation measured in metres (m). Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table B3: The relationship between temperature and violent crimes: climate terciles

	Crimes (per 100,000 people)					
	Violent (1)	Assault			Rape (5)	Homicide (6)
		Intimidation (2)	Simple (3)	Aggravated (4)		
Temperature × Cold	0.0267*** (0.0016)	0.0038*** (0.0004)	0.0167*** (0.0009)	0.0059*** (0.0006)	0.0006*** (0.00006)	0.0001*** (0.00002)
Temperature × Temperate	0.0301*** (0.0016)	0.0054*** (0.0007)	0.0192*** (0.0009)	0.0052*** (0.0004)	0.0006*** (0.00007)	0.0001*** (0.00003)
Temperature × Hot	0.0451*** (0.0026)	0.0073*** (0.0007)	0.0268*** (0.0014)	0.0105*** (0.0007)	0.0009*** (0.00009)	0.0001*** (0.00003)
Precipitation × Cold	-2.023*** (0.2196)	-0.3793*** (0.0662)	-1.121*** (0.1450)	-0.5094*** (0.0840)	-0.0527* (0.0286)	-0.0109 (0.0068)
Precipitation × Temperate	-1.718*** (0.2197)	-0.3232*** (0.0729)	-1.090*** (0.1683)	-0.2312*** (0.0600)	-0.0864*** (0.0304)	-0.0063 (0.0110)
Precipitation × Hot	-1.984*** (0.2125)	-0.3099*** (0.0824)	-1.175*** (0.1599)	-0.4969*** (0.0939)	-0.0402 (0.0340)	-0.0135 (0.0111)
Mean Outcome	3.2737	0.6094	2.0078	0.5685	0.1137	0.0123
Observations	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663
Jurisdiction-Month-Year	✓	✓	✓	✓	✓	✓
Day-of-Year FE × Climate FE	✓	✓	✓	✓	✓	✓
Week-of-Year FE × Climate FE	✓	✓	✓	✓	✓	✓
Day-of-Week FE × Climate FE	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily crimes per 100,000 people. Temperature is the average daily temperature measured in degrees Celsius (°C). Precipitation is the total daily precipitation measured in metres (m). Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table B4: The relationship between temperature and violent crime: excluding Covid-19 years

	Crimes (per 100,000 people)						
			Assault				
	Crime (1)	Violent (2)	Intimidation (3)	Simple (4)	Aggravated (5)	Rape (6)	Homicide (7)
Panel A. 1991–2019							
Temperature (°C)	0.1068*** (0.0030)	0.0338*** (0.0013)	0.0054*** (0.0004)	0.0212*** (0.0007)	0.0067*** (0.0004)	0.0007*** (0.00005)	0.0001*** (0.00001)
Precipitation (m)	-3.483*** (0.3853)	-2.149*** (0.1473)	-0.3849*** (0.0495)	-1.251*** (0.1126)	-0.5376*** (0.0559)	-0.0215 (0.0216)	-0.0026 (0.0067)
Mean Outcome	15.3457	3.4290	0.6433	2.1279	0.5658	0.1179	0.0109
Observations	37,675,406	37,675,406	37,675,406	37,675,406	37,675,406	37,675,406	37,675,406
Panel B. Excluding 2020–2021							
Temperature (°C)	0.0988*** (0.0028)	0.0322*** (0.0012)	0.0051*** (0.0003)	0.0201*** (0.0007)	0.0066*** (0.0004)	0.0007*** (0.00005)	0.0001*** (0.00001)
Precipitation (m)	-3.573*** (0.3413)	-2.122*** (0.1397)	-0.4034*** (0.0460)	-1.229*** (0.1037)	-0.4829*** (0.0513)	-0.0437** (0.0193)	-0.0061 (0.0062)
Mean Outcome	14.5530	3.3293	0.6244	2.0507	0.5662	0.1140	0.0116
Observations	44,585,586	44,585,586	44,585,586	44,585,586	44,585,586	44,585,586	44,585,586
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily crimes per 100,000 people. Panel A estimates the regression in a sample from 1991 to 2019. Panel B uses the full sample period (1991–2023) excluding years 2020 and 2021. Temperature is the daily average temperature measured in degrees Celsius (°C). Precipitation is the total daily precipitation measured in metres (m). Regressions are weighted by jurisdiction population. Standard errors clustered at the county level. Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table B5: The relationship between temperature and crime: agencies reporting in all twelve months

	Crimes (per 100,000 people)						
	Crime (1)	Violent (2)	Assault			Rape (6)	Homicide (7)
			Intimidation (3)	Simple (4)	Aggravated (5)		
Temperature (°C)	0.1001*** (0.0027)	0.0331*** (0.0012)	0.0052*** (0.0003)	0.0205*** (0.0007)	0.0070*** (0.0004)	0.0007*** (0.00005)	0.0001*** (0.00001)
Precipitation (m)	-3.746*** (0.3294)	-2.062*** (0.1332)	-0.3639*** (0.0452)	-1.216*** (0.0955)	-0.4608*** (0.0507)	-0.0558*** (0.0192)	-0.0108* (0.0059)
Mean Outcome	14.7635	3.4114	0.6348	2.0925	0.5922	0.1187	0.0127
Observations	43,804,517	43,804,517	43,804,517	43,804,517	43,804,517	43,804,517	43,804,517
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily crimes per 100,000 people. Temperature is the daily average temperature measured in degrees Celsius (°C). Precipitation is the total daily precipitation measured in metres (m). Regressions are weighted by jurisdiction population. Sample is restricted to agencies reporting crimes for 12 months. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table B6: Relationship between temperature and crime: alternative fixed effects

	Crimes (per 100,000 people)						
	Crime (1)	Violent (2)	Assault			Rape (6)	Homicide (7)
			Intimidation (3)	Simple (4)	Aggravated (5)		
Panel A. State × month-year FE + day-of-year, week-of-year, and day-of-week FE							
Temperature (°C)	0.1673*** (0.0167)	0.0489*** (0.0049)	0.0094*** (0.0013)	0.0287*** (0.0028)	0.0102*** (0.0012)	0.0008*** (0.0001)	0.0003*** (0.0001)
Precipitation (m)	1.457 (1.923)	-1.040** (0.4855)	-0.0077 (0.1473)	-0.8238*** (0.2926)	-0.1948 (0.1202)	-0.0448* (0.0232)	-0.0010 (0.0061)
Panel B. Jurisdiction FE + day-of-year, week-of-year, and day-of-week FE							
Temperature (°C)	0.0850*** (0.0039)	0.0259*** (0.0014)	0.0042*** (0.0004)	0.0158*** (0.0007)	0.0054*** (0.0005)	0.0007*** (0.00004)	0.0001*** (0.00002)
Precipitation (m)	-4.295*** (0.8877)	-2.822*** (0.2746)	-0.5472*** (0.0970)	-1.765*** (0.1617)	-0.5206*** (0.0713)	-0.0321* (0.0182)	-0.0058 (0.0056)
Panel C. Jurisdiction × day-of-year FE and Jurisdiction × month-year FE							
Temperature (°C)	0.0981*** (0.0027)	0.0323*** (0.0012)	0.0051*** (0.0003)	0.0200*** (0.0007)	0.0068*** (0.0004)	0.0007*** (0.00004)	0.0001*** (0.00001)
Precipitation (m)	-3.838*** (0.3439)	-2.053*** (0.1348)	-0.3728*** (0.0452)	-1.183*** (0.0962)	-0.4750*** (0.0520)	-0.0587*** (0.0191)	-0.0090 (0.0059)
Panel D. Jurisdiction × day-of-year and Jurisdiction × month-year FE + date FE							
Temperature (°C)	0.0888*** (0.0022)	0.0293*** (0.0009)	0.0042*** (0.0002)	0.0183*** (0.0006)	0.0064*** (0.0003)	0.0008*** (0.00005)	0.0001*** (0.00002)
Precipitation (m)	-2.389*** (0.3572)	-1.831*** (0.1431)	-0.3568*** (0.0465)	-1.037*** (0.1024)	-0.4225*** (0.0539)	-0.0537*** (0.0203)	-0.0089 (0.0061)
Mean Outcome	14.1566	3.2737	0.6094	2.0078	0.5685	0.1137	0.0123
Observations	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily crimes per 100,000 people. Panel A includes state $\times$ month-year fixed effects and day-of-year, week-of-year, and day-of-week fixed effects. Panel B includes jurisdiction fixed effects and day-of-year, week-of-year, and day-of-week fixed effects. Panel C includes jurisdiction $\times$ day-of-year and jurisdiction $\times$ month-year fixed effects. Panel D adds date fixed effects to the Panel C specification. Temperature is the daily average temperature measured in degrees Celsius ($^{\circ}$ C). Precipitation is the total daily precipitation measured in metres (m). Regressions are weighted by jurisdiction population. Standard errors clustered at the county level. Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table B7: The relationship between temperature and crime: state-level standard errors

	Crimes (per 100,000 people)						
	Crime (1)	Violent (2)	Assault			Rape (6)	Homicide (7)
			Intimidation (3)	Simple (4)	Aggravated (5)		
Temperature (°C)	0.0959*** (0.0044)	0.0317*** (0.0024)	0.0050*** (0.0009)	0.0196*** (0.0013)	0.0067*** (0.0007)	0.0007*** (0.00007)	0.0001*** (0.00002)
Precipitation (m)	-3.543*** (0.4690)	-2.001*** (0.1997)	-0.3475*** (0.0695)	-1.187*** (0.1241)	-0.4458*** (0.0558)	-0.0534** (0.0214)	-0.0095** (0.0045)
Mean Outcome	14.1566	3.2737	0.6094	2.0078	0.5685	0.1137	0.0123
Observations	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily crimes per 100,000 people. Temperature is the daily average temperature measured in degrees Celsius (°C). Precipitation is the total daily precipitation measured in metres (m). Regressions are weighted by jurisdiction population. Standard errors are clustered at the state level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table B8: The relationship between temperature and crime: unweighted

	Crimes (per 100,000 people)						
	Crime (1)	Violent (2)	Assault			Rape (6)	Homicide (7)
			Intimidation (3)	Simple (4)	Aggravated (5)		
Temperature (°C)	0.0863*** (0.0022)	0.0248*** (0.0010)	0.0042*** (0.0002)	0.0157*** (0.0011)	0.0046*** (0.0003)	0.0005*** (0.00007)	0.00007*** (0.00002)
Precipitation (m)	-2.037 (2.060)	-1.221*** (0.4470)	-0.1669 (0.1385)	-0.7621* (0.4339)	-0.2010* (0.1079)	-0.0857** (0.0396)	-0.0131 (0.0111)
Mean Outcome	12.927	2.7181	0.53483	1.6784	0.43614	0.09298	0.00715
Observations	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily crimes per 100,000 people. Temperature is the daily average temperature measured in degrees Celsius (°C). Precipitation is the total daily precipitation measured in metres (m). Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table B9: The relationship between temperature and violent crime: count dependent variable

	Crimes						
	Assault						Homicide (7)
	Crime (1)	Violent (2)	Intimidation (3)	Simple (4)	Aggravated (5)	Rape (6)	
Panel A. OLS (Count, jurisdiction-month-year FE)							
Temperature (°C)	0.1943*** (0.0269)	0.0759*** (0.0125)	0.0135*** (0.0030)	0.0426*** (0.0064)	0.0186*** (0.0035)	0.0016*** (0.0005)	0.0006*** (0.0001)
Precipitation (m)	-9.498** (4.653)	-5.273*** (1.138)	-1.439* (0.7688)	-2.609*** (0.7744)	-1.155*** (0.4352)	-0.1095 (0.0910)	-0.0771** (0.0390)
Mean Outcome	34.3644	9.2704	2.3872	4.9672	1.7069	0.2494	0.0453
Observations	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663
Panel B. PPML (state-month-year FE)							
Temperature (°C)	0.0306*** (0.0047)	0.0389*** (0.0057)	0.0528*** (0.0090)	0.0332*** (0.0055)	0.0431*** (0.0050)	0.0261*** (0.0043)	0.0332*** (0.0056)
Precipitation (m)	2.418** (1.115)	2.325* (1.189)	3.110** (1.540)	1.895* (1.028)	2.802* (1.484)	1.550 (1.067)	1.189 (1.367)
Mean Outcome	34.3715	9.2733	2.3909	4.9701	1.7090	0.2498	0.0469
Observations	51,060,438	51,057,563	51,011,886	51,054,555	51,030,506	51,002,670	48,050,193
Panel C. PPML (jurisdiction-month-year FE)							
Temperature (°C)	0.0065*** (0.0002)	0.0095*** (0.0004)	0.0074*** (0.0007)	0.0097*** (0.0003)	0.0129*** (0.0003)	0.0075*** (0.0011)	0.0132*** (0.0020)
Precipitation (m)	-0.2105*** (0.0745)	-0.5160*** (0.0636)	-0.4901*** (0.1547)	-0.4400*** (0.1190)	-0.6698*** (0.1306)	-0.8026** (0.3553)	-1.581** (0.6513)
Mean Outcome	35.5432	9.8892	3.0801	5.4474	2.1536	0.3928	0.1788
Observations	46,080,037	39,094,801	22,596,092	35,213,659	22,430,880	12,006,163	1,949,483
Day-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily crimes. Panel A uses OLS regression, Panel B uses Poisson Pseudo-Maximum Likelihood (PPML) with state-month-year fixed effects, and Panel C uses PPML with jurisdiction-month-year fixed effects. Temperature is the daily average temperature measured in degrees Celsius (°C). Precipitation is the total daily precipitation measured in metres (m). Regressions are weighted by jurisdiction population. Standard errors clustered at the county level. Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table B10: The relationship between temperature and crime: fourth-order polynomial

Panel A. All and violent crimes	Crimes (per 100,000 people)						
	Crime (1)	Violent (2)	Assault			Rape (6)	Homicide (7)
			Intimidation (3)	Simple (4)	Aggravated (5)		
Temperature (°C)	0.1233*** (0.0037)	0.0246*** (0.0010)	0.0048*** (0.0004)	0.0146*** (0.0006)	0.0048*** (0.0003)	0.0005*** (0.0000732)	0.0001*** (0.0000169)
Temperature ²	-0.0005*** (0.0001)	0.0007*** (0.000038)	0.0000942*** (0.0000105)	0.0005*** (0.000023)	0.0002*** (0.0000109)	0.0000166*** (0.00000306)	0.00000121 (0.000000796)
Temperature ³	-0.0000539*** (0.00000394)	0.00000365*** (0.00000115)	-0.00000136*** (0.000000358)	0.00000348*** (0.000000808)	0.00000153*** (0.000000371)	0.000000105 (0.000000175)	-0.0000000217 (0.0000000323)
Temperature ⁴	0.000000635*** (0.000000129)	-0.000000574*** (0.0000000433)	-0.000000539*** (0.0000000123)	-0.000000378*** (0.0000000301)	-0.000000136*** (0.0000000122)	-0.0000000141** (0.00000000554)	-0.00000000027 (0.00000000124)
Precipitation (m)	-14.09*** (1.036)	-4.597*** (0.4207)	-0.5221*** (0.1441)	-2.717*** (0.2969)	-1.265*** (0.1432)	-0.1315** (0.0568)	-0.0346* (0.0178)
Precipitation ²	451.1*** (51.63)	147.0*** (20.56)	8.001 (7.152)	92.38*** (15.40)	42.55*** (6.878)	3.909 (3.065)	1.748* (0.9661)
Precipitation ³	-4,890.8*** (743.2)	-1,686.8*** (287.4)	-65.56 (94.84)	-1,133.0*** (226.0)	-434.7*** (91.68)	-41.72 (43.90)	-26.04* (13.46)
Precipitation ⁴	14,671.5*** (2,706.1)	5,223.5*** (1,047.7)	63.14 (327.0)	3,718.8*** (855.4)	1,244.2*** (323.2)	143.4 (167.9)	86.48* (51.64)
Mean Outcome	14.1566	3.2737	0.6094	2.0078	0.5685	0.1137	0.0123
Observations	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663
Panel B. Monetary crimes	Monetary (8)	Robbery (9)	Motor Vehicle Theft (10)	Burglary (11)	Larceny (12)	Gambling (13)	Fraud (14)
Temperature (°C)	0.0583*** (0.0020)	0.0020*** (0.0002)	0.0026*** (0.0004)	0.0137*** (0.0006)	0.0380*** (0.0012)	0.0000177** (0.00000823)	0.0020*** (0.0002)
Temperature ²	-0.0006*** (0.0000721)	-0.00000512 (0.00000666)	0.00000837 (0.0000159)	-0.0000816*** (0.0000197)	-0.0005*** (0.0000455)	-0.0000000143 (0.000000404)	-0.0000397*** (0.00000894)
Temperature ³	-0.0000323*** (0.00000241)	-0.00000119*** (0.000000241)	-0.00000153** (0.000000597)	-0.00000802*** (0.00000065)	-0.0000202*** (0.0000017)	-0.000000018 (0.0000000196)	-0.00000139*** (0.000000445)
Temperature ⁴	0.000000566*** (0.0000000907)	0.0000000158* (0.00000000838)	0.000000045** (0.0000000219)	0.000000108*** (0.0000000238)	0.000000363*** (0.000000061)	-0.000000000205 (0.000000000641)	0.0000000346** (0.000000015)
Precipitation (m)	-1.456** (0.7008)	-0.0444 (0.0831)	0.2082 (0.1474)	0.7679*** (0.2433)	-2.394*** (0.4947)	-0.0035 (0.0087)	0.0093 (0.1640)
Precipitation ²	26.22 (35.26)	-0.9832 (4.265)	-8.583 (7.878)	-46.47*** (13.57)	83.63*** (25.57)	-0.1497 (0.3910)	-1.235 (8.905)
Precipitation ³	-112.5 (511.9)	30.87 (63.96)	137.2 (110.8)	799.6*** (203.6)	-1,055.4*** (383.1)	4.925 (5.730)	-29.59 (131.5)
Precipitation ⁴	89.33 (1,867.3)	-138.2 (234.7)	-465.1 (396.6)	-2,485.7*** (738.5)	3,145.5** (1,441.3)	-22.50 (22.12)	54.78 (504.0)
Mean Outcome	7.6168	0.2045	0.6704	1.2832	4.7574	0.0023	0.6990
Observations	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily crimes per 100,000 people. Panel A reports estimates for violent crime outcomes and Panel B for monetary crime outcomes. Temperature is the daily average temperature measured in degrees Celsius (°C). Precipitation is the total daily precipitation measured in meters (m). Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table B11: The relationship between temperature and crime: temperature bins

Panel A. All and violent crimes	Crimes (per 100,000 people)						
	Crime (1)	Violent (2)	Assault				Homicide (7)
			Intimidation (3)	Simple (4)	Aggravated (5)	Rape (6)	
Temperature (°C)							
≤ 5	-1.369*** (0.0396)	-0.5052*** (0.0180)	-0.0802*** (0.0053)	-0.3126*** (0.0102)	-0.1067*** (0.0052)	-0.0104*** (0.0008)	-0.0018*** (0.0003)
5-10	-0.7177*** (0.0219)	-0.3429*** (0.0118)	-0.0513*** (0.0034)	-0.2135*** (0.0066)	-0.0745*** (0.0036)	-0.0067*** (0.0006)	-0.0013*** (0.0002)
10-15	-0.3691*** (0.0126)	-0.1947*** (0.0071)	-0.0276*** (0.0021)	-0.1223*** (0.0042)	-0.0425*** (0.0021)	-0.0039*** (0.0005)	-0.0008*** (0.0002)
20-25	0.2970*** (0.0129)	0.1770*** (0.0068)	0.0240*** (0.0019)	0.1129*** (0.0041)	0.0387*** (0.0023)	0.0033*** (0.0005)	0.0005*** (0.0002)
25-30	0.4296*** (0.0210)	0.2736*** (0.0107)	0.0322*** (0.0025)	0.1752*** (0.0066)	0.0643*** (0.0036)	0.0054*** (0.0007)	0.0008*** (0.0002)
≥ 30	0.3637*** (0.0279)	0.3188*** (0.0148)	0.0354*** (0.0048)	0.2087*** (0.0100)	0.0732*** (0.0052)	0.0054*** (0.0019)	0.0008 (0.0005)
Precipitation (m)	-9.163*** (0.5989)	-3.129*** (0.2272)	-0.4662*** (0.0752)	-1.805*** (0.1568)	-0.8051*** (0.0822)	-0.1021*** (0.0285)	-0.0075 (0.0093)
Precipitation ²	94.07*** (12.89)	27.24*** (4.452)	1.738 (1.492)	15.25*** (3.138)	9.543*** (1.557)	1.192* (0.6168)	-0.0812 (0.2226)
Mean Outcome	14.1566	3.2737	0.6094	2.0078	0.5685	0.1137	0.0123
Observations	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663
Panel B. Monetary crimes	Monetary (8)	Robbery (9)	Motor Vehicle Theft (10)	Burglary (11)	Larceny (12)	Gambling (13)	Fraud (14)
≤ 5	-0.5451*** (0.0178)	-0.0211*** (0.0021)	-0.0315*** (0.0022)	-0.1412*** (0.0061)	-0.3381*** (0.0110)	-0.0002* (0.0001)	-0.0129*** (0.0021)
5-10	-0.2414*** (0.0106)	-0.0115*** (0.0014)	-0.0180*** (0.0015)	-0.0691*** (0.0040)	-0.1385*** (0.0066)	-0.0001 (0.0001)	-0.0042** (0.0018)
10-15	-0.1154*** (0.0064)	-0.0062*** (0.0008)	-0.0107*** (0.0013)	-0.0329*** (0.0023)	-0.0653*** (0.0043)	-0.00007 (0.00009)	-0.0003 (0.0016)
20-25	0.0856*** (0.0076)	0.0052*** (0.0007)	0.0129*** (0.0014)	0.0197*** (0.0022)	0.0485*** (0.0054)	-0.00008 (0.0001)	-0.0006 (0.0017)
25-30	0.1170*** (0.0115)	0.0067*** (0.0011)	0.0261*** (0.0023)	0.0247*** (0.0031)	0.0630*** (0.0078)	-0.0002 (0.0002)	-0.0033 (0.0025)
≥ 30	0.0399** (0.0184)	0.0038* (0.0021)	0.0261*** (0.0042)	0.0176*** (0.0062)	-0.0071 (0.0139)	-0.0003 (0.0004)	-0.00008 (0.0053)
Precipitation (m)	-1.660*** (0.3750)	-0.0838 (0.0521)	0.0088 (0.0738)	-0.3384** (0.1464)	-1.290*** (0.2631)	-0.0060 (0.0053)	0.0498 (0.0857)
Precipitation ²	20.68*** (7.601)	0.9143 (1.019)	1.687 (1.704)	15.30*** (3.829)	6.579 (5.656)	0.0934 (0.1335)	-3.888** (1.685)
Mean Outcome	7.6168	0.2045	0.6704	1.2832	4.7574	0.0023	0.6990
Observations	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily crimes per 100,000 people. Temperature is the daily average temperature measured in degrees Celsius (°C). Reference category is (15,20] °C. Precipitation is the total daily precipitation measured in metres (m). Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table B12: The relationship between temperature and violent crime: share of hours

	Crimes (per 100,000 people)						
	Crime (1)	Violent (2)	Assault			Rape (6)	Homicide (7)
			Intimidation (3)	Simple (4)	Aggravated (5)		
Temperature (°C)							
≤ 5	-1.733*** (0.0475)	-0.5897*** (0.0204)	-0.0954*** (0.0062)	-0.3637*** (0.0114)	-0.1236*** (0.0060)	-0.0118*** (0.0010)	-0.0021*** (0.0004)
5-10	-0.6538*** (0.0253)	-0.3653*** (0.0133)	-0.0534*** (0.0039)	-0.2287*** (0.0077)	-0.0805*** (0.0041)	-0.0060*** (0.0009)	-0.0013*** (0.0003)
10-15	-0.4632*** (0.0177)	-0.2379*** (0.0088)	-0.0346*** (0.0030)	-0.1490*** (0.0056)	-0.0516*** (0.0030)	-0.0042*** (0.0009)	-0.0008** (0.0003)
20-25	0.3810*** (0.0217)	0.2178*** (0.0112)	0.0307*** (0.0030)	0.1343*** (0.0082)	0.0483*** (0.0037)	0.0077*** (0.0010)	0.0005 (0.0004)
25-30	0.7455*** (0.0335)	0.4425*** (0.0175)	0.0550*** (0.0046)	0.2831*** (0.0104)	0.1006*** (0.0067)	0.0086*** (0.0013)	0.0016*** (0.0005)
≥ 30	0.5453*** (0.0409)	0.4671*** (0.0223)	0.0520*** (0.0057)	0.3017*** (0.0141)	0.1084*** (0.0066)	0.0096*** (0.0019)	0.0020*** (0.0006)
Precipitation (m)	-8.130*** (0.5598)	-2.372*** (0.2100)	-0.3695*** (0.0732)	-1.329*** (0.1493)	-0.6306*** (0.0759)	-0.0894*** (0.0280)	-0.0035 (0.0091)
Precipitation ²	85.43*** (12.29)	20.64*** (4.259)	1.001 (1.471)	11.09*** (3.077)	8.054*** (1.498)	1.022* (0.6090)	-0.1165 (0.2152)
Mean Outcome	14.1566	3.2737	0.6094	2.0078	0.5685	0.1137	0.0123
Observations	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663	51,066,663
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓	✓

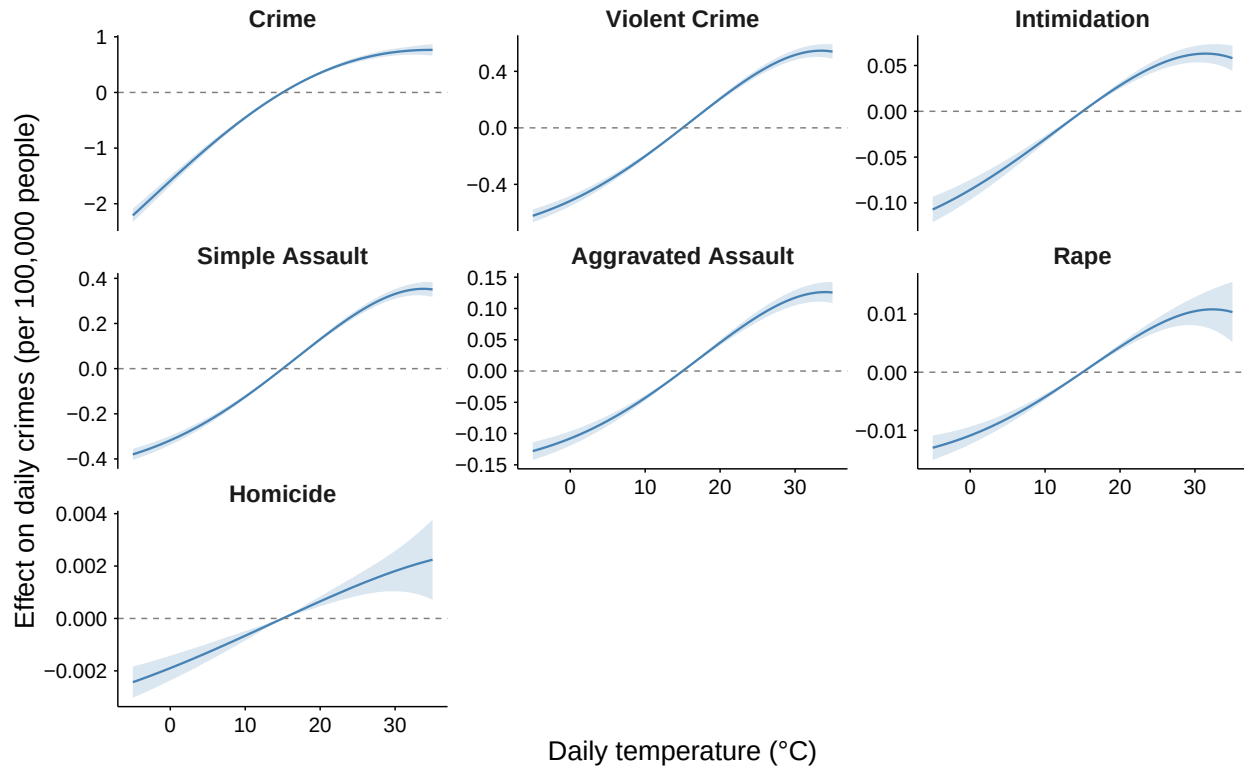
Notes: The dependent variable is the number of daily crimes per 100,000 people. Temperature is specified as the daily share of hours in each temperature interval. Reference category is (15,20] °C. Precipitation is the total daily precipitation measured in metres (m). Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table B13: The relationship between temperature and crime: displacement effects

	Crimes (per 100,000 people)						
	Crime (1)	Violent (2)	Assault			Rape (6)	Homicide (7)
			Intimidation (3)	Simple (4)	Aggravated (5)		
Contemporaneous Temperature (°C)	0.1099*** (0.0032)	0.0350*** (0.0013)	0.0056*** (0.0004)	0.0215*** (0.0007)	0.0076*** (0.0004)	0.0007*** (0.00005)	0.0001*** (0.00002)
Precipitation (m)	-2.471*** (0.3198)	-1.754*** (0.1229)	-0.3144*** (0.0428)	-1.046*** (0.0925)	-0.3696*** (0.0468)	-0.0556*** (0.0186)	-0.0079 (0.0058)
Cumulative Effect of Temperature	0.0870*** (0.0028)	0.0253*** (0.0010)	0.0043*** (0.0003)	0.0152*** (0.0005)	0.0052*** (0.0003)	0.0007*** (0.0001)	0.0001*** (0.00002)
Mean Outcome	14.1710	3.2773	0.6102	2.0099	0.5691	0.1138	0.0123
Observations	50,958,016	50,958,016	50,958,016	50,958,016	50,958,016	50,958,016	50,958,016
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓	✓

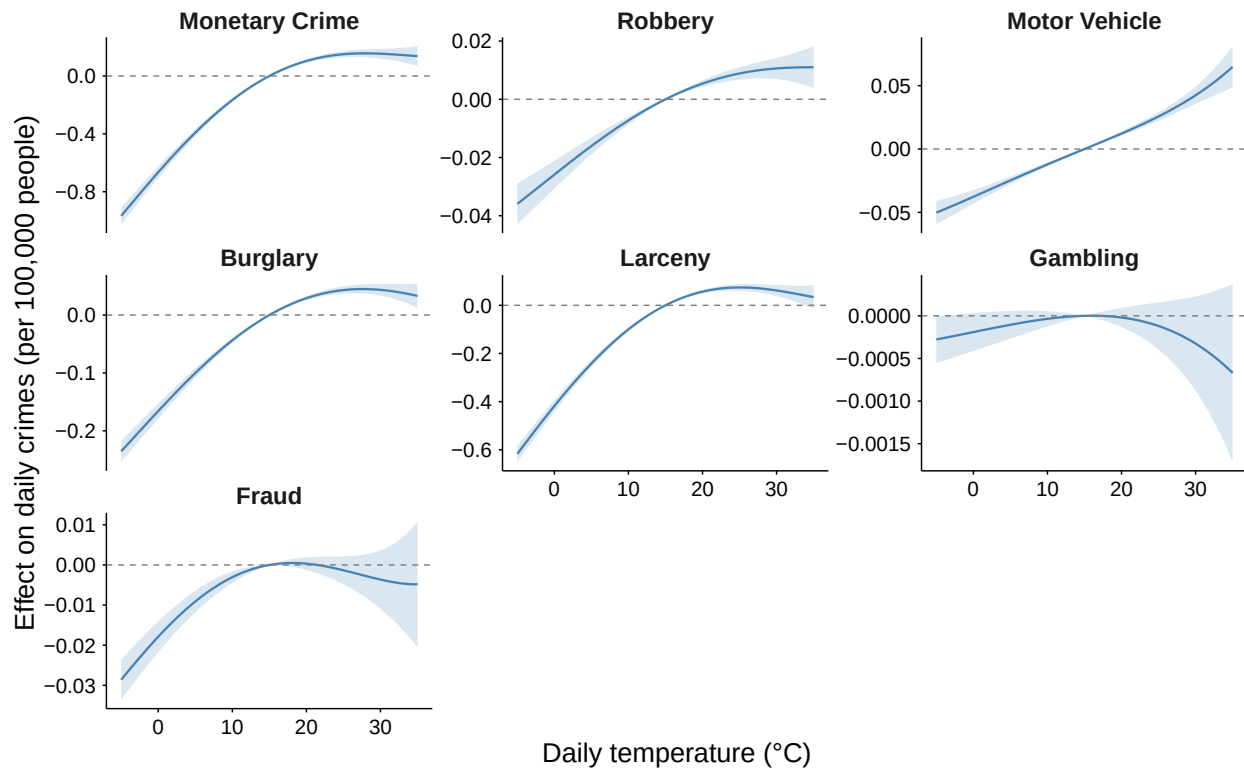
Notes: The dependent variable is the number of daily crimes per 100,000 people. We include seven lags of temperature and precipitation. The cumulative effect of temperature is the sum of the contemporaneous coefficient and its lags. Only the contemporaneous coefficients are shown, while lag coefficients are estimated but omitted for brevity. Temperature is the daily average temperature measured in degrees Celsius (°C). Precipitation is the total daily precipitation measured in metres (m). Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Figure B1: The temperature-violent crime response function: fourth-order polynomial



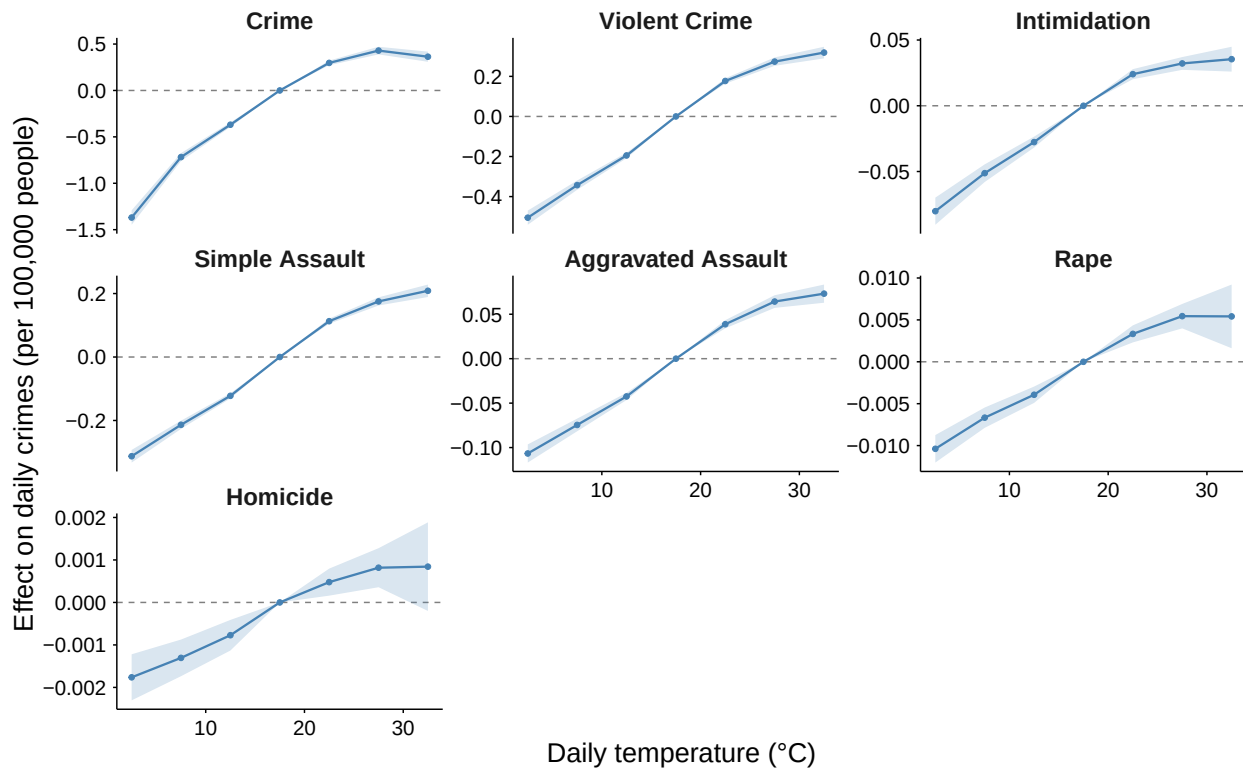
Notes: The figure plots the temperature-crime response functions across different categories of crimes. The response function compares the predicted value of a crime across various temperature levels to its predicted value at 15 °C. The regressions control for daily total precipitation, and jurisdiction-month-year, day-of-year, week-of-year and day-of-week fixed effects. Regressions are weighted by jurisdiction-location population. Confidence intervals at 95% are reported with standard errors clustered at the county-level.

Figure B2: The temperature-monetary crime response function: fourth-order polynomial



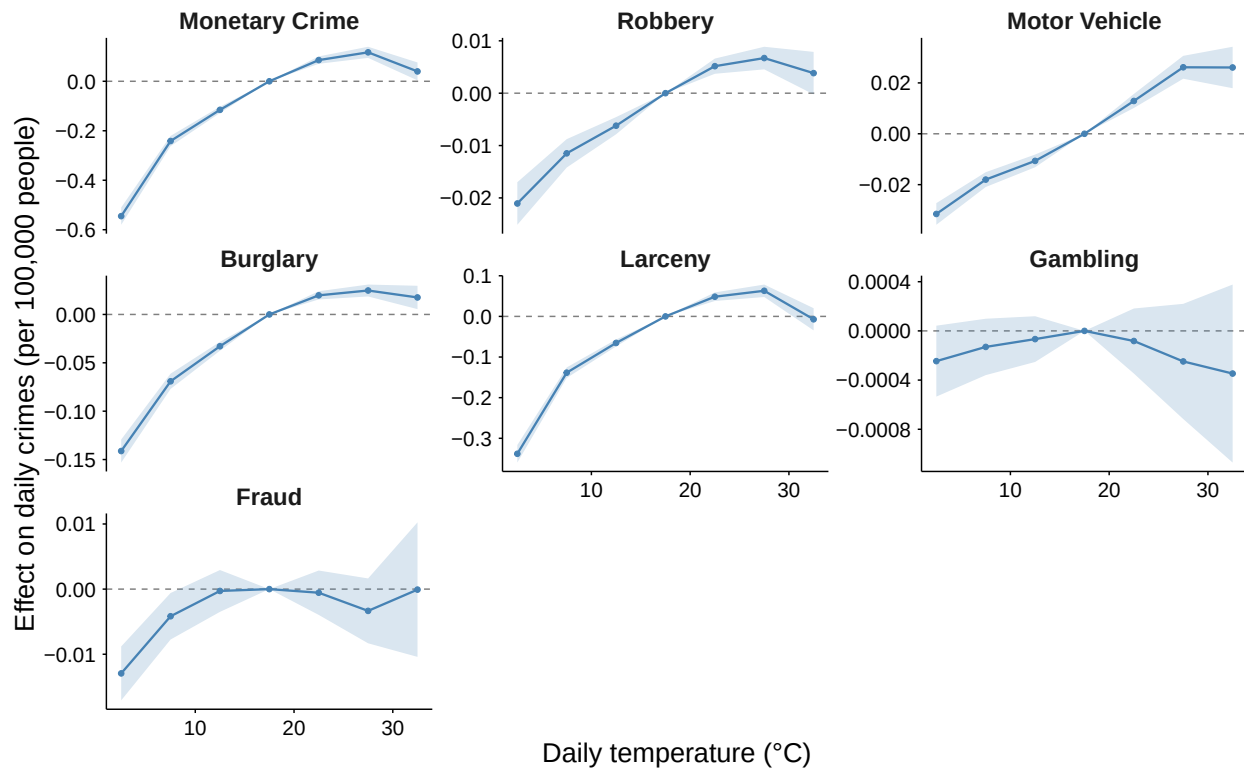
Notes: The figure plots the temperature-crime response functions across different categories of crimes. The response function compare the predicted value of a crime across various temperature levels to its predicted value at 15 °C. The regressions control for daily total precipitation, and jurisdiction-month-year, day-of-year, week-of-year and day-of-week fixed effects. Regressions are weighted by jurisdiction-location population. Confidence intervals at 95% are reported with standard errors clustered at the county-level.

Figure B3: The temperature-violent crime response function: temperature bins



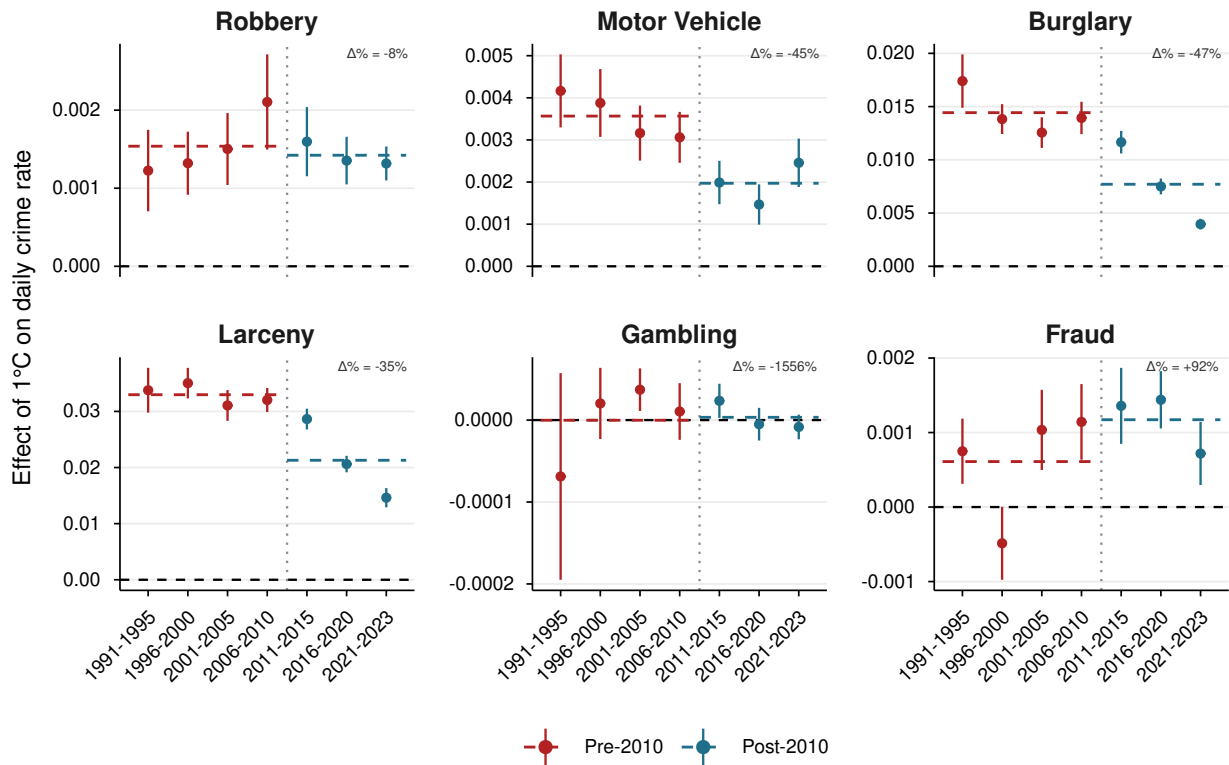
Notes: The figure plots the temperature-crime response functions across different categories of crimes using temperature bins (reference category: (15-20] °C). The regressions control for daily total precipitation, and jurisdiction-month-year, day-of-year, week-of-year and day-of-week fixed effects. Regressions are weighted by jurisdiction-location population. Confidence intervals at 95% are reported with standard errors clustered at the county-level.

Figure B4: The temperature-monetary crime response function: temperature bins



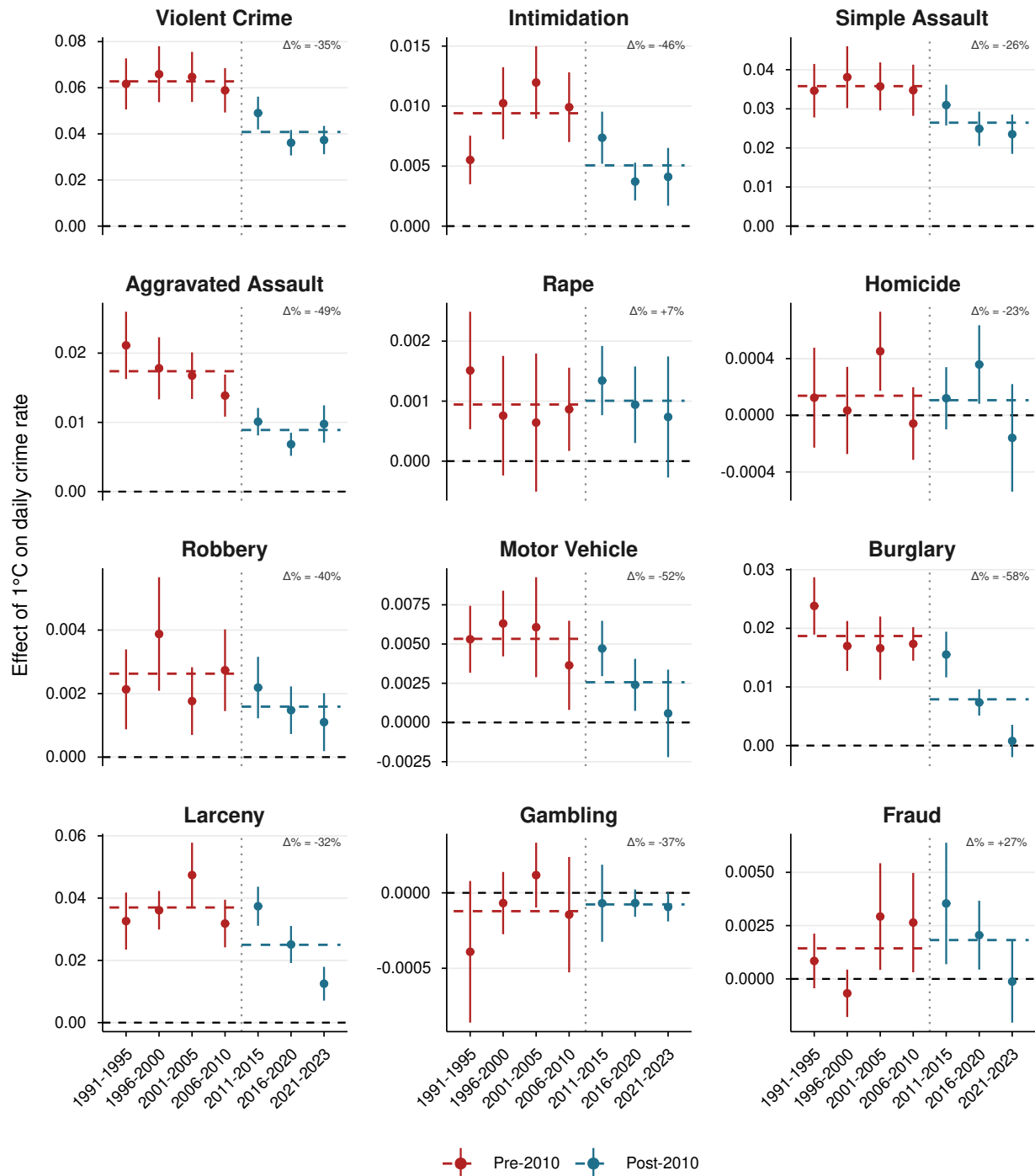
Notes: The figure plots the temperature-crime response functions across different categories of crimes using temperature bins (reference category: (15-20] °C). The regressions control for daily total precipitation, and jurisdiction-month-year, day-of-year, week-of-year and day-of-week fixed effects. Regressions are weighted by jurisdiction-location population. Confidence intervals at 95% are reported with standard errors clustered at the county-level.

Figure B5: The relationship between temperature and other crimes over time



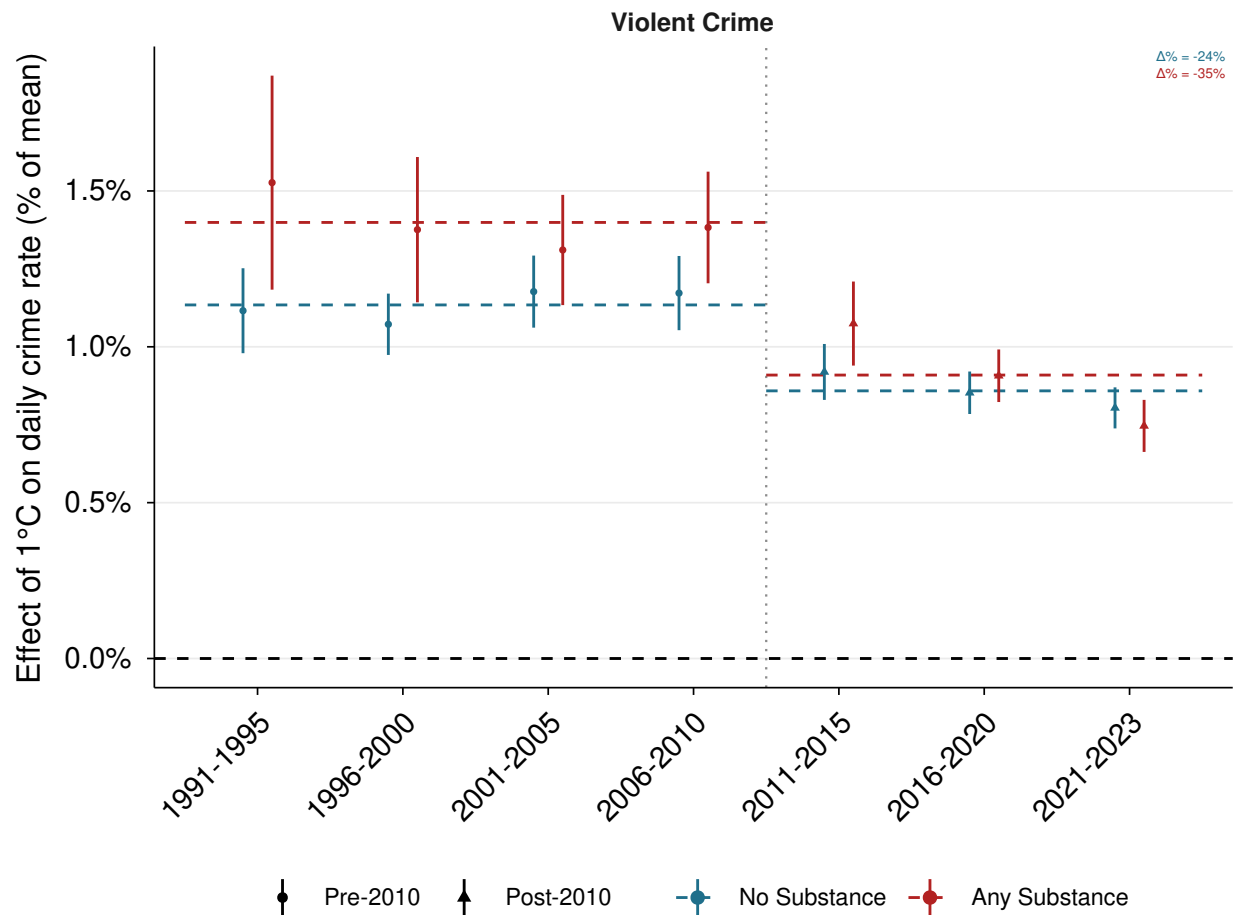
Notes: The figure plots temperature–crime response functions for categories of crimes across time periods. For each crime category, the coefficients are estimated from a single regression in which daily average temperature is interacted with time-period indicators. The regressions control for daily total precipitation, and jurisdiction-month-year, day-of-year, week-of-year and day-of-week fixed effects. Regressions are weighted by jurisdiction-location population. Confidence intervals at 95% are reported with standard errors clustered at the county-level. The dashed lines denote the average of the coefficients in the pre-2010 (red) and post-2010 (blue) periods, and $\Delta\%$ reports the percentage change between the two averages.

Figure B6: The relationship between temperature and violent crimes over time: balanced panel of agencies



Notes: The figure plots temperature–crime response functions for categories of crimes across time periods. The sample is restricted to a balanced panel of jurisdictions reporting to NIBRS continuously since 1991. For each crime category, the coefficients are estimated from a single regression in which daily average temperature is interacted with time-period indicators. The regressions control for daily total precipitation, and jurisdiction-month-year, day-of-year, week-of-year and day-of-week fixed effects. Regressions are weighted by jurisdiction-location population. Confidence intervals at 95% are reported with standard errors clustered at the county-level. The dashed lines denote the average of the coefficients in the pre-2010 (red) and post-2010 (blue) periods, and $\Delta\%$ reports the percentage change between the two averages.

Figure B7: The relationship between temperature and substance and non-substance related violent crimes over time



Notes: The figure plots the effect of a 1°C increase in daily mean temperature on the daily violent crime rate, estimated separately for offenses with (red) and without (blue) reported offender substance involvement, by five-year period between 1991 and 2023. For each category, the coefficients are estimated from a single regression in which daily average temperature is interacted with the period indicators. Coefficients are expressed as a percent of each category's mean, computed over the full 1991–2023 sample. The regressions control for daily total precipitation, and jurisdiction-by-month-by-year, day-of-year, week-of-year, and day-of-week fixed effects. Regressions are weighted by jurisdiction-location population. Vertical bars show 95% confidence intervals based on standard errors clustered at the county level. Dashed lines denote the average of the period-specific coefficients before and after 2010 for each category, and $\Delta\%$ reports the percentage change between the two averages.

C Temperature and substance abuse: Survey evidence

We provide evidence in support of the relationship between temperature and alcohol consumption using individual data from the BRFSS. We estimate the following regression:

$$Y_{idmy} = \beta_1 T_{c(i)dmy}^{30d} + \beta_2 T_{c(i)dmy}^{30d} \times H_{c(i)} + \beta_3 P_{c(i)dmy}^{30d} + \mu_{c(i)} + \phi_{dmy} + \delta_{s(i)m} + \varepsilon_{idmy}$$

where Y_{idmy} is a dummy equal to one if individual i drank heavily in the month before the interview. Heavy drinking is defined based on BRFSS definition: whether in the last month an individual has consumed more than 56 drinks if male, and 28 drinks if female. $T_{c(i)dmy}^{30d}$ and $P_{c(i)dmy}^{30d}$ are the 30-day mean of daily average and the 30-day sum of daily total precipitation prior to the interview date; and $H_{c(i)}$ is a binary indicator equal to one if a county is above 75th percentile of the pre-2010 opioid exposure. We also account for county ($\mu_{c(i)}$), calendar date (ϕ_{dmy}), and state-month fixed effects ($\delta_{s(i)m}$). We cluster standard errors at the county level, and we estimate the regression using individual sample weights.

Other than the results presented in [Table 1](#), we also find a positive and significant relationship when we substitute the dependent variable with an indicator for alcohol consumption and with the number of alcoholic drinks in the last month ([Table C1](#)). In [Table C2](#) we restrict the sample to the years 2006-2012 to match the analysis period of the OxyContin reformulation analysis.

Table C1: Impact of temperature on alcohol consumption: other outcomes

	Drink (0/1)			# of Drinks		
	(1)	(2)	(3)	(4)	(5)	(6)
Temperature (°C)	0.000660*** (0.0002)	0.00100*** (0.0004)	0.000853** (0.0004)	0.0560*** (0.015)	0.0806*** (0.028)	0.0597** (0.027)
Precipitation (m)	-0.0268** (0.011)	0.00461 (0.014)	0.00145 (0.013)	0.655 (1.055)	2.141 (1.396)	2.589* (1.385)
Mean Outcome	0.5439	0.5439	0.5439	11.8653	11.8653	11.8653
Observations	4,046,249	4,046,249	3,842,234	3,994,304	3,994,304	3,793,862
BRFSS Controls			✓			✓
County FE	✓	✓	✓	✓	✓	✓
Date FE	✓	✓	✓	✓	✓	✓
State-Month FE		✓	✓		✓	✓
Sample Weights	✓	✓	✓	✓	✓	✓

Notes: In Columns 1-3 the dependent variable is a dummy variable indicating whether the individual has consumed alcohol in the last month. In Columns 4-6 the dependent variable is the number of alcoholic drinks consumed in the last month. The regressions are weighted using BRFSS sample weights. Standard errors are clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table C2: Impact of temperature on alcohol consumption: period 2006-2012

	Any Drink (0/1)	# of Drinks	Heavy Drinking (0/1)
	(1)	(2)	(3)
Temperature (°C)	0.000298 (0.0005)	0.0512 (0.036)	0.000608** (0.0003)
Precipitation (m)	0.00981 (0.015)	-1.584 (1.355)	-0.00602 (0.008)
Mean Outcome	0.5293	12.2559	0.0789
Observations	2,291,605	2,268,459	2,268,459
BRFSS Controls	✓	✓	✓
County FE	✓	✓	✓
Date FE	✓	✓	✓
State-Month FE	✓	✓	✓
Sample Weights	✓	✓	✓

Notes: The sample is restricted to the period 2006-2012. In Column 1 the dependent variable is a dummy variable indicating whether the individual has consumed alcohol in the last month. In Columns 2 the dependent variable is the number of alcoholic drinks consumed in the last month. In Columns 3 the dependent variable is a dummy variable indicating whether the individual was a heavy drinker in the last month. Heavy drinking indicates whether in the last month an individual has consumed more than 56 drinks if male, and 28 drinks if female. The regressions are weighted using BRFSS sample weights. Standard errors are clustered at the county level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

D Opioid reformulation: Additional results and robustness

Table D1: Effects of the opioid reformulation on the temperature–crime relationship: without location- and time-specific weather effects

	Violent crimes (per 100,000 people)					
	Total	Assault			Homicide	Rape
		Intimidation	Simple	Aggravated		
	(1)	(2)	(3)	(4)	(5)	(6)
Temperature	0.0184*** (0.0031)	0.0011 (0.0010)	0.0127*** (0.0019)	0.0042*** (0.0008)	0.0000495 (0.0000614)	0.0005** (0.0002)
Temperature × Opioid exposure	0.0251*** (0.0031)	0.0063*** (0.0012)	0.0143*** (0.0019)	0.0042*** (0.0009)	0.0001 (0.000066)	0.0004 (0.0003)
Temperature × Post-2010	-0.0025 (0.0024)	0.0007 (0.0008)	-0.0027* (0.0016)	-0.0004 (0.0007)	0.0000648 (0.0000658)	-0.0002 (0.0002)
Temperature × Opioid exposure × Post-2010	-0.0105*** (0.0026)	-0.0030*** (0.0009)	-0.0059*** (0.0016)	-0.0014* (0.0007)	-0.0000832 (0.0000717)	-0.0001 (0.0003)
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓
Pre-policy outcome mean	3.909	0.799	2.406	0.607	0.012	0.124
Observations	38,013,909	38,013,909	38,013,909	38,013,909	38,013,909	38,013,909

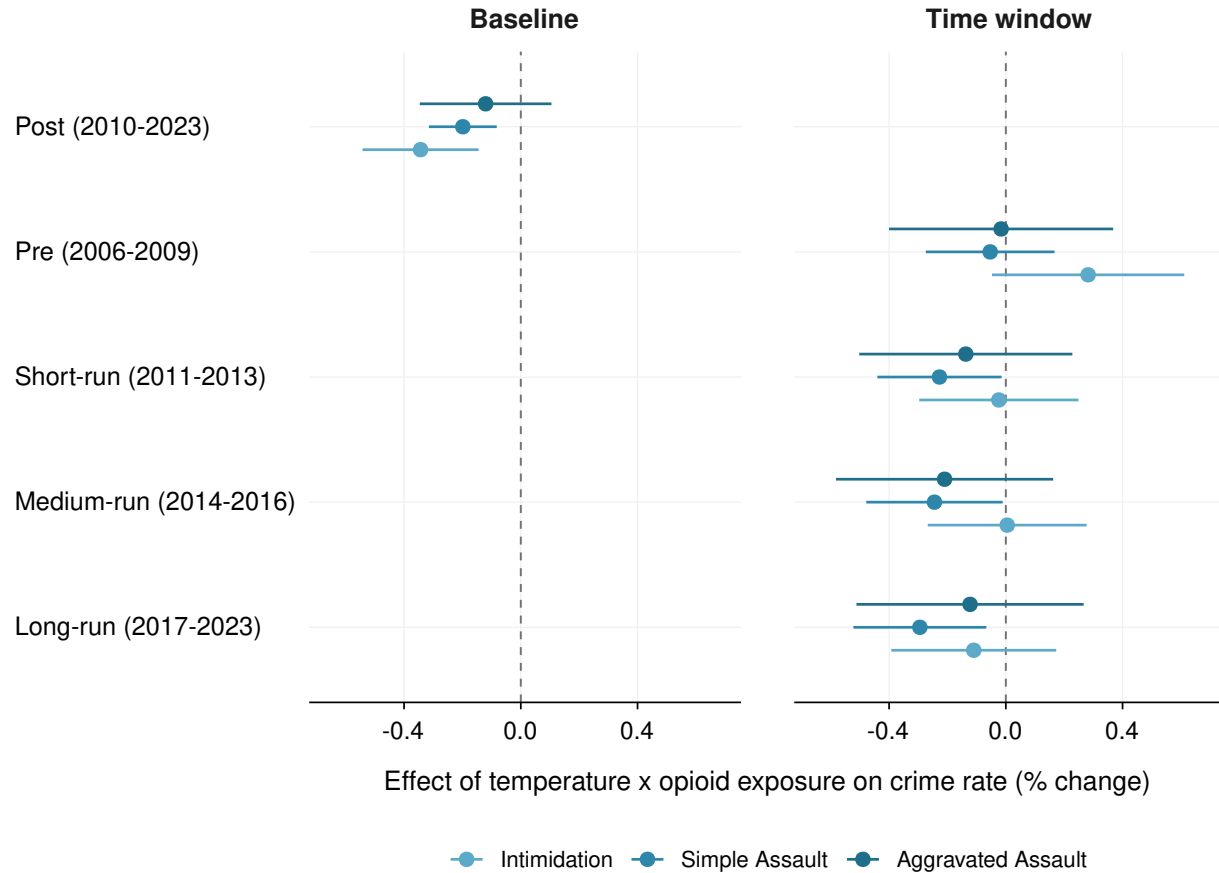
Notes: The table reports estimates of Equation (1) omitting the state- and county-specific and year-specific temperature and precipitation effects included in Table 2, so that the level effect of temperature and its interaction with opioid exposure are identified. The dependent variable is the daily number of crimes per 100,000 people. *Post-2010* is a dummy variable equal to one for the post-reformulation period, starting from 2010. *Opioid exposure* is the county-level average of Schedule II opioid prescriptions per capita over 2006–2009. All regressions also include precipitation, its interactions with opioid exposure and with the post-2010 indicator, and their triple interaction. Standard errors are clustered at the county level. All regressions are weighted using jurisdiction population weights. Significance levels are indicated as follows: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table D2: Effects of the opioid reformulation on the temperature–crime relationship: monetary crimes

	Monetary crimes (per 100,000 people)						
	Total (1)	Burglary (2)	Fraud (3)	Gambling (4)	Larceny (5)	Motor Vehicle Theft (6)	Robbery (7)
Panel A							
Temperature × Opioid exposure	0.0258*** (0.0046)	0.0074*** (0.0014)	0.0009 (0.0008)	0.0000658 (0.0000589)	0.0153*** (0.0028)	0.0016*** (0.0006)	0.0006 (0.0006)
Temperature × Opioid exposure × Post-2010	-0.0114*** (0.0034)	-0.0038*** (0.0011)	-0.0003 (0.0008)	-0.0000939 (0.0000573)	-0.0062*** (0.0023)	-0.0004 (0.0006)	-0.0005 (0.0005)
Temperature × State FE	✓	✓	✓	✓	✓	✓	✓
Precipitation × State FE	✓	✓	✓	✓	✓	✓	✓
Panel B							
Temperature × Opioid exposure × Post-2010	-0.0110*** (0.0033)	-0.0039*** (0.0011)	-0.0001 (0.0008)	-0.0000875 (0.0000587)	-0.0059*** (0.0023)	-0.0004 (0.0006)	-0.0006 (0.0004)
Temperature × County FE	✓	✓	✓	✓	✓	✓	✓
Precipitation × County FE	✓	✓	✓	✓	✓	✓	✓
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓	✓
Temperature × Year FE	✓	✓	✓	✓	✓	✓	✓
Precipitation × Year FE	✓	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓	✓
Observations	38,013,909	38,013,909	38,013,909	38,013,909	38,013,909	38,013,909	38,013,909
Pre-policy outcome mean	9.3864	1.8537	0.7969	0.0037	5.7067	0.7317	0.2936

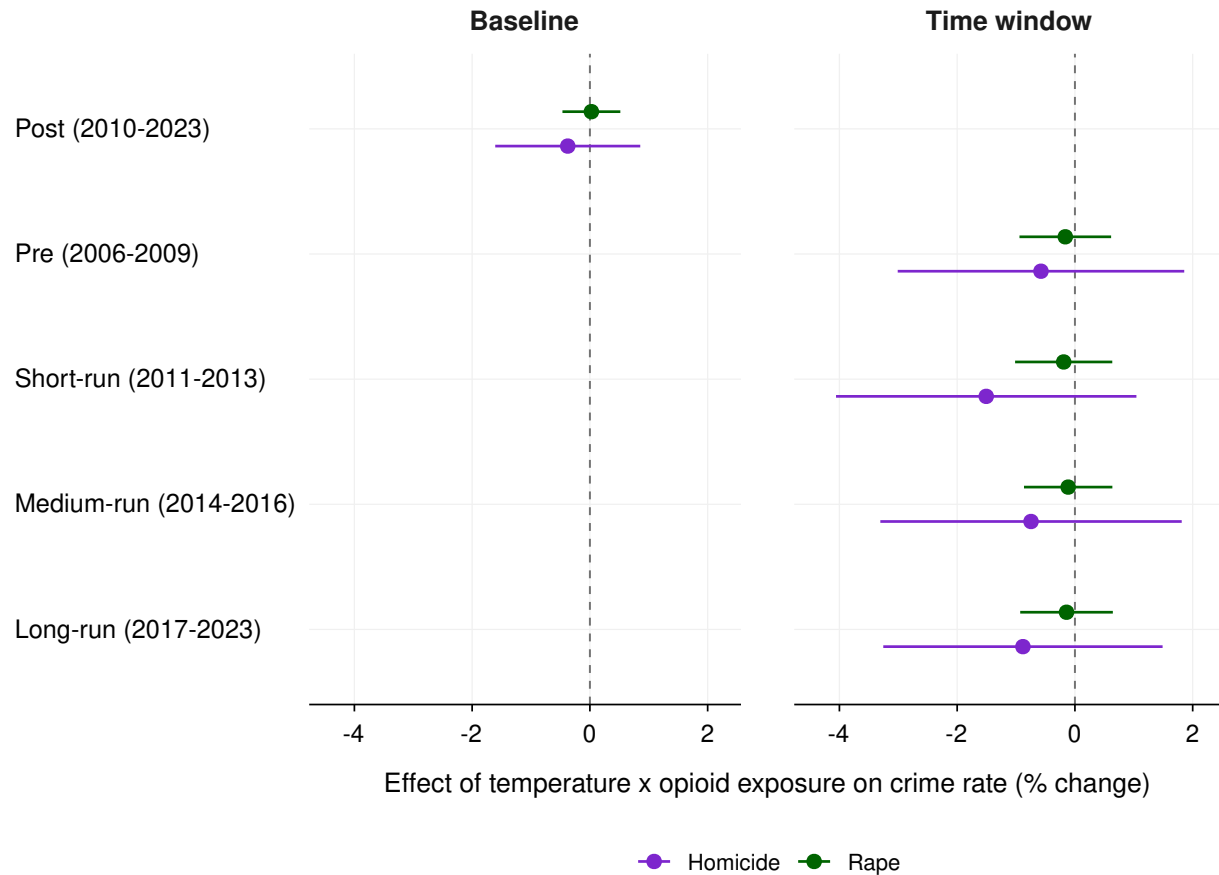
Notes: The dependent variable is the number of daily monetary crimes per 100,000 people. *Post-2010* is a dummy variable equal to 1 for the post-reformulation period, starting from 2010. *Opioid exposure* is the county-level pre-2010 exposure to opioids. Standard errors are clustered at the county level. All regressions are weighted using jurisdiction population weights. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Figure D1: Differential effects of opioid reformulation on temperature-assault types over time



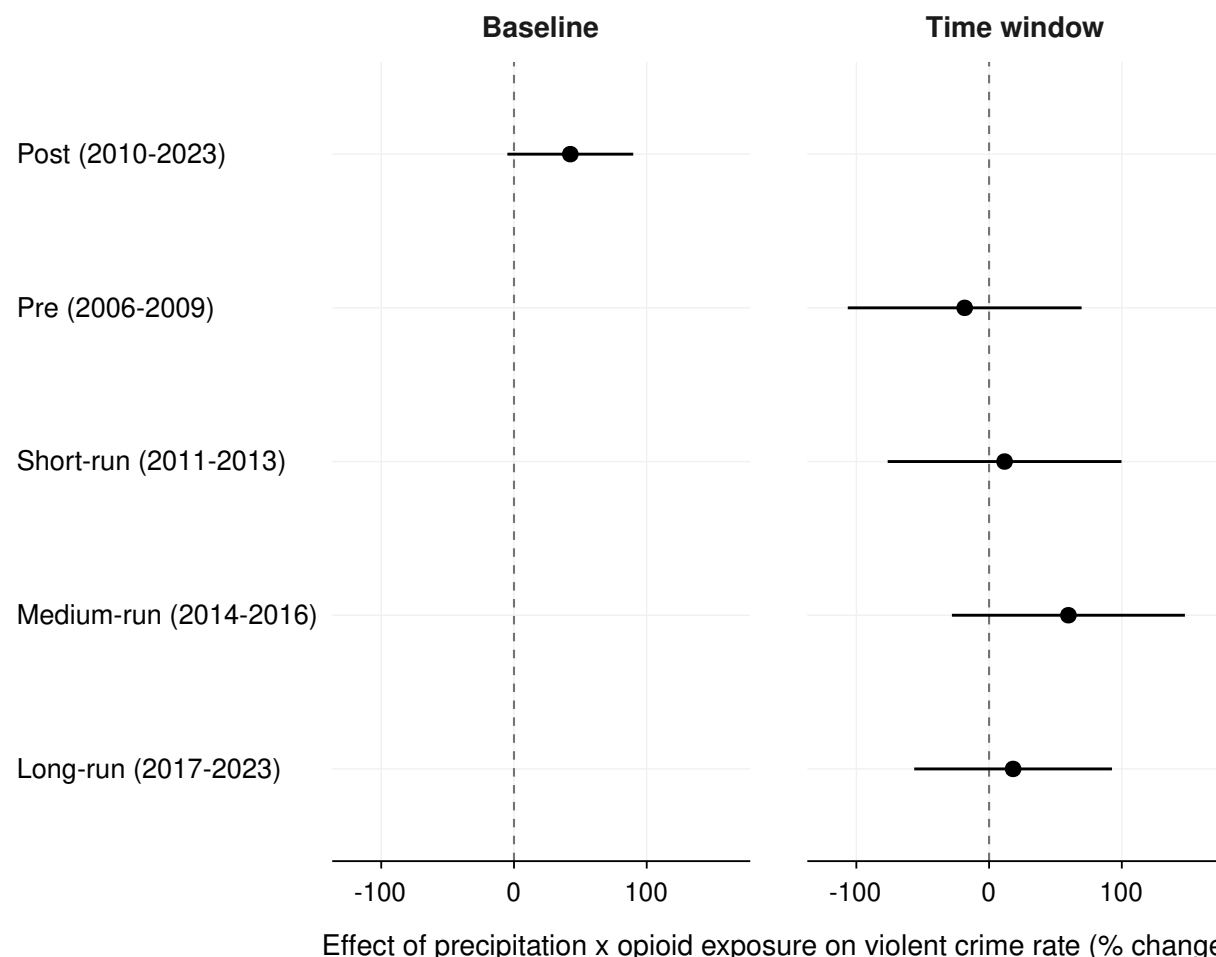
Notes: Each panel and color corresponds to a separate regression. The dependent variable is the number of daily crimes per 100,000 inhabitants, normalized on the y-axis according to the average of each category before 2010. The figure plots the coefficients of the triple interaction term between average daily county-level temperature, pre-intervention exposure as the population-weighted mean per capita opioid prescription, and in the left graph an indicator variable that takes the value of one for the post-reformulation years, 2010-2023, and in the right graph an indicator variable (short-run) equal to one for the years immediately following reformulation, 2011 to 2013, another one (medium-run) equal to one for the years 2014 to 2016, and another indicator variable (long-run) for the years 2017-2023 on the jurisdiction-level rate of crimes. The regression also controls for year-specific and county-specific temperature and precipitation coefficients, jurisdiction-month-year, week-of-year, day-of-year and day-of-week fixed effects. The regressions are weighted using jurisdiction population weights. Bins represent the 95% confidence intervals with standard errors clustered at the county-level.

Figure D2: Differential effects of opioid reformulation on temperature-homicide and rape over time



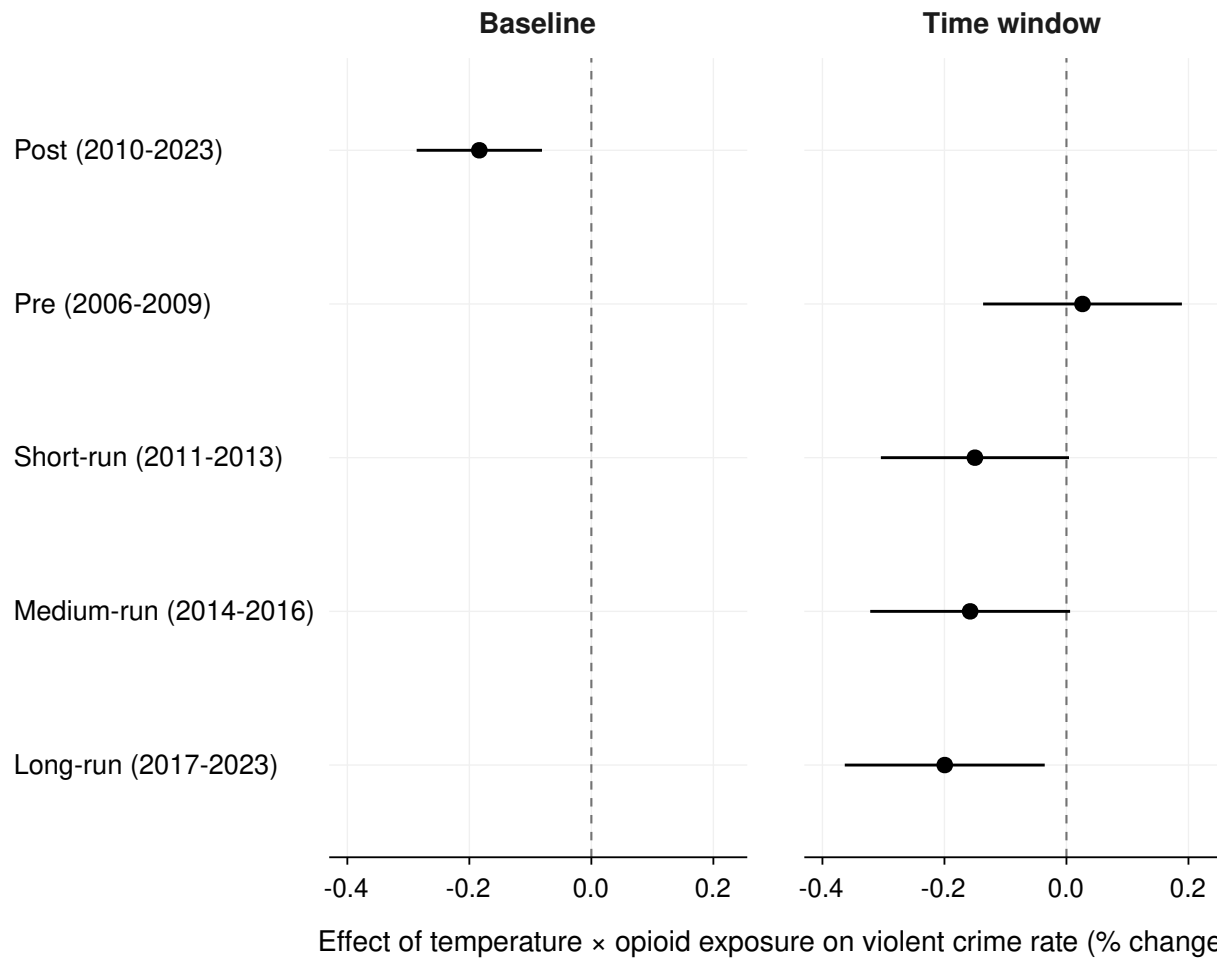
Notes: Each panel and color corresponds to a separate regression. The dependent variable is the number of daily crimes per 100,000 inhabitants, normalized on the y-axis according to the average of each category before 2010. The figure plots the coefficients of the triple interaction term between average daily county-level temperature, pre-intervention exposure as the population-weighted mean per capita opioid prescription, and in the left graph an indicator variable that takes the value of one for the post-reformulation years, 2010-2023, and in the right graph an indicator variable (short-run) equal to one for the years immediately following reformulation, 2011 to 2013, another one (medium-run) equal to one for the years 2014 to 2016, and another indicator variable (long-run) for the years 2017-2023 on the jurisdiction-level rate of crimes. The regression also controls for year-specific and county-specific temperature and precipitation coefficients, jurisdiction-month-year, week-of-year, day-of-year and day-of-week fixed effects. The regressions are weighted using jurisdiction population weights. Bins represent the 95% confidence intervals with standard errors clustered at the county-level.

Figure D3: Differential effects of opioid reformulation on precipitation-violent crimes over time



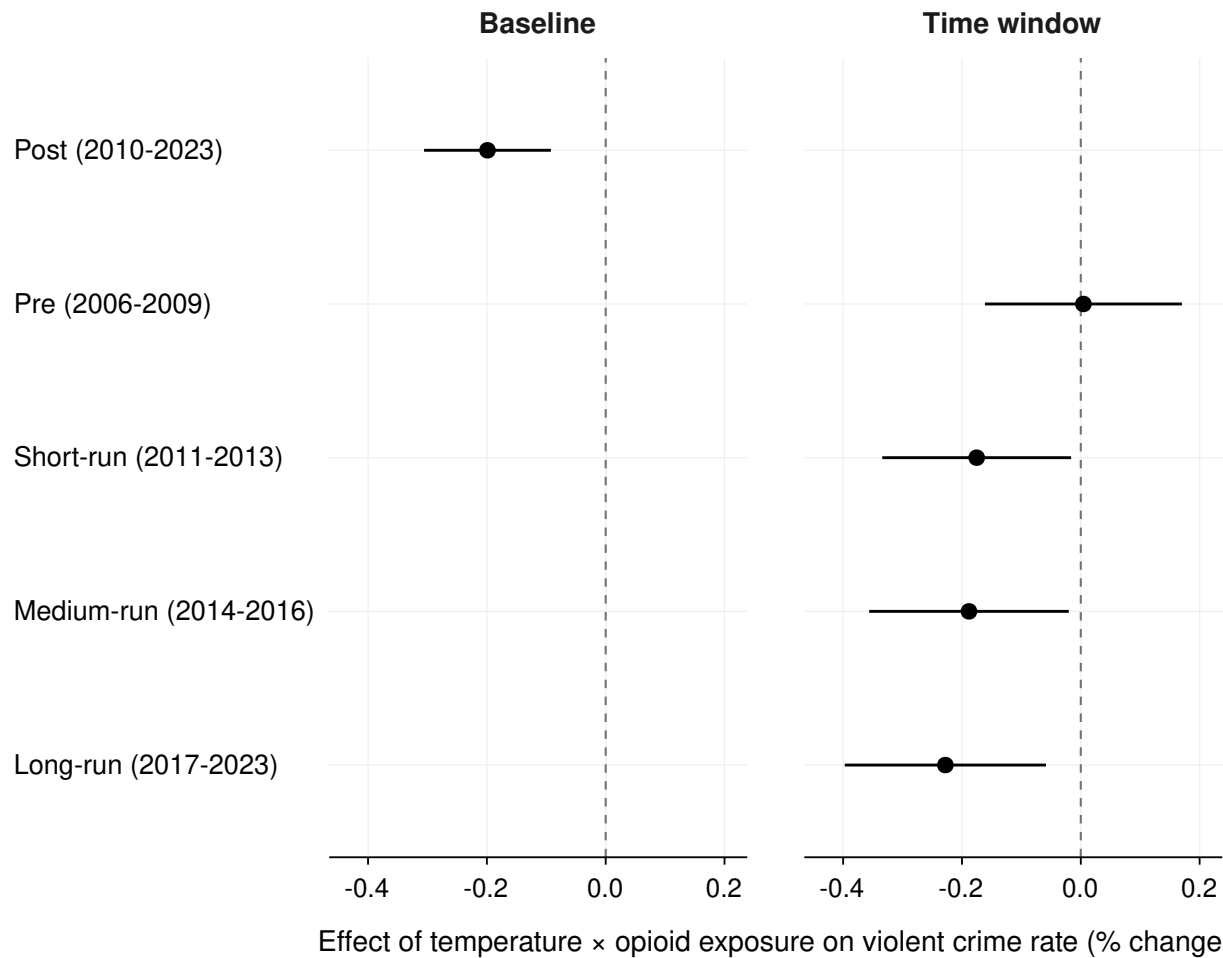
Notes: The figure plots the coefficients from the triple interaction term between total daily county-level precipitation, pre-intervention exposure as the population-weighted mean per capita opioid prescription, and in the left graph an indicator variable that takes the value of one for the post-reformulation years, 2010-2023, and in the right graph an indicator variable (short-run) equal to one for the years immediately following reformulation, 2011 to 2013, another one (medium-run) equal to one for the years 2014 to 2016, and another indicator variable (long-run) for the years 2017-2023 on the jurisdiction-level rate of violent crimes. The regression also controls for year-specific and state-specific temperature and precipitation coefficients, jurisdiction-month-year, week-of-year, day-of-year and day-of-week fixed effects. Regressions are weighted by the population in each jurisdiction. Bins represent the 95% confidence intervals with standard errors clustered at the county-level.

Figure D4: Differential effects of opioid reformulation on temperature-violent crimes over time: excluding Covid-19 period



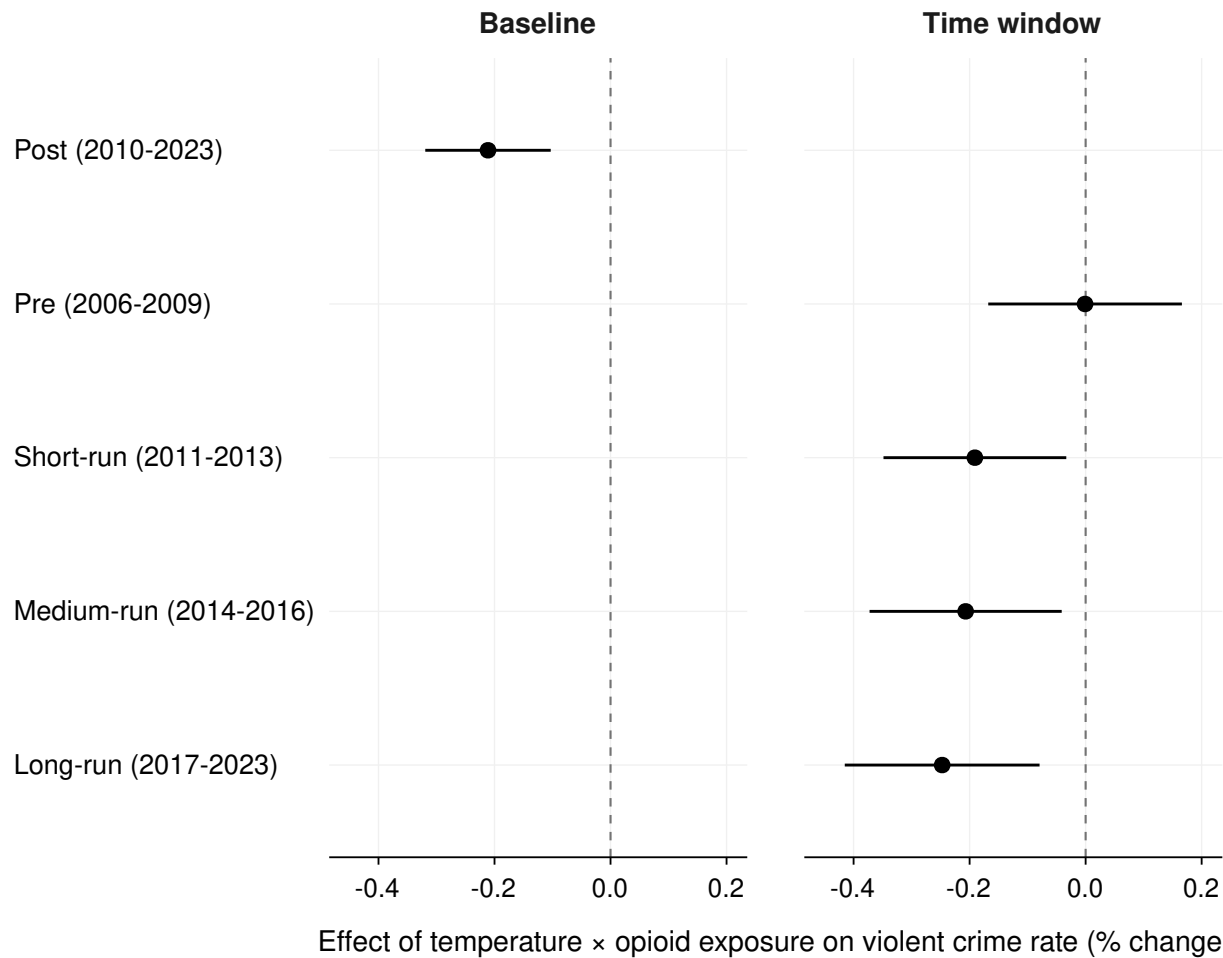
Notes: The figure plots the coefficients from the triple interaction term between average daily county-level temperature, pre-intervention exposure as the population-weighted mean per capita opioid prescription, and in the left graph an indicator variable that takes the value of one for the post-reformulation years, 2010-2023, and in the right graph an indicator variable (short-run) equal to one for the years immediately following reformulation, 2011 to 2013, another one (medium-run) equal to one for the years 2014 to 2016, and another indicator variable (long-run) for the years 2017-2023 on the jurisdiction-level rate of violent crimes. The regression also controls for year-specific and state-specific temperature and precipitation coefficients, jurisdiction-month-year, week-of-year, day-of-year and day-of-week fixed effects. Regressions are weighted by the population in each jurisdiction. Bins represent the 95% confidence intervals with standard errors clustered at the county-level.

Figure D5: Differential effects of opioid reformulation on temperature-violent crimes over time: time-invariant population weights



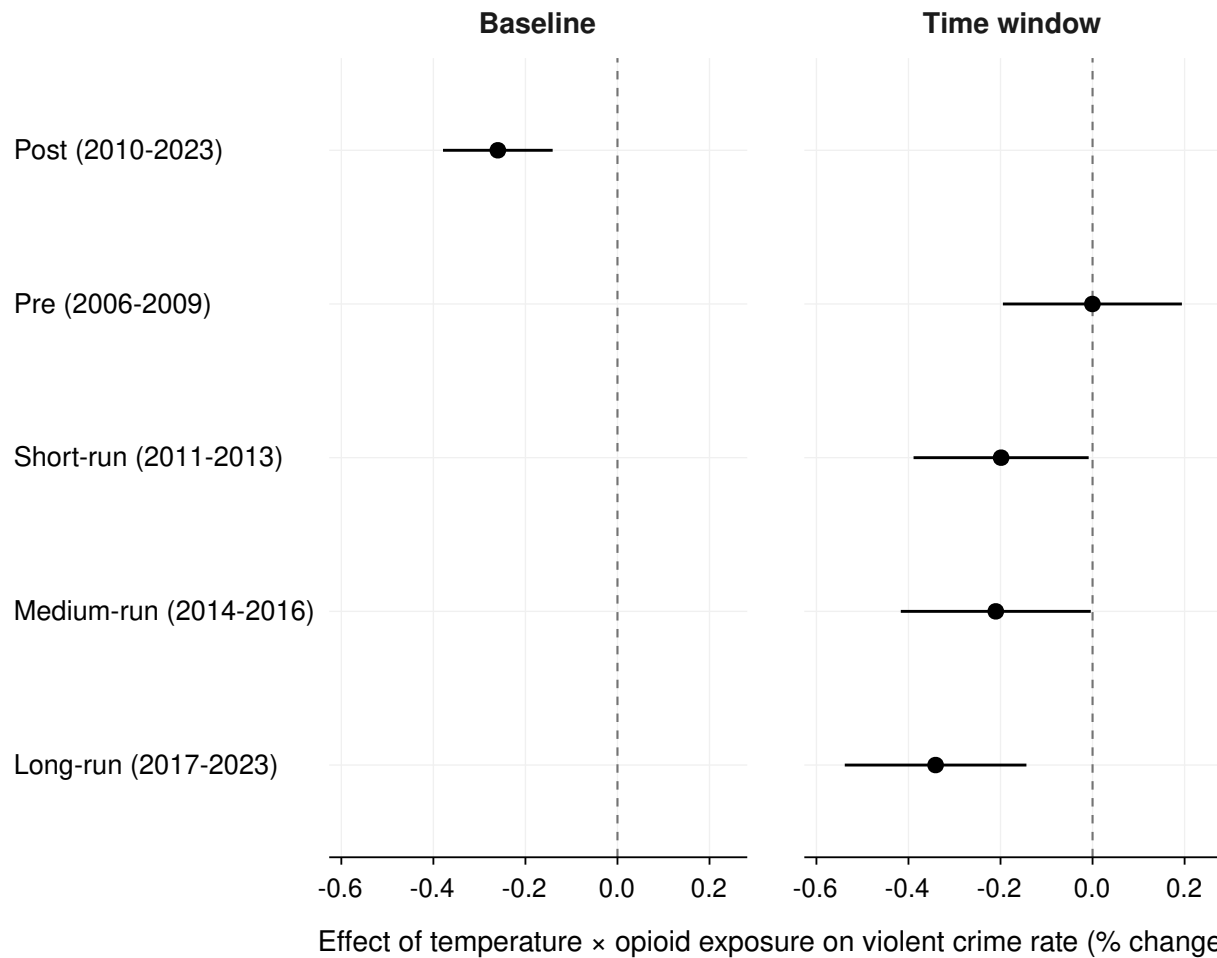
Notes: The figure plots the coefficients from the triple interaction term between average daily county-level temperature, pre-intervention exposure as the population-weighted mean per capita opioid prescription, and in the left graph an indicator variable that takes the value of one for the post-reformulation years, 2010-2023, and in the right graph an indicator variable (short-run) equal to one for the years immediately following reformulation, 2011 to 2013, another one (medium-run) equal to one for the years 2014 to 2016, and another indicator variable (long-run) for the years 2017-2023 on the jurisdiction-level rate of violent crimes. The regression also controls for year-specific and state-specific temperature and precipitation coefficients, jurisdiction-month-year, week-of-year, day-of-year and day-of-week fixed effects. The regressions are weighted using time-invariant jurisdiction population weights, specifically the average population between 2006 and 2023. Bins represent the 95% confidence intervals with standard errors clustered at the county-level.

Figure D6: Differential effects of opioid reformulation on temperature-violent crimes over time: agencies reporting in all twelve months



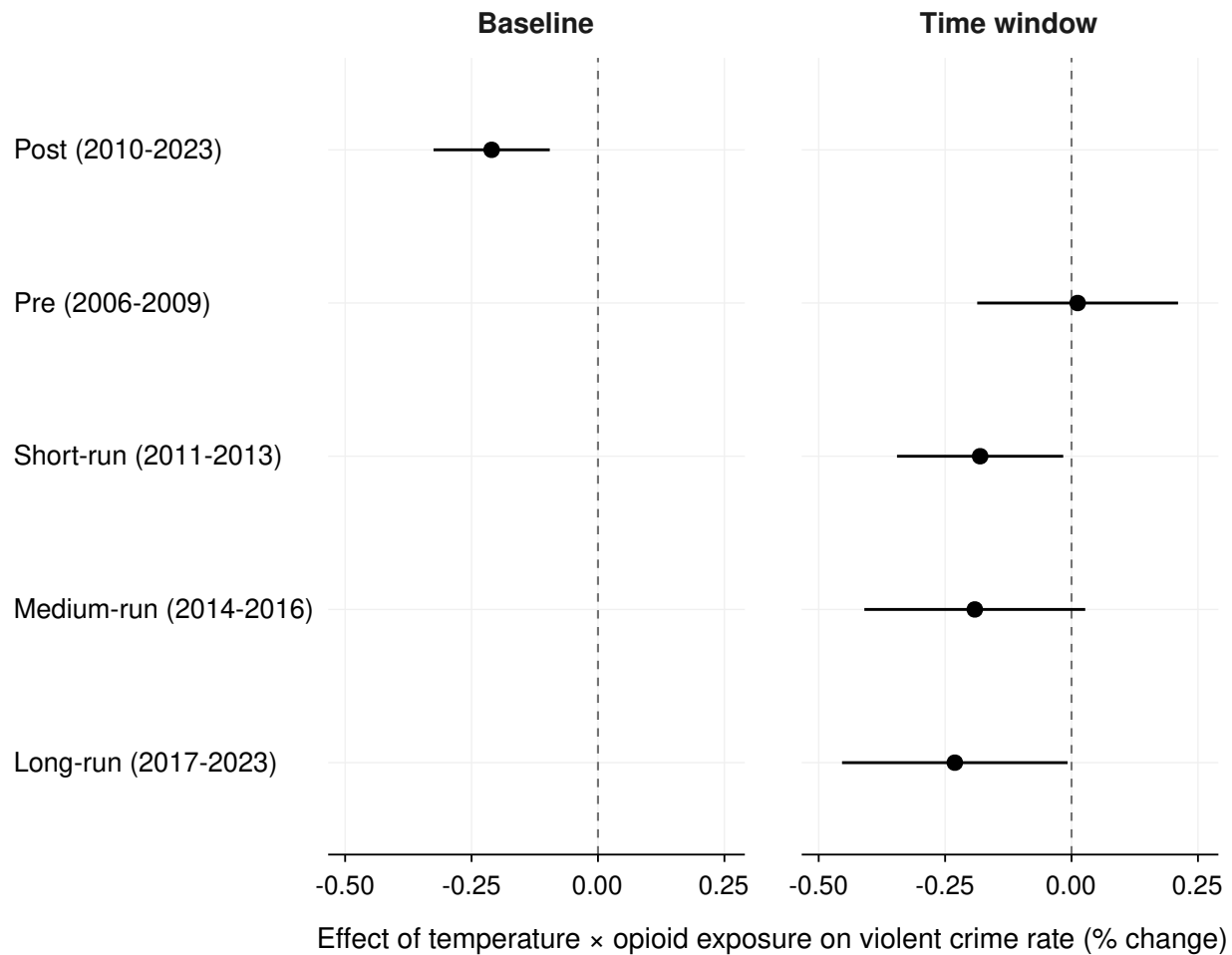
Notes: The figure plots the coefficients from the triple interaction term between average daily county-level temperature, pre-intervention exposure as the population-weighted mean per capita opioid prescription, and in the left graph an indicator variable that takes the value of one for the post-reformulation years, 2010-2023, and in the right graph an indicator variable (short-run) equal to one for the years immediately following reformulation, 2011 to 2013, another one (medium-run) equal to one for the years 2014 to 2016, and another indicator variable (long-run) for the years 2017-2023 on the jurisdiction-level rate of violent crimes. The regression also controls for year-specific and state-specific temperature and precipitation coefficients, jurisdiction-month-year, week-of-year, day-of-year and day-of-week fixed effects. Regressions are weighted by the population in each jurisdiction. Sample is restricted to agencies reporting crimes for 12 months. Bins represent the 95% confidence intervals with standard errors clustered at the county-level.

Figure D7: Differential effects of opioid reformulation on temperature-violent crimes over time: balanced sample of agencies



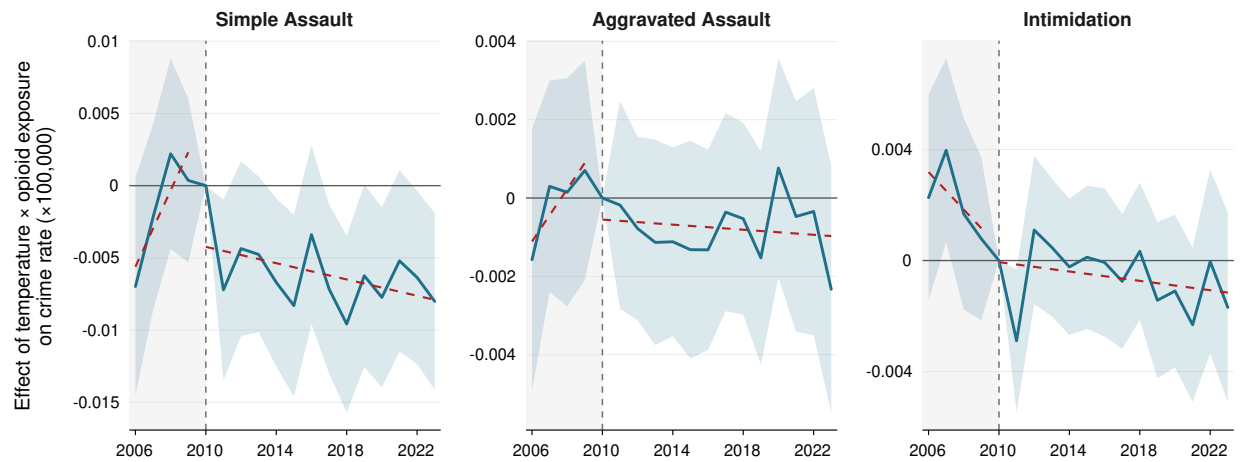
Notes: The figure plots the coefficients from the triple interaction term between average daily county-level temperature, pre-intervention exposure as the population-weighted mean per capita opioid prescription, and in the left graph an indicator variable that takes the value of one for the post-reformulation years, 2010-2023, and in the right graph an indicator variable (short-run) equal to one for the years immediately following reformulation, 2011 to 2013, another one (medium-run) equal to one for the years 2014 to 2016, and another indicator variable (long-run) for the years 2017-2023 on the jurisdiction-level rate of violent crimes. The regression also controls for year-specific and state-specific temperature and precipitation coefficients, jurisdiction-month-year, week-of-year, day-of-year and day-of-week fixed effects. Regressions are weighted by the population in each jurisdiction. Bins represent the 95% confidence intervals with standard errors clustered at the county-level.

Figure D8: Differential effects of opioid reformulation on temperature-violent crimes over time: state-level standard errors



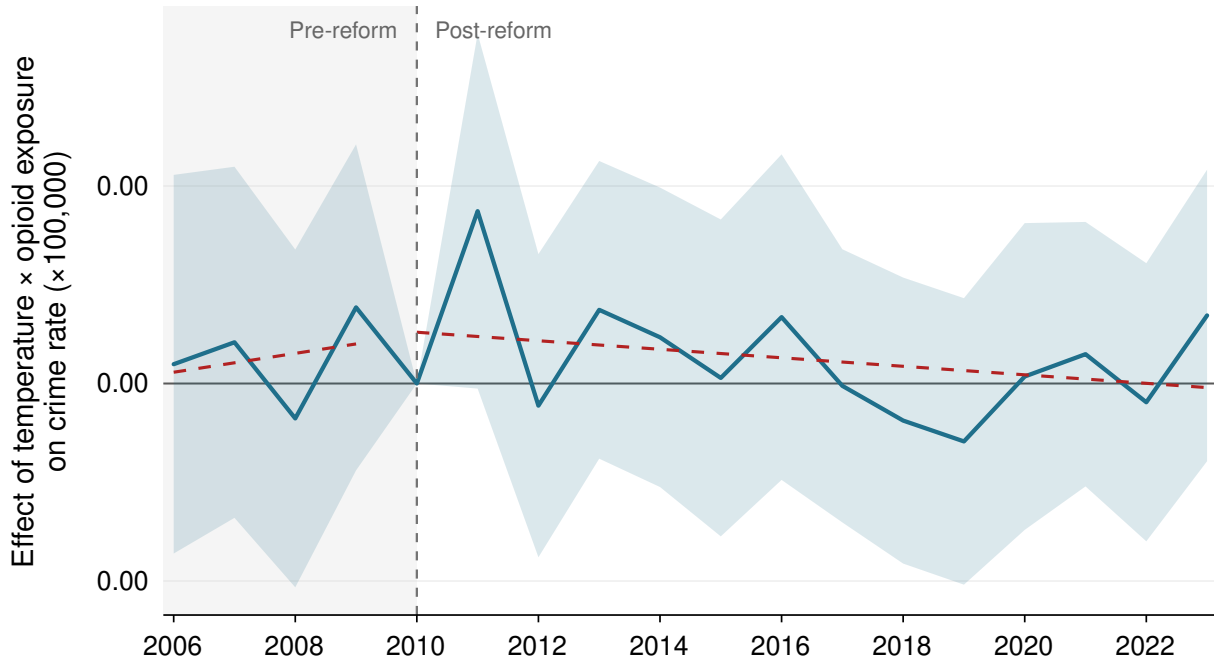
Notes: The figure plots the coefficients from the triple interaction term between average daily county-level temperature, pre-intervention exposure as the population-weighted mean per capita opioid prescription, and in the left graph an indicator variable that takes the value of one for the post-reformulation years, 2010-2023, and in the right graph an indicator variable (short-run) equal to one for the years immediately following reformulation, 2011 to 2013, another one (medium-run) equal to one for the years 2014 to 2016, and another indicator variable (long-run) for the years 2017-2023, several years post-reformulation. The regression also controls for year-specific and state-specific temperature and precipitation coefficients, jurisdiction-month-year, week-of-year, day-of-year and day-of-week fixed effects. Regressions are weighted by the population in each jurisdiction. Bins represent the 95% confidence intervals with standard errors clustered at the state-level.

Figure D9: Event study of the differential effect of opioid reformulation on temperature-assault types



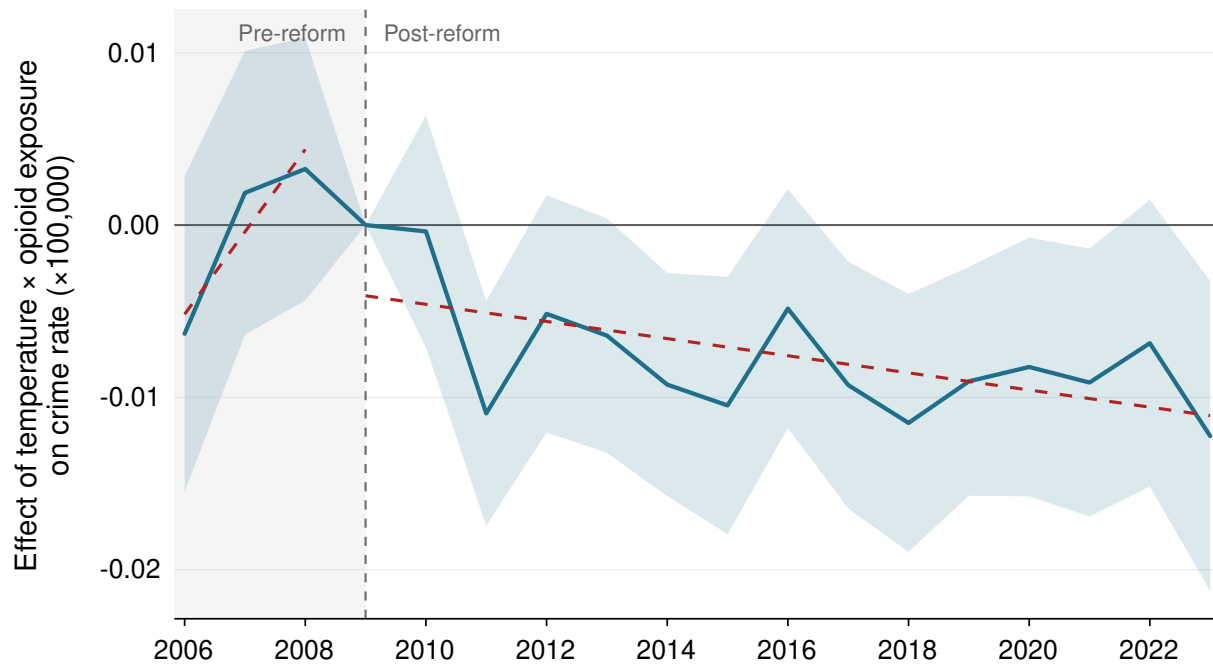
Notes: The figure plots the coefficients associated with the triple interaction term between daily temperature, pre-intervention opioid exposure and year dummies in a regression where the outcome variable is the number of daily crimes (per 100,000 people) for each of the three categories of assault crimes. The regression also controls for jurisdiction-month-year, day-of-year, day-of-week, and week-of-year fixed effects, and county- and year-specific temperature and precipitation effects. Dashed red lines show fitted lines in the annual coefficients before and after the reform. The joint test of the four pre-reformulation coefficients (2006–2009) gives $F(4) = 1.62$ ($p = 0.165$) for simple assault, $F(4) = 0.62$ ($p = 0.647$) for aggravated assault and $F(4) = 1.63$ ($p = 0.163$) for intimidation. The regressions are weighted using jurisdiction population weights. Shaded area represent the 95% confidence intervals with standard errors clustered at the county-level.

Figure D10: Event study of the differential effect of opioid reformulation on temperature-fraud



Notes: The figure plots the coefficients associated with the triple interaction term between daily temperature, pre-intervention opioid exposure and year dummies in a regression where the outcome variable is the number of daily crimes (per 100,000 people) for fraud crimes. The regression also controls for jurisdiction-month-year, day-of-year, day-of-week, and week-of-year fixed effects, and county- and year-specific temperature and precipitation effects. Dashed red lines show fitted lines in the annual coefficients before and after the reform. The joint test of the four pre-reformulation coefficients (2006–2009) gives $F(4) = 0.39$ ($p = 0.813$). The regression is weighted using jurisdiction population weights. Shaded area represent the 95% confidence intervals with standard errors clustered at the county-level.

Figure D11: Event study of the differential effect of opioid reformulation on temperature-violent crime: 2009 as reference year



Notes: The figure plots the coefficients associated with the triple interaction term between daily temperature, pre-intervention opioid exposure and year dummies in a regression where the outcome variable is the number of daily violent crimes (per 100,000 people). The regression also controls for jurisdiction-month-year, day-of-year, day-of-week, and week-of-year fixed effects, and county- and year-specific temperature and precipitation effects. Unlike in Figure 9, we set 2009 rather than 2010 as omitted reference year. Dashed red lines show fitted lines in the annual coefficients before and after the reform. The joint test of the three pre-reformulation coefficients (2006–2008) gives $F(4) = 0.15$ ($p = 0.227$). The regression is weighted using jurisdiction population weights. Shaded area represent the 95% confidence intervals with standard errors clustered at the county-level.

Table D3: Effects of opioid reformulation on temperature and violent crimes: robustness checks

	Violent crimes (per 100,000 people)				
	No Covid years (1)	Alternative pop. weights (2)	12-month reporters (3)	Balanced panel (4)	State-clustered SE (5)
Temperature × Opioid exposure	0.0162*** (0.0041)	0.0168*** (0.0041)	0.0173*** (0.0042)	0.0184*** (0.0052)	0.0170*** (0.0047)
Temperature × Opioid exposure × Post	-0.0080*** (0.0021)	-0.0082*** (0.0023)	-0.0089*** (0.0024)	-0.0108*** (0.0025)	-0.0085*** (0.0019)
Jurisdiction × Month-Year FE	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓
Temperature × State FE	✓	✓	✓	✓	✓
Precipitation × State FE	✓	✓	✓	✓	✓
Temperature × Year FE	✓	✓	✓	✓	✓
Precipitation × Year FE	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓
Pre-policy mean outcome	3.9086	3.8916	4.0158	4.0778	3.9086
Observations	25,102,974	38,013,909	33,423,058	18,912,045	38,013,909

Notes: The dependent variable is the number of daily violent crimes per 100,000 people. Each column reports the baseline triple-difference estimated on a different sample, with state- and year-specific temperature and precipitation trends. Column 1 excludes the Covid years (sample stops in 2019). Column 2 replaces the time-varying jurisdiction population with its 2006-2023 average, both as the denominator of the crime rate and as the regression weight. Column 3 restricts the sample to jurisdictions reporting crimes for 12 months. Column 4 restricts the sample to jurisdictions observed in every year between 2006 and 2023. Column 5 uses the full sample and clusters standard errors at the state level. Temperature is the average daily temperature measured in degrees Celsius (°C). Opioid exposure is the county average of Schedule II prescriptions per capita over 2006-2009. Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level, except in Column 5. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table D4: Effects of opioid reformulation on temperature and violent crimes: alternative exposure and aggregation

	Violent crimes (per 100,000 people)			
	(1)	(2)	(3)	(4)
Temperature × Opioid exposure	0.0002** (0.0001)	0.0740 (0.0470)	0.5488*** (0.1141)	0.0184*** (0.0040)
Temperature × Opioid exposure × Post	-0.0001** (0.00005)	-0.0463* (0.0246)	-0.3048*** (0.0709)	-0.0095*** (0.0023)
Exposure measure	Pills	Shipments	Prescriptions	Prescriptions
Unit of observation	Jurisdiction-day	Jurisdiction-day	Jurisdiction-month	County-day
Jurisdiction × Month-Year FE	✓	✓		
Week-of-Year FE	✓	✓		✓
Day-of-Week FE	✓	✓		✓
Day-of-Year FE	✓	✓		✓
Jurisdiction × Year FE			✓	
Month-Year FE			✓	
County × Month-Year FE				✓
Temperature × State FE	✓	✓	✓	✓
Precipitation × State FE	✓	✓	✓	✓
Temperature × Year FE	✓	✓	✓	✓
Precipitation × Year FE	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓
Pre-policy mean outcome	3.8970	3.8970	118.9679	3.9936
Observations	39,085,853	39,085,853	1,249,092	10,338,612

Notes: The dependent variable is the number of violent crimes per 100,000 people, measured daily in columns 1, 2 and 4 and monthly in column 3. Temperature is the average daily temperature in degrees Celsius (°C), and in column 3 it is the average of daily mean temperatures within the month. In columns 1 and 2 opioid exposure is measured as the county-specific 2006-2009 average of opioid pills and opioid shipments per capita from the ARCOS database, respectively; elsewhere it is the county average of Schedule II prescriptions per capita over 2006-2009. In column 3 the data are aggregated at the jurisdiction-month level, summing crime counts and precipitation within jurisdiction and month, so that jurisdiction-by-month-year fixed effects are replaced by jurisdiction-by-year and month-year fixed effects. In column 4 the data are aggregated at the county-day level, summing crimes and population across the jurisdictions of a county. Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table D5: Effects of opioid reformulation on temperature and violent crimes: leave-one-season out

	Violent crimes (per 100,000 people)			
	No winter (1)	No spring (2)	No summer (3)	No fall (4)
Temperature $\times$ Opioid exposure	0.0216*** (0.0044)	0.0143*** (0.0040)	0.0164*** (0.0043)	0.0165*** (0.0039)
Temperature $\times$ Opioid exposure $\times$ Post	-0.0120*** (0.0029)	-0.0066*** (0.0024)	-0.0081*** (0.0024)	-0.0078*** (0.0023)
Jurisdiction $\times$ Month-Year FE	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓
Temperature $\times$ State FE	✓	✓	✓	✓
Precipitation $\times$ State FE	✓	✓	✓	✓
Temperature $\times$ Year FE	✓	✓	✓	✓
Precipitation $\times$ Year FE	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓
Pre-policy mean outcome	4.0430	3.8519	3.8252	3.9134
Observations	28,623,925	28,437,905	28,437,905	28,541,992

Notes: The dependent variable is the number of daily violent crimes per 100,000 people. Each column estimates Equation 1 excluding one meteorological season: winter (December-February), spring (March-May), summer (June-August) and fall (September-November). Temperature is the average daily temperature measured in degrees Celsius (°C). Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table D6: Effects of opioid reformulation on temperature and violent crimes: alternative fixed effects

	Violent crimes (per 100,000 people)		
	(1)	(2)	(3)
Temperature $\times$ Opioid exposure	0.0174*** (0.0040)	0.0183*** (0.0040)	0.0186*** (0.0040)
Temperature $\times$ Opioid exposure $\times$ Post	-0.0087*** (0.0023)	-0.0091*** (0.0022)	-0.0091*** (0.0022)
Jurisdiction $\times$ Month-Year FE	✓	✓	✓
Jurisdiction $\times$ Day-of-Year FE		✓	✓
Date FE	✓		✓
Week-of-Year FE		✓	
Day-of-Week FE		✓	
Temperature $\times$ State FE	✓	✓	✓
Precipitation $\times$ State FE	✓	✓	✓
Temperature $\times$ Year FE	✓	✓	✓
Precipitation $\times$ Year FE	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓
Pre-policy mean outcome	3.9086	3.9086	3.9086
Observations	38,013,909	38,013,909	38,013,909

Notes: The dependent variable is the number of daily violent crimes per 100,000 people. Column 1 replaces the day-of-year, week-of-year and day-of-week fixed effects with date fixed effects. Column 2 adds jurisdiction-by-day-of-year fixed effects to the baseline specification. Column 3 combines jurisdiction-by-day-of-year and date fixed effects. Temperature is the average daily temperature measured in degrees Celsius (°C). Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table D7: Effects of opioid reformulation on temperature and violent crimes: controlling for county-level characteristics

	Violent crimes (per 100,000 people)			
	(1)	(2)	(3)	(4)
Temperature × Opioid exposure	0.0163*** (0.0037)	0.0116*** (0.0036)	0.0152*** (0.0029)	0.0146*** (0.0029)
Temperature × Opioid exposure × Post-2010	-0.0083*** (0.0024)	-0.0062** (0.0025)	-0.0064*** (0.0024)	-0.0064*** (0.0024)
Pre-policy mean outcome	3.9090	3.9090	3.9090	3.9090
Observations	38,013,909	38,013,909	38,013,909	37,001,957
Jurisdiction-Location × Month-Year FE	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓
Temperature × State FE	✓	✓	✓	✓
Precipitation × State FE	✓	✓	✓	✓
Temperature × Year FE	✓	✓	✓	✓
Precipitation × Year FE	✓	✓	✓	✓
Unemployment rate, labor force	✓	✓	✓	✓
Rural population share		✓	✓	✓
Demographic shares			✓	✓
Police officers per capita				✓
Jurisdiction Population Weights	✓	✓	✓	✓

Notes: The dependent variable is the number of violent crimes per 100,000 people. Temperature is the average daily temperature measured in degrees Celsius (°C). Each column adds a set of pre-determined county characteristics, entered as interactions with temperature and with precipitation and as interactions of both with the post-2010 indicator, so that the weather–crime gradient is allowed to shift differentially after the reformulation with each characteristic. Column (1) controls for the unemployment rate and the labor force; column (2) adds the rural population share; column (3) adds the shares of female, white, Black and Hispanic residents and of residents aged 0–19 and 65 and over; column (4) adds sworn officers per capita in the jurisdiction from LEOKA. All characteristics are time-invariant and measured pre-reformulation. The unemployment rate and the labor force are county averages over 2006–2009, officers per capita is the jurisdiction-average over the same period, and the remaining demographic shares are from the 2010 Census. *Opioid exposure* is the county-level pre-2010 exposure to opioids. Standard errors are clustered at the county level. All regressions are weighted using jurisdiction population weights. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table D8: Effects of opioid reformulation on temperature and violent crimes: controlling for other substance abuse regulation policies

	Violent crimes (per 100,000 people)		
	Total (1)	No substance (2)	Any substance (3)
Temperature × Opioid exposure	0.0140*** (0.0045)	0.0105** (0.0041)	0.0036*** (0.0009)
Temperature × Opioid exposure × Post-2010	-0.0057*** (0.0022)	-0.0032 (0.0020)	-0.0024*** (0.0006)
Temperature × SAT	0.0003* (0.0002)	0.0004** (0.0002)	0.00004 (0.00004)
Temperature × PDMP	-0.00007 (0.0020)	-0.0001 (0.0018)	0.00006 (0.0007)
Temperature × Pill Mill	-0.0043** (0.0018)	-0.0037** (0.0018)	-0.0006 (0.0005)
Temperature × DSL	-0.0053 (0.0039)	-0.0044 (0.0038)	-0.0009 (0.0013)
Mean Outcome	3.8213	3.2991	0.5276
Observations	17,399,437	17,399,437	17,399,437
Jurisdiction-Month-Year FE	✓	✓	✓
Day-of-Year FE	✓	✓	✓
Week-of-the-Year FE	✓	✓	✓
Day-of-Week FE	✓	✓	✓
Temperature × State FE	✓	✓	✓
Temperature × Year FE	✓	✓	✓
Precipitation × State FE	✓	✓	✓
Precipitation × Year FE	✓	✓	✓
Precipitation × All policies	✓	✓	✓

Notes: The dependent variable is the number of daily monetary crimes per 100,000 people. *Post-2010* is a dummy variable equal to 1 for the post-reformulation period, starting from 2010. *Opioid exposure* is the county-level pre-2010 exposure to opioids. *SAT* indicates whether the number of open SAT per 100,000 people in the previous year. The sample period is 1999-2016. Standard errors are clustered at the county level. All regressions are weighted using jurisdiction population weights. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table D9: Effects of opioid reformulation on temperature-violent crimes by climate

	Violent crimes (per 100,000 people)	
	(1)	(2)
Temperature	0.0037 (0.0048)	
Temperature × Temperate	0.0182*** (0.0063)	
Temperature × Hot	0.0459*** (0.0106)	
Temperature × Post-2010	0.0023 (0.0042)	
Temperature × Post-2010 × Temperate	-0.0069 (0.0056)	
Temperature × Post-2010 × Hot	-0.0274*** (0.0079)	
Temperature × Opioid exposure	0.0373*** (0.0064)	0.0318*** (0.0066)
Temperature × Opioid exposure × Temperate	-0.0195*** (0.0074)	-0.0138* (0.0076)
Temperature × Opioid exposure × Hot	-0.0239** (0.0095)	-0.0281*** (0.0107)
Temperature × Opioid exposure × Post-2010	-0.0118** (0.0051)	-0.0111*** (0.0035)
Temperature × Opioid exposure × Post-2010 × Temperate	0.0033 (0.0060)	0.0030 (0.0046)
Temperature × Opioid exposure × Post-2010 × Hot	0.0125* (0.0076)	0.0139** (0.0064)
Pre-policy mean outcome	3.9090	3.9090
Observations	38,013,909	38,013,909
Jurisdiction-Month-Year FE	✓	✓
Day-of-Year FE × Climate FE	✓	✓
Week-of-Year-× Climate FE	✓	✓
Day-of-Week × Climate FE	✓	✓
Temperature × Year × Climate FE		✓
Temperature × State × Climate FE		✓
Precipitation × Year × Climate FE		✓
Precipitation × State × Climate FE		✓
Jurisdiction Population Weights	✓	✓

Notes: The dependent variable is the number of crimes per 100,000 people. *Post-2010* is a dummy variable equal to one for the post-reformulation period, starting from 2010, and zero, otherwise. *Opioid exposure* is the county-level pre-2010 exposure to opioids. Terciles are defined on the nationwide distribution of county mean temperature. The reference group is *Cold*. All regressions are weighted using jurisdiction population weights. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table D10: Effects of opioid reformulation on temperature-violent crimes by time of the day and weekday

	Violent crimes (per 100,000 people)					
	Time of the day				Day of the week	
	Morning (1)	Afternoon (2)	Evening (3)	Night (4)	Weekdays (5)	Weekends (6)
Temperature × Opioid exposure	0.0020** (0.0009)	0.0036*** (0.0012)	0.0065*** (0.0016)	0.0044*** (0.0013)	0.0154*** (0.0040)	0.0245*** (0.0053)
Temperature × Opioid exposure × Post-2010	-0.0002 (0.0006)	-0.0014 (0.0010)	-0.0041*** (0.0011)	-0.0025*** (0.0008)	-0.0072*** (0.0025)	-0.0153*** (0.0035)
Pre-policy mean outcome	0.6064	1.1269	1.3356	0.7965	3.7475	4.3119
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓
Temperature × Year FE	✓	✓	✓	✓	✓	✓
Temperature × State FE	✓	✓	✓	✓	✓	✓
Precipitation × Year FE	✓	✓	✓	✓	✓	✓
Precipitation × State FE	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓
Observations	37,755,212	37,755,212	37,755,212	37,755,212	27,146,315	10,867,594

Notes: The dependent variable is the number of daily violent crimes per 100,000 people. Columns (1)–(4) split violent crime by time of day. Columns (5)–(6) use split the sample between crimes occurred on weekdays and weekends, respectively. All regressions are weighted using jurisdiction population weights. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table D11: Effects of opioid reformulation on temperature-crimes by crime characteristics

	Violent crimes (per 100,000 people)					
	No firearm (1)	Firearm (2)	Away from home (3)	Home (4)	Indoor (5)	Outdoor (6)
Temperature × Opioid exposure	0.0137*** (0.0028)	0.0003 (0.0005)	0.0066*** (0.0019)	0.0087*** (0.0023)	0.0100*** (0.0028)	0.0019*** (0.0005)
Temperature × Opioid exposure × Post-2010	-0.0059*** (0.0018)	0.0001 (0.0003)	-0.0029** (0.0012)	-0.0042*** (0.0014)	-0.0049*** (0.0017)	-0.0009** (0.0004)
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓	✓
Temperature × Year FE	✓	✓	✓	✓	✓	✓
Temperature × State FE	✓	✓	✓	✓	✓	✓
Precipitation × Year FE	✓	✓	✓	✓	✓	✓
Precipitation × State FE	✓	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓	✓
Pre-policy mean outcome	2.9925	0.1352	1.2574	2.4160	3.0045	0.2052
Observations	38,013,909	38,013,909	38,013,909	38,013,909	38,013,909	38,013,909

Notes: The dependent variable is the number of daily violent crimes per 100,000 people by specific characteristics. All regressions are weighted using jurisdiction population weights. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

E Substance abuse treatment facilities: Methods and results

We measure access to treatment as the number of SAT establishments per 100,000 people in a county, lagged one year, and we only consider counties hosting at least one establishment at some point between 1998 and 2016 (Bondurant et al., 2018; Swensen, 2015). We estimate a specification that is similar to Equation 1, where we interact temperature with access to treatment facilities. Our estimating equation is written as

$$\begin{aligned}
 Y_{idmt} = & \beta T_{c(i)dm} \times SAT_{c(i)t-1} + \gamma P_{c(i)dm} \times SAT_{c(i)t-1} + \\
 & + \delta_{c(i)} T_{c(i)dm} + \theta_{c(i)} P_{c(i)dm} + \delta_t T_{c(i)dm} + \theta_t P_{c(i)dm} + \\
 & + \mu_{imt} + \phi_{dt} + \kappa_{dw} + \lambda_{wt} + \varepsilon_{idmt},
 \end{aligned} \tag{E.1}$$

where Y_{idmt} denotes the number of reported crimes per 100,000 people in jurisdiction i on day d , month m , and year t . Daily average temperature ($T_{c(i)dm}$) and total precipitation ($P_{c(i)dm}$) are at the county level c . $SAT_{c(i)t-1}$ captures the number of SAT facilities per 100,000 people in a county, lagged one year to ensure that changes in treatment capacity precede observed crime outcomes. As in Equation 1, we include jurisdiction-month-year (μ_{imt}), day-of-year (ϕ_{dt}), day-of-week (κ_{dw}), and week-of-year (λ_{wt}) fixed effects. All regressions are weighted by population in the jurisdiction, and we cluster standard errors at the county level.

Since here the coefficient of interest varies at the county-day-month-year level, we can include both year- and county-specific temperature and precipitation slopes in the baseline specification. This is not possible in Equation 1, where opioid exposure varies across counties but not within them, so that county-specific slopes absorb the pre-period interaction. Here, instead, permanent differences in the temperature-crime gradient across counties ($\delta_{c(i)}$) and movements in that gradient common to all counties in a given year (δ_t) are absorbed rather than attributed to treatment capacity, and β is identified from within-county changes in the facility rate.

Altogether, identification of β comes from within-jurisdiction-month-year variation in weather, comparing how crimes in a given jurisdiction respond to daily temperatures as county treatment capacity changes over time, accounting for other temporal factors that are common across jurisdictions and correlate with variation within month-year in the daily temperature-crime relationship by day of the week, day of the year, and week of the year.

Our baseline model does not allow us to identify a direct effect of SAT expansion, that is absorbed by jurisdiction-month-year fixed effects. We relax our specification to estimate the effect of opening an additional SAT facility per 100,000 people on violent crime, irrespective of daily temperatures. Our specification is written as follows:

$$\begin{aligned}
Y_{idmt} = & \alpha SAT_{c(i)t-1} + \beta_1 T_{c(i)dmt} \times SAT_{c(i)t-1} + \beta_2 T_{c(i)dmt} + \\
& + \gamma_1 P_{c(i)dmt} \times SAT_{c(i)t-1} + \gamma_2 P_{c(i)dmt} + \\
& + \mu_{idm} + \lambda_{mt} + \kappa_{dw} + \varepsilon_{idmt},
\end{aligned} \tag{E.2}$$

where μ_{idm} denotes jurisdiction-day-of-year fixed effects, λ_{mt} month-year fixed effects, and κ_{dw} day-of-week fixed effects. Because SAT varies only at the county-year level, the main effect α is identified from the remaining cross-county-year variation, and it represents the effect of SAT when temperature and precipitation are both at zero. Daily variation in temperature and precipitation within a county-year identifies the composite slope $\beta_2 + \beta_1 SAT_{c(i)t-1}$, so separating β_1 from β_2 requires comparing these slopes across county-years with different treatment capacity. [Equation E.2](#) corresponds to Panel A of [Table E2](#), which reports four fixed-effect structures of increasing restrictiveness, the last of which adds the county- and year-specific weather slopes of [Equation E.1](#). In the first three structures that comparison is largely across counties, and it requires that counties with different treatment capacity would display the same temperature slope in the absence of the facilities. The county- and year-specific slopes of the last structure absorb permanent differences across counties in temperature sensitivity and changes common to all counties in a given year, so that β_1 is identified from changes in capacity within a county over time. What remains is the possibility that these changes coincide with other local changes in temperature sensitivity, which is one of the reasons we read the SAT results as corroborating evidence.

Table E1: Substance abuse treatment facilities and the temperature-crime gradient

	Violent crimes (per 100,000 people)				
	All (1)	No substance (2)	Any substance (3)	Alcohol (4)	Drugs (5)
Temperature × SAT	-0.0002 (0.0001)	0.00006 (0.0001)	-0.0001*** (0.00004)	-0.0001*** (0.00004)	0.000003 (0.00001)
Precipitation × SAT	-0.0459 (0.0697)	-0.0603 (0.0636)	0.0160 (0.0249)	0.0184 (0.0236)	0.0035 (0.0097)
Mean Outcome	3.057	2.569	0.4929	0.4618	0.0572
Observations	26,867,494	26,867,494	26,867,494	26,867,494	26,867,494
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓
Temperature × County	✓	✓	✓	✓	✓
Temperature × Year	✓	✓	✓	✓	✓
Precipitation × County	✓	✓	✓	✓	✓
Precipitation × Year	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily violent crimes per 100,000 people. Temperature is the average daily temperature measured in degrees Celsius (°C). Precipitation is the total daily precipitation measured in metres (m). SAT indicates the number of open SAT facilities per 100,000 people in the previous year. Facility counts cover 1998-2016. The sample is restricted to the period 1999-2017 and to counties with at least one establishment opened in the same period. “Mean Outcome” is the average number of crimes in counties without any SAT establishments during the sample period. Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table E2: Substance abuse treatment facilities and the temperature-crime gradient: alternative fixed effects

	Violent crimes (per 100,000 people)				
	All (1)	No substance (2)	Any substance (3)	Alcohol (4)	Drugs (5)
Panel A: Jurisdiction-Day-of-Year + Month-Year + Day-of-Week					
Temperature	0.0369*** (0.0017)	0.0317*** (0.0015)	0.0052*** (0.0004)	0.0048*** (0.0004)	0.0008*** (0.0001)
SAT	0.0069 (0.0043)	0.0072* (0.0038)	-0.0004 (0.0013)	-0.0004 (0.0013)	-0.0000028 (0.0002)
Temperature × SAT	-0.0003* (0.0002)	-0.0002 (0.0002)	-0.000092* (0.000049)	-0.000086** (0.000043)	-0.000021 (0.000014)
Panel B: Jurisdiction-Month-Year + Day-of-Year + Week-of-Year + Day-of-Week					
Temperature	0.0394*** (0.0020)	0.0337*** (0.0018)	0.0058*** (0.0004)	0.0054*** (0.0003)	0.0007*** (0.0001)
Temperature × SAT	-0.0006*** (0.0002)	-0.0005*** (0.0002)	-0.0001*** (0.000036)	-0.0001*** (0.000034)	-0.0000037 (0.000009)
Panel C: Jurisdiction-Day-of-Year + Jurisdiction-Month-Year + Date					
Temperature	0.0356*** (0.0017)	0.0302*** (0.0016)	0.0055*** (0.0003)	0.0051*** (0.0003)	0.0007*** (0.000092)
Temperature × SAT	-0.0005*** (0.0002)	-0.0004*** (0.0002)	-0.000097*** (0.000036)	-0.000098*** (0.000034)	0.00000035 (0.000009)
Panel D: Jurisdiction-Day-of-Year + Jurisdiction-Month-Year + Date + Temp/Precip × Year/County					
Temperature × SAT	-0.0002* (0.0001)	-0.00008 (0.0001)	-0.0001*** (0.000049)	-0.0001*** (0.000047)	0.00000051 (0.000015)
Jurisdiction Population Weights	✓	✓	✓	✓	✓
Mean Outcome	3.057	2.569	0.4929	0.4618	0.0572
Observations	26,867,494	26,867,494	26,867,494	26,867,494	26,867,494

Notes: The dependent variable is the number of daily violent crimes per 100,000 people. Temperature is the average daily temperature measured in degrees Celsius (°C). Precipitation is the total daily precipitation measured in metres (m). SAT indicates the number of open SAT facilities per 100,000 people in the previous year. Facility counts cover 1998-2016. The sample is restricted to the period 1999-2017 and to counties with at least one establishment opened in the same period. "Mean Outcome" is the average number of crimes in counties without any SAT establishments during the sample period. Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, **p<0.05, ***p<0.01.

Table E3: Substance abuse treatment facilities and the temperature-crime gradient: including counties that never host a facility

	Violent crimes (per 100,000 people)				
	All (1)	No substance (2)	Any substance (3)	Alcohol (4)	Drugs (5)
Temperature $\times$ SAT	-0.0002** (0.0001)	-0.0001 (0.00009)	-0.0001*** (0.00004)	-0.0001*** (0.00004)	0.000004 (0.00001)
Precipitation $\times$ SAT	-0.0474 (0.0658)	-0.0519 (0.0603)	0.0067 (0.0245)	0.0092 (0.0232)	0.0032 (0.0098)
Mean Outcome	3.057	2.569	0.4929	0.4618	0.0572
Observations	30,233,200	30,233,200	30,233,200	30,233,200	30,233,200
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓
Temperature $\times$ County	✓	✓	✓	✓	✓
Temperature $\times$ Year	✓	✓	✓	✓	✓
Precipitation $\times$ County	✓	✓	✓	✓	✓
Precipitation $\times$ Year	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily violent crimes per 100,000 people. Temperature is the average daily temperature measured in degrees Celsius ($^{\circ}$ C). Precipitation is the total daily precipitation measured in metres (m). SAT indicates the number of open SAT facilities per 100,000 people in the previous year. Facility counts cover 1998-2016. The sample period is 1999-2017. The sample includes counties that never have a SAT according to the CBP data set in the sample period. "Mean Outcome" is the average number of crimes in counties without any SAT establishments during the sample period. Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E4: Substance abuse treatment facilities and the temperature-crime gradient: agencies reporting in all twelve months

	Violent crimes (per 100,000 people)				
	All (1)	No substance (2)	Any substance (3)	Alcohol (4)	Drugs (5)
Temperature × SAT	-0.0002* (0.0001)	0.00007 (0.0001)	-0.0001*** (0.00005)	-0.0001*** (0.00004)	0.000004 (0.00001)
Precipitation × SAT	-0.0622 (0.0776)	-0.0753 (0.0704)	0.0144 (0.0268)	0.0178 (0.0253)	0.0044 (0.0101)
Mean Outcome	3.232	2.718	0.5192	0.4865	0.0600
Observations	23,615,655	23,615,655	23,615,655	23,615,655	23,615,655
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓
Temperature × County	✓	✓	✓	✓	✓
Temperature × Year	✓	✓	✓	✓	✓
Precipitation × County	✓	✓	✓	✓	✓
Precipitation × Year	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily violent crimes per 100,000 people. Temperature is the average daily temperature measured in degrees Celsius (°C). Precipitation is the total daily precipitation measured in metres (m). SAT indicates the number of open SAT facilities per 100,000 people in the previous year. Facility counts cover 1998-2016. The sample period is 1999-2017. The sample includes only jurisdictions that reported for twelve consecutive months to NIBRS and counties with at least one establishment opened in the sample period. "Mean Outcome" is the average number of crimes in counties without any SAT establishments during the sample period. Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, **p<0.05, *** p<0.01.

Table E5: Substance abuse treatment facilities and the temperature-crime gradient: balanced panel of agencies

	Violent crimes (per 100,000 people)				
	All (1)	No substance (2)	Any substance (3)	Alcohol (4)	Drugs (5)
Temperature $\times$ SAT	-0.0002 (0.0002)	-0.000079 (0.0001)	-0.0001** (0.000064)	-0.0001** (0.000061)	-0.000018 (0.000018)
Precipitation $\times$ SAT	-0.0239 (0.0943)	-0.0539 (0.0864)	0.0319 (0.0340)	0.0301 (0.0321)	0.0188 (0.0115)
Mean Outcome	3.6179	3.1327	0.4907	0.4531	0.0641
Observations	11,036,786	11,036,786	11,036,786	11,036,786	11,036,786
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓
Temperature $\times$ County	✓	✓	✓	✓	✓
Temperature $\times$ Year	✓	✓	✓	✓	✓
Precipitation $\times$ County	✓	✓	✓	✓	✓
Precipitation $\times$ Year	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily violent crimes per 100,000 people. Temperature is the average daily temperature measured in degrees Celsius ($^{\circ}\text{C}$). SAT indicates the number of open SAT per 100,000 people in the previous year. Facility counts cover 1998-2016. The sample is restricted to the period 1999-2017 and to counties with at least one establishment opened in the same period. Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table E6: Substance abuse treatment facilities and the temperature-crime gradient: contemporaneous and one-year-ahead facility rates

	Violent crimes (per 100,000 people)				
	All (1)	No substance (2)	Any substance (3)	Alcohol (4)	Drugs (5)
Panel A: Including contemporaneous					
Temperature $\times$ SAT _{<i>t</i>-1}	-0.00006 (0.0001)	0.000075 (0.0001)	-0.0001** (0.000068)	-0.0001* (0.000068)	-0.000018 (0.000021)
Temperature $\times$ SAT _{<i>t</i>}	-0.0001 (0.0002)	-0.0001 (0.0001)	0.000036 (0.000063)	0.000025 (0.000063)	0.000027 (0.000021)
Mean Outcome	3.057	2.569	0.4929	0.4618	0.0572
Observations	24,977,889	24,977,889	24,977,889	24,977,889	24,977,889
Panel B: Including contemporaneous and lead					
Temperature $\times$ SAT _{<i>t</i>-1}	-0.000003 (0.0002)	0.0001 (0.0001)	-0.0001* (0.000072)	-0.0001* (0.000072)	-0.000017 (0.000021)
Temperature $\times$ SAT _{<i>t</i>}	-0.0001 (0.0002)	-0.0002 (0.0002)	0.000024 (0.000081)	0.000019 (0.000082)	0.000026 (0.000025)
Temperature $\times$ SAT _{<i>t</i>+1}	-0.000047 (0.0002)	-0.00006 (0.0001)	0.000015 (0.000072)	0.000014 (0.000073)	-0.000009 (0.000022)
Mean Outcome	3.057	2.569	0.4929	0.4618	0.0572
Observations	23,147,157	23,147,157	23,147,157	23,147,157	23,147,157
Jurisdiction-Month-Year FE	✓	✓	✓	✓	✓
Day-of-Year FE	✓	✓	✓	✓	✓
Week-of-Year FE	✓	✓	✓	✓	✓
Day-of-Week FE	✓	✓	✓	✓	✓
Temperature $\times$ County	✓	✓	✓	✓	✓
Temperature $\times$ Year	✓	✓	✓	✓	✓
Precipitation $\times$ County	✓	✓	✓	✓	✓
Precipitation $\times$ Year	✓	✓	✓	✓	✓
Jurisdiction Population Weights	✓	✓	✓	✓	✓

Notes: The dependent variable is the number of daily violent crimes per 100,000 people. Temperature is the average daily temperature measured in degrees Celsius (°C). Precipitation is the total daily precipitation measured in metres (m). SAT indicates the number of open SAT facilities per 100,000 people in the previous year. Facility counts cover 1998-2016. The sample period is restricted to the period 1999-2017 and counties with at least one establishment opened in the same period. “Mean Outcome” is the average number of crimes in counties without any SAT establishments during the sample period. Regressions are weighted by jurisdiction population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

Table E7: Temperature, opioid exposure, and the number of SAT facilities

	# SAT (1)	# SAT per 100,000s (2)	# SAT (3)	# SAT per 100,000s (4)
Temperature	0.4732 (0.5219)	-0.0936 (0.1104)	0.6571 (0.5550)	-0.0217 (0.1174)
Temperature × Opioid exposure	0.1229 (0.2501)	0.0962 (0.0636)	-0.0751 (0.3117)	-0.0143 (0.0730)
Temperature × Post-2010			-0.6778 (0.4586)	-0.0224 (0.0735)
Temperature × Opioid exposure × Post-2010			0.3114 (0.3585)	0.0838 (0.0519)
Mean Outcome	33.529	5.803	33.529	5.803
Observations	13,000	13,000	13,000	13,000
Precipitation	✓	✓	✓	✓
Precipitation × Opioid exposure	✓	✓	✓	✓
Precipitation × Post 2010			✓	✓
Precipitation × Opioid exposure × Post 2010			✓	✓
County FE	✓	✓	✓	✓
State-Year FE	✓	✓	✓	✓
County Population Weights	✓	✓	✓	✓

Notes: Regressions are at the county-year level over the sample period 2006-2016. The dependent variable is the number of SAT facilities (columns 1, 3) or the number of SAT facilities per 100,000 people (columns 2, 4). Temperature is the average daily temperature measured in degrees Celsius (°C). *Opioid exposure* indicates the average number of opioid prescriptions per capita in the pre-policy period 2006-2009. *Post 2010* is an indicator variable equal to 1 from 2010 onwards. Regressions are weighted by county population. Standard errors are clustered at the county level. Significance levels are indicated as follows: *p<0.10, ** p<0.05, *** p<0.01.

F Monetizing social benefits of substance-abuse policies: Methods and results

This section describes how we translate the estimated temperature–crime relationship into the monetary costs reported in [Figure 12](#) and [Table F1](#). For each violent crime type (simple assault, intimidation, aggravated assault, rape, homicide) we estimate two specifications on the 2006–2023 sample, weighted by agency population and with standard errors clustered at the county level. Differently than our baseline specification in [Equation 1](#), we do not include location- or time-specific temperature and precipitation effects, so that the level effect of temperature and its interaction with opioid exposure are both identified. Our two estimated equations are

- the opioid reformulation specification in [Table D1](#);
- the same specification interacted with climate terciles, reported in column (1) in [Table D9](#).

From these two regressions, we construct four scenarios. All are evaluated in the post-reformulation period. We only distinguish whether the opioid-attributable moderation is accounted for (III and IV), and whether the temperature response is allowed to vary by climate (II and IV). Our four scenarios read as follows:

- (I) *No adaptation*: no opioid moderation, common temperature response;
- (II) *Climate heterogeneity*: no opioid moderation, temperature response varying by climate tercile;
- (III) *OxyContin reformulation*: opioid moderation, common temperature response;
- (IV) *Reformulation and climate heterogeneity*: both.

To compute the monetary costs associated with violent crimes in each scenario, let m_{cj} denote the effect of a $+1^\circ\text{C}$ increase on the daily rate of crime j per 100,000 people in county c , and let Opioid exposure_c denote its pre-reformulation opioid exposure. Writing β_j^T for the temperature coefficient, β_j^{Tp} for its interaction with the post-reformulation indicator, β_j^{To} for its interaction with opioid exposure and β_j^{Tpo} for the triple interaction, scenarios (I) and (III) are

$$m_{cj}^{(I)} = \beta_j^T + \beta_j^{Tp} + \beta_j^{To} \text{Opioid exposure}_c, \quad (\text{F.1})$$

$$m_{cj}^{(III)} = \beta_j^T + \beta_j^{Tp} + (\beta_j^{To} + \beta_j^{Tpo}) \text{Opioid exposure}_c. \quad (\text{F.2})$$

Scenarios (II) and (IV) replace each coefficient with its tercile-specific counterpart, $\beta_{jk(c)} = \beta_j + \beta_j^{(k)}$, where $k(c)$ is the climate tercile of county c and the coldest tercile is the omitted category, so that $\beta_j^{(\text{Cold})} = 0$. Each county therefore receives an effect determined jointly by its own opioid exposure and by the tercile to which it belongs.

Because the dependent variable is a daily rate per 100,000 people, we obtain the annual national cost of a uniform increase in temperature ΔT by multiplying each county-specific effect with county population (P_c), the number of days in a year (D), and the unit cost of the crime u_j , such that

$$C_j = u_j \frac{D \Delta T}{10^5} \sum_c P_c m_{cj}, \quad (\text{F.3})$$

where $D = 365$, $\Delta T = 1^\circ\text{C}$, and population P_c is the county average over the 2006–2009 pre-reformulation period. Each C_j is a linear combination of the coefficients of a single regression, so we obtain its confidence interval by the delta method using the county-clustered variance–covariance matrix.

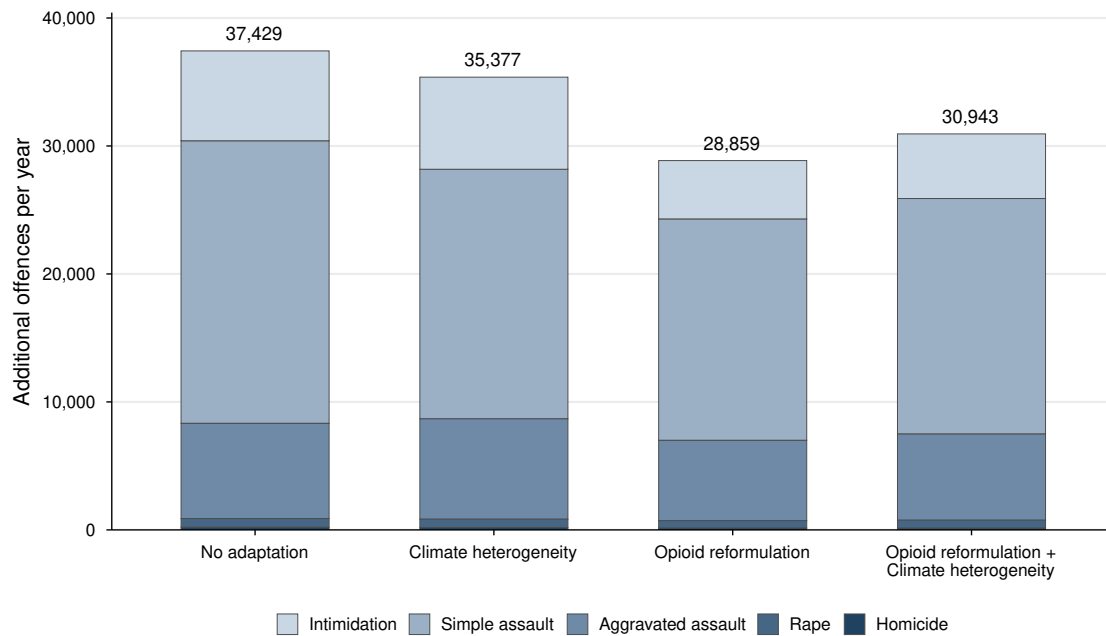
Unit costs u_j reflect the social monetary cost per crime expressed in \$2023 and reported in [Table F1](#). For aggravated assault, rape and homicide we use the police-reported information from [Miller et al. \(2021\)](#) in Table 5. For simple assault, we use [Roman et al. \(2023\)](#), who estimate average harm per victim from linked court, police, and medical claims records ([Miller et al. \(2021\)](#) report a single assault row without distinguishing costs for single and aggravated assault cases). For intimidation, for which no published unit cost exists, we scale the assault row in [Miller et al. \(2021\)](#) by the ratio of the cost of an assault with no injury to that of the average assault in [Miller et al. \(1996\)](#) in Table 2. All figures are converted to \$2023 using the CPI-U (annual average factors of 1.2431 from 2017 and 1.0412 from 2022). Unit costs are treated as constants with no uncertainty, so the reported confidence intervals reflect uncertainty in the temperature response only.

Table F1: Annual social cost of violent crimes for a +1°C increase in temperature

	Pre-policy mean	Effect of +1°C on the daily rate per 100,000			
		No adaptation (I)	Climate heterogeneity (II)	Opioid reformulation (III)	Reformulation + climate het. (IV)
Simple assault	2.406	0.0212 [0.0185, 0.0239]	0.0187 [0.0158, 0.0217]	0.0166 [0.0155, 0.0177]	0.0177 [0.0165, 0.0188]
Intimidation	0.799	0.0067 [0.0052, 0.0083]	0.0069 [0.0050, 0.0088]	0.0044 [0.0038, 0.0050]	0.0048 [0.0041, 0.0056]
Aggravated assault	0.607	0.0071 [0.0058, 0.0085]	0.0075 [0.0058, 0.0092]	0.0060 [0.0054, 0.0067]	0.0065 [0.0058, 0.0071]
Rape	0.124	0.0007 [0.0002, 0.0011]	0.0006 [0.0001, 0.0011]	0.0006 [0.0005, 0.0007]	0.0006 [0.0005, 0.0007]
Homicide	0.012	0.0002 [0.0001, 0.0003]	0.0002 [0.0000, 0.0003]	0.0001 [0.0001, 0.0002]	0.0001 [0.0001, 0.0002]
	Unit cost	Annual cost (million \$2023)			
		No adaptation (I)	Climate heterogeneity (II)	Opioid reformulation (III)	Reformulation + climate het. (IV)
Simple assault	10,531	233 [203, 262]	205 [173, 238]	182 [170, 194]	194 [181, 206]
Intimidation	10,258	72 [55, 89]	74 [54, 94]	47 [41, 53]	52 [44, 60]
Aggravated assault	48,212	359 [291, 426]	378 [293, 464]	302 [269, 336]	324 [292, 357]
Rape	498,226	342 [105, 579]	325 [58, 592]	294 [244, 344]	312 [257, 367]
Homicide	9,707,561	1,956 [776, 3,137]	1,761 [310, 3,211]	1,298 [951, 1,644]	1,278 [911, 1,644]
Non-fatal violent crimes		1,005	982	825	882
All violent crimes		2,961	2,743	2,123	2,160

Notes: The estimation sample is between 2006 and 2023 on the 2,694 counties for which we have pre-policy opioid exposure information. Pre-policy means are population-weighted averages of the daily rate per 100,000 over 2006–2009. All four scenarios come from specifications without location- or time-specific temperature and precipitation effects, and all are evaluated in the post-reformulation regime at the county's own pre-policy opioid exposure. Columns (I) and (III) come from one regression and (II) and (IV) from the same regression fully interacted with climate terciles; within each pair, the first column switches off the opioid-attributable moderation (the temperature $\times$ exposure $\times$ post interaction) and the second leaves it in, so the difference is the avoided cost attributable to the reformulation. Each entry is a population-weighted average of the county-specific marginal effect. Annual costs multiply county-specific marginal effects by county population, 365 days and the unit cost, then aggregate. 95% confidence intervals in brackets, obtained by the delta method on the aggregated gradient with standard errors clustered at the county level. Confidence intervals are reported for each crime type separately; the aggregate rows sum estimates from distinct regressions and are reported as point estimates only.

Figure F1: Number of additional violent crimes for a 1°C increase in temperature



Notes: The figure reports estimated annual number of additional violent crimes induced by a 1°C increase in temperature based on the estimates for each category of crime reported in [Table F1](#). Annual counts multiply each county-specific effect by county population and 365 days, and sum across counties. The “No adaptation” bar switches off the moderation by the reformulation and reports the uniform temperature response interacted with opioid exposure. The “Climate heterogeneity” bar allows the temperature-crime relationship to vary by terciles of long-run average temperature. The “Opioid reformulation” bar uses the estimates reported in [Table D1](#). The “Opioid reformulation + climate heterogeneity” bar allows for the temperature-crime relationship to vary by terciles of long-run average temperature and by opioid exposure before and after the reformulation, using estimates reported in [Table D9](#), column 1, estimated for each crime category. Bar totals sum across separate regressions; confidence intervals for the underlying estimates by crime category are reported in [Table F1](#).